

Mathematical Companion

to

Introduction to Formal and Computational Argumentation

Chapters 0–15

Scholarly approved continuous reading edition

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The author approves the complete companion's content, organization and exercise correspondence within its declared scope. This edition completes the bounded Libra source-status wording correction. The exact AXPLR published-version comparison remains open. Scholarly approval does not establish publication readiness or close that source reservation.

Reading this companion

This text develops the mathematical objects behind the textbook’s reasoning methods: sources and agents, reconstructions and structured reasons, semantics and representation, public interaction and choices, computation and explanation, systems and bounded action. Definitions and declared assumptions precede the results that use them. The mathematical preparation begins in Chapter 0. The chapter numbers correspond approximately to textbook Chapters 0–15.

Finite constructions are the default where so declared; they do not restrict a source theorem that explicitly has a larger domain. Classical propositional consequence is the opening convention; Chapter 7 separately states the classical/SP source claim’s first-order source assumptions. A correct calculation concerns its specified input and query. It does not by itself establish interpretive adequacy, source truth, public authority, successful execution or human usefulness.

Notation and navigation. Results use a shared chapter-local counter: Definition 6.1 and Theorem 6.6, for example, belong to Chapter 6. Subsections and displayed equations have their own chapter-local numbers. Existing group numbering is retained; the supporting results 7.5a–d follow the reported source claim 7.5. Blue references lead to the relevant chapter, subsection or result. The common symbol $\text{Stable}(F) = \text{Stable}(F)$ denotes the stable extension family; the opening group’s abbreviation $\text{St}(F)$ has this same meaning. Chapter 2’s D_F and Chapter 6’s Def_F denote the same defense operator. **Cred** and **Skept** retain the nonempty-family convention of Definition 0.2. Local notation is typed at its point of use: a public commitment, an extension, an issue and a permitted option are distinct objects, even when letters recur. Different recurring graphs retain their names and edge sets.

Exercises and sources. Every chapter ends with the exact *Related textbook exercises* section from its delivered group, with its identifiers and subparts. These are connections to the approved textbook, not replacement exercise statements or a second solution collection. Contents entries lead to these sections. The textbook remains self-contained without the companion. Point-of-use source locators and proof-status notes are retained, with a common source guide at the end.

Status of this edition. The author approves the complete companion’s content, organization and exercise correspondence within its declared scope. This is whole-companion scholarly approval, including the developed classical/SP–PDL obstruction (Counterexample 7.5d). Reported claim 7.5 is refuted under the recorded construction through its full-conclusion requirement; existence or uniqueness separately is not disproved. No replacement theorem or changed domain is adopted. The wider Dung correspondences, general ASPIC+ rationality results and redundant-default invariance remain explicitly conditional imports, as authorized. This edition completes the bounded Libra source-status wording correction; approval does not establish publication readiness. AXPLR’s probability identity remains source-attributed, not developed or used as a premise; its full classification clause is not imported. Its zero-strength/tie question is specific to the checked corrected-proof version. The published-version PDF was identified but was not retrieved in the earlier or renewed bounded check. The exact published-version comparison remains open. The recurring-example index after Chapter 15 links the principal returning cases.

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0 Formal preparation: objects, consequences and finite constructions

Textbook correspondence: Chapter 0, Making a reasoning claim checkable. An arrow, a premise occurrence and a selected set have different types. Keeping those types visible prepares later representation claims.

0.1 Relations, models and output families

For a relation $R \subseteq A \times A$, let $R[X] = \{y \in A : \exists x \in X (x, y) \in R\}$. A path uses a sequence of R -edges; it need not be an R -edge. Thus $(a, b), (b, c) \in R$ does not imply $(a, c) \in R$. For a family $\mathcal{E} \subseteq 2^A$, the union forgets how members were grouped: $\bigcup \emptyset = \bigcup \{\emptyset\} = \emptyset$, although the first family has no output and the second one empty output.

Fix a propositional language L with classical Boolean valuations V . A formula set Γ is consistent when some $v \in V$ satisfies it. Put

$$\text{Mod}(\Gamma) = \{v \in V : v \models \Gamma\}, \quad \text{Th}(W) = \{\varphi \in L : \forall v \in W \ v \models \varphi\}, \quad \text{Cn}(\Gamma) = \text{Th}(\text{Mod}(\Gamma)).$$

Write $\Gamma \vdash \varphi$ for $\varphi \in \text{Cn}(\Gamma)$. Then $\text{Th}(\emptyset) = L$, while $\text{Cn}(\emptyset)$ contains just the tautologies. No models and no premises are different inputs.

Proposition 0.1 (Model intersection and monotonicity). *For arbitrary formula sets Γ, Δ ,*

$$\text{Mod}(\Gamma \cup \Delta) = \text{Mod}(\Gamma) \cap \text{Mod}(\Delta), \quad \Gamma \subseteq \Delta \implies \text{Cn}(\Gamma) \subseteq \text{Cn}(\Delta).$$

Proof. A valuation satisfies the union exactly when it satisfies every member of each set, proving both inclusions of the equality. If $\Gamma \subseteq \Delta$, every model of Δ is a model of Γ . A formula true in every model of Γ remains true in every model of Δ , including when there is no latter model. $\square$

Definition 0.2 (Guarded output queries). *For $\mathcal{E} \subseteq 2^A$, put*

$$\text{Cred}_{\mathcal{E}}(a) \iff \exists E \in \mathcal{E} \ a \in E, \quad \text{Skept}_{\mathcal{E}}(a) \iff \mathcal{E} \neq \emptyset \wedge \forall E \in \mathcal{E} \ a \in E.$$

The nonempty-family clause is a query convention; it does not change classical universal quantification over an empty domain.

Example 0.3. For $\mathcal{E}_0 = \emptyset$ and $\mathcal{E}_1 = \{\emptyset\}$, both guarded membership queries are false for every object. The unguarded universal statement is true for $\mathcal{E}_0$ and false for $\mathcal{E}_1$.

0.2 Orders, witnesses and quantifiers

For $\mathcal{X} \subseteq 2^A$, a member X is *minimal* if no member is its proper subset; it is *least* if it is a subset of every member. Maximal and greatest reverse inclusion. A least member is unique by antisymmetry. The two members $\{a\}$ and $\{b, c\}$ of their family are incomparable and both minimal, although their cardinalities differ.

Lemma 0.4 (Quantifier order). *For $Q \subseteq D \times B$,*

$$\exists d \in D \ \forall b \in B \ dQb \implies \forall b \in B \ \exists d \in D \ dQb.$$

The converse fails for finite nonempty sets. If $B = \emptyset$, the right side is true; the left side still requires $D \neq \emptyset$.

Proof. A witness for all b can be reused for each one. Take $B = \{b_1, b_2\}$, $D = \{d_1, d_2\}$ and $Q = \{(d_1, b_1), (d_2, b_2)\}$. Each task has an answer, but neither answer covers both. When B is empty the universal clauses are vacuous, while the existential clause still requires an element of D . $\square$

The negation of $\forall b \in B \exists d \in D dQb$ is $\exists b \in B \forall d \in D \neg dQb$. A failed candidate answer is not a complete failure certificate; every eligible answer must fail for one task.

0.3 Finite fixed-point constructions

Let A be finite and $F : 2^A \rightarrow 2^A$ be *monotone*: $X \subseteq Y$ implies $F(X) \subseteq F(Y)$. This differs from *inflationary*: $X \subseteq F(X)$ for every X .

Theorem 0.5 (Iteration from the bottom). *Set $X_0 = \emptyset$, $X_{n+1} = F(X_n)$. The sequence increases, stabilizes after at most $|A|$ strict increases, and its final value X_* is the least fixed point of F under inclusion.*

Proof. $X_0 \subseteq X_1$ since X_0 is empty. If $X_n \subseteq X_{n+1}$, monotonicity gives $X_{n+1} = F(X_n) \subseteq F(X_{n+1}) = X_{n+2}$. Each strict increase adds an element, so finiteness forces $X_m = X_{m+1} = F(X_m)$ after at most $|A|$ strict increases. For any fixed point Y , $X_0 \subseteq Y$. Inductively, $X_{n+1} = F(X_n) \subseteq F(Y) = Y$. Thus $X_m \subseteq Y$. $\square$

For reachability from $a \in A$ take $H(X) = X \cup \{a\} \cup R[X]$. The theorem gives the least successor-closed set containing a . Every added node has a path from a by induction on stages; every finite path ends at a member of the final set by induction on path length. These prove both directions of the reachability claim.

Counterexample 0.6. On nonempty A , the constant map $F(X) = \emptyset$ is monotone but not inflationary on nonempty X . Later defense operators need their own properties proved; bottom iteration can grow without global inflationarity.

Related textbook exercises

0.1–0.2 relation image, paths and loops; **0.3–0.4** countermodels and classical monotonicity; **0.5, 0.9** maximal consistency and inclusion orders; **0.6–0.8** certificates, empty families and quantifier negation; **0.10–0.11** finite closure and two-inclusion proof. Optional external **0.L1** compares specified readings and is not a mathematical prerequisite here.

1 Source methods, selected models and exact representation

Textbook correspondence: Chapter 1, Choosing and representing a reasoning method. The mathematical question is which operation on a source base an argument construction represents. A source method, an argumentative output family and a readback map are separate data.

1.1 Repairs and distinct pivots

Let K be a finite set of revisable formula occurrences and P a finite set of hard occurrences, with disjoint roles and $\text{Mod}(P) \neq \emptyset$. Logical tests ignore occurrence labels, but extraction preserves them. If an underlying formula appears in both roles, use distinct labels ℓ and a projection $\pi(\ell, \varphi) = \varphi$. In unlabelled notation assume $K \cap P = \emptyset$.

Definition 1.1 (Maximal compatible repair).

$$\text{MCS}(K, P) = \{T \subseteq K : \text{Mod}(T \cup P) \neq \emptyset, \nexists T' (T \subsetneq T' \subseteq K, \text{Mod}(T' \cup P) \neq \emptyset)\}.$$

Fixed-pivot consequence is $P \vdash_K \varphi$ iff $K \cup P \vdash \varphi$. Cautious and credulous repair consequence quantify respectively over every and some $T \in \text{MCS}(K, P)$ such that $T \cup P \vdash \varphi$, with a nonempty-family guard.

Proposition 1.2 (Finite existence and closure output). *$\text{MCS}(K, P)$ is nonempty. Its cautious conclusion set is exactly $\bigcap_{T \in \text{MCS}(K, P)} \text{Cn}(T \cup P)$.*

Proof. $\emptyset \cup P$ is consistent. Among finitely many compatible subsets of K , choose one of maximum cardinality. It has no compatible proper superset and is an MCS. The second equality expands membership in every repair's consequence set; consequence is taken after adjoining P in each member. $\square$

Example 1.3. For $K_0 = \{p, \neg p, q\}$ and $P = \emptyset$, $\text{MCS}(K_0, P) = \{\{p, q\}, \{\neg p, q\}\}$. Both repairs entail q ; only one entails p . Their separate closures must be intersected after consequence is taken.

Makinson's pivotal assumptions (Def. 2.1, p. 24) and default assumptions (Def. 2.4, p. 31) motivate these different operations. Our finite propositional repair and nonvacuous guard are the textbook's restricted setting, not a claim about every variant. If P is inconsistent, no compatible subset exists: $\text{MCS}(K, P) = \emptyset$. The guarded cautious and credulous repair queries are then rejected (false under the textbook query convention), whereas classical $\text{Cn}(K \cup P) = L$. The finite-existence proposition above deliberately excludes this input.

Fix $W_0 \subseteq V$ and let $C_{W_0}^{\text{fix}}(P) = \text{Th}(W_0 \cap \text{Mod}(P))$.

Proposition 1.4 (Fixed valuation and assumption identity). *For arbitrary K, P ,*

$$C_{\text{Mod}(K)}^{\text{fix}}(P) = \text{Cn}(K \cup P).$$

For fixed W_0 , $P \subseteq P'$ implies $C_{W_0}^{\text{fix}}(P) \subseteq C_{W_0}^{\text{fix}}(P')$.

Proof. $\text{Mod}(K) \cap \text{Mod}(P) = \text{Mod}(K \cup P)$ by model intersection. Apply Th, including when the set is empty. If $P \subseteq P'$, the tested valuations for P' are a subset of those for P ; universal truth over the larger set persists over the smaller. $\square$

This is the finite-book form of Makinson's fixed-valuation comparison (Def. 3.1, Lemma 3.2, pp. 60–61). It contains no preference relation. A rule pivot is different again: an inference pair (p, q) authorizes a forward inference without automatically making $p \rightarrow q$ a premise. From $p \rightarrow q$ and $\neg q$ classical reasoning gives $\neg p$; the inference pair alone gives no such backward application. Later chapters specify their particular rule-based constructions.

1.2 Selection can withdraw a conclusion

For valuation selection in this subsection restrict L to a finite atom vocabulary, so its full valuation set V is finite. Earlier fixed-valuation identities did not need this restriction. Let $<$ be a fixed strict partial order on V , where $u < v$ means u is preferred to v . Define

$$\text{Best}_{<}(P) = \{v \in \text{Mod}(P) : \nexists u \in \text{Mod}(P) (u < v)\}, \quad C_{<}(P) = \text{Th}(\text{Best}_{<}(P)).$$

Every nonempty finite $\text{Mod}(P)$ has a minimal member: following a strictly improving chain terminates. This selects valuations, not subsets of a source base. Separately, a preorder on the nonempty finite MCS family selects its maximal members. A strict-total lexicographic priority on source occurrences is a specific textbook policy, not a canonical argument preference. More precisely, order the occurrences in K strictly and compare two distinct repairs at the highest-priority occurrence in their symmetric difference; the repair containing it wins. This is a strict total order on the finite repair family, since a priority list gives each subset a unique Boolean membership vector and lexicographic comparison is transitive. For K_0 with $p \succ \neg p \succ q$, this rule selects $\{p, q\}$ rather than $\{\neg p, q\}$; reversing the first two priorities selects the other repair. Neither mutation changes the unprioritized MCS family.

Counterexample 1.5 (Change of admissible valuations). Write $f = p \rightarrow q$, $K = \{f\}$, $P = \{p\}$ and $P' = \{p, \neg q\}$. Rank v_{00}, v_{01}, v_{11} at 0 and v_{10} at 1, with smaller ranks preferred. Then $\text{Best}_{<}(P) = \{v_{11}\}$ but $\text{Best}_{<}(P') = \{v_{10}\}$: $q \in C_{<}(P)$ and $q \notin C_{<}(P')$. Meanwhile $\text{MCS}(K, P) = \{\{f\}\}$ and $\text{MCS}(K, P') = \{\emptyset\}$, so repair consequence also withdraws q . Fixed-pivot consequence for P' is all of L , because $K \cup P'$ is inconsistent. These operations have different definitions even where two answers agree.

If all four valuations tie, $\text{Best}(P) = \{v_{10}, v_{11}\}$, so preferential consequence already lacks q at P . The MCS family is unchanged. This rank variation refutes a general equation between preferential valuation selection and MCS repair selection. The repaired family $\{\emptyset\}$ at P' cannot be written $\emptyset$.

1.3 Base arguments and a full correspondence proof

Let B be a finite set of source-labelled formula occurrences. Write $\text{Cn}(\Gamma)$ for the closure of their projected formulas $\pi[\Gamma]$; entailment and consistency use the same projection. Define

$$\rho(B) = \{\Gamma \subseteq B : \Gamma \text{ consistent and } \nexists \Gamma' \subsetneq \Gamma \text{ } \text{Cn}(\Gamma') = \text{Cn}(\Gamma)\}.$$

The empty set belongs to $\rho(B)$. This is irredundancy for the whole closure, not minimal support for a designated conclusion. For $\Gamma, \Delta \in \rho(B)$ put

$$\Gamma \rightarrow_B \Delta \iff \exists \psi \in \Delta \text{ } \Gamma \vdash \neg \pi(\psi).$$

Let $F_B = (\rho(B), \rightarrow_B)$. A stable $E \subseteq \rho(B)$ has no internal attack and attacks every outside argument. The construction matches Rienstra, van der Torre and Yu, supplied manuscript Sec. 3.2, Defs. 3–5, pp. 10–11; stability uses Dung’s 1995 Defs. 5 and 13, pp. 326, 328.

Lemma 1.6 (Minimal entailment supplies a base argument). *If consistent $S \subseteq B$ entails α , then some $\Gamma \in \rho(B)$ has $\Gamma \subseteq S$ and $\Gamma \vdash \alpha$.*

Proof. Choose an inclusion-minimal subset $\Gamma \subseteq S$ entailing α , possible by finiteness. It is consistent. If a proper Γ' had the same closure, it would still entail α , contradicting minimality. $\square$

Theorem 1.7 (Stable base sets and maximal consistent sources). *For finite B and classical consequence, the map $S \mapsto \rho(S)$ maps $\text{MC}(B) = \text{MCS}(B, \emptyset)$ onto the stable sets of F_B :*

$$S \in \text{MC}(B) \implies \rho(S) \in \text{Stable}(F_B), \quad E \in \text{Stable}(F_B) \implies \exists S \in \text{MC}(B) \ E = \rho(S).$$

No premise-union extraction or injectivity is asserted.

Proof. If S is maximal consistent, members of $\rho(S)$ cannot attack each other: their premises lie in consistent S . For $\Delta \in \rho(B) \setminus \rho(S)$, irredundancy is intrinsic to Δ , so $\Delta \not\subseteq S$. Choose $\psi \in \Delta \setminus S$. Maximality makes $S \cup \{\psi\}$ inconsistent, hence $S \vdash \neg\pi(\psi)$. The lemma supplies $\Gamma \in \rho(S)$ entailing this negation, which attacks Δ . Thus $\rho(S)$ is stable.

Conversely let E be stable and $U = \bigcup_{\Delta \in E} \Delta$. Every $\Gamma \in \rho(B)$ with $\Gamma \subseteq U$ belongs to E : otherwise some $\Lambda \in E$ attacks Γ at $\psi \in \Gamma$; since $\psi \in U$, it occurs in a $\Delta \in E$, so Λ also attacks Δ internally.

The set U is consistent. Otherwise choose an inclusion-minimal inconsistent $C \subseteq U$ and any $\psi \in C$. $C \setminus \{\psi\}$ is consistent and entails $\neg\pi(\psi)$. By the lemma a base argument $\Gamma \subseteq C \setminus \{\psi\}$ entails this negation. The preceding paragraph puts Γ in E ; some $\Delta \in E$ contains ψ , an internal attack.

Extend U to a maximal consistent $S \subseteq B$. Every member of E belongs to $\rho(S)$. If $\Delta \in \rho(S) \setminus E$, stability of E gives $\Gamma \in E$ attacking Δ . Both sets lie in consistent S , impossible. Hence $E = \rho(S)$. $\square$

This is the statement of the supplied manuscript's Proposition 1, p. 11, which credits Chen, Liao and van der Torre (2024) and does not supply a proof there. The proof above is the companion's finite classical derivation, with source occurrences preserved as labels.

Example 1.8 (The two-sided base graph). For unlabelled $K_0 = \{p, \neg p, q\}$, the six base arguments are $\emptyset, \{p\}, \{\neg p\}, \{q\}, \{p, q\}$ and $\{\neg p, q\}$. Every argument containing p attacks each argument containing $\neg p$, and conversely: eight directed attacks. There are no other attacks. The theorem yields the two stable sets $\rho(\{p, q\})$ and $\rho(\{\neg p, q\})$. Their paired source bases are the two repairs; adding a tautological labelled premise to the source illustrates why an unpaired argument-set readback loses information.

Remark 1.9 (Readback is data). S need not equal $\bigcup \rho(S)$. A tautological source occurrence belongs to a maximal consistent S but to no closure-irredundant argument: deleting it preserves the closure. To preserve source identity retain $(S, \rho(S))$ or a specified extraction object. A one-node-per-formula graph also fails on $\{p, q, \neg(p \wedge q)\}$: every pair is consistent while the triple is not. Set-of-premises arguments expose this higher-order conflict.

Corollary 1.10 (Hard-premise policy filter). *For finite role-disjoint K, P with consistent P , let $B = K \cup P$ and*

$$\text{Pair}_P(B) = \{(S, \rho(S)) : S \in \text{MC}(B), \ P \subseteq S\}.$$

Then $\{S \cap K : (S, \rho(S)) \in \text{Pair}_P(B)\} = \text{MCS}(K, P)$.

Proof. For a retained pair, $T = S \cap K$ is compatible with P . A larger compatible $T' \subseteq K$ would make $T' \cup P$ a consistent proper extension of $S = T \cup P$, contradicting maximality. Conversely, for $T \in \text{MCS}(K, P)$, $S = T \cup P$ is consistent. A consistent proper extension within B would add a member of $K \setminus T$, contradicting repair maximality. Thus $S \in \text{MC}(B)$, the theorem supplies $\rho(S)$, and role disjointness gives $S \cap K = T$. $\square$

Consequences are evaluated on $(S \cap K) \cup P$. The filter $P \subseteq S$ belongs to readback; nothing here represents every valuation preference or makes unprioritized stability select lexicographically preferred repairs.

Related textbook exercises

1.1–1.3 repair witnesses, cautious and credulous status, and source priority; **1.4, 1.5** base construction, stable sets and readback; **1.6** inconsistent hard-input guard; **1.7** higher-order conflict; **1.8** priority mutation; **1.9(a–d)** flat valuation rank, fixed-pivot identity, nonempty empty repair and labelled stable pair.

2 Sources, defense and collective formation

Textbook correspondence: Chapter 2, Reasons across agents, including its two supplied studies. Association of an argument with a source is an input relation, not a certification of public assertion, endorsement or expertise.

2.1 Three defense predicates

Definition 2.1 (Agent framework and defense). *An agent argumentation framework is $F = (A, R, S, \text{src})$, with finite argument set A , attack $R \subseteq A \times A$, source set S and association $\text{src} \subseteq A \times S$. The whole S need not be finite. Let $A_i = \{d \in A : (d, i) \in \text{src}\}$, $S_x = \{i \in S : (x, i) \in \text{src}\}$ and $\text{Att}(x) = \{b \in A : (b, x) \in R\}$. For $E \subseteq A$ put*

$$\text{OrdDef}(E, x) \iff \forall b \in \text{Att}(x) \exists d \in E (d, b) \in R,$$

$$\text{IndDef}(i, E, x) \iff i \in S_x \wedge \forall b \in \text{Att}(x) \exists d \in E \cap A_i (d, b) \in R,$$

$$\text{IndDef}^\exists(E, x) \iff \exists i \in S_x \text{IndDef}(i, E, x),$$

$$\text{CollDef}(E, x) \iff \forall b \in \text{Att}(x) \exists i \in S_x \exists d \in E \cap A_i (d, b) \in R.$$

The ordinary predicate is Dung’s defense by an explicit set (1995, Def. 6(1), p. 326); it need not have conflict-free E . The agent framework and latter two source tests follow the supplied 52-page agent-defense proof, Defs. 5–6, internal pp. 4–5. Its published 2026 article has different pagination. The source’s collective predicate has no coalition parameter.

Proposition 2.2 (Implications and strictness). *For every F, E, x ,*

$$\text{IndDef}^\exists(E, x) \implies \text{CollDef}(E, x) \implies \text{OrdDef}(E, x).$$

Both converses fail for finite frameworks. If $\text{Att}(x) = \emptyset = S_x$, collective and ordinary defense hold, while existential individual defense fails.

Proof. For the first implication reuse one witnessing agent for each attacker. For the second discard the agent component. For failure of the first converse, take two attackers b_1, b_2 of x , $d_1 \rightarrow b_1$, $d_2 \rightarrow b_2$, $E = \{d_1, d_2\}$ and $S_x = \{i_1, i_2\}$, where only (d_1, i_1) and (d_2, i_2) are defender associations. Each attack has a target-associated answer, but neither agent covers both. For failure of the second converse, take $b \rightarrow x$, $d \rightarrow b$, $E = \{d\}$, $S_x = \{i\}$, and $d \notin A_i$. Ordinary defense holds but no eligible source supplies the counterattack. All unlisted edges and associations are absent. In the stated empty case the attacker clauses are vacuous, while an individual witness requires an element of empty S_x . $\square$

The first counterexample is the pattern of the textbook’s five-argument case: Alice supplies c_1 against b_1 and Bob supplies c_2 against b_2 , both associated with target a . A defender in $A \setminus E$ remains unavailable.

2.2 Coalitions as derived certificates

Definition 2.3 (Witness and coverage). *A coalition witness for E, x is $C \subseteq S_x$ with*

$$\text{CWit}(C, E, x) \iff \forall b \in \text{Att}(x) \exists i \in C \exists d \in E \cap A_i (d, b) \in R.$$

It is inclusion-minimal when no proper subset is a witness. For $i \in S_x$ let $U_i = \{b \in \text{Att}(x) : \exists d \in E \cap A_i (d, b) \in R\}$.

Coalitions are explanatory certificates derived from the collective predicate, not a replacement definition in the source.

Proposition 2.4 (Coverage normal form). *For $C \subseteq S_x$,*

$$\text{CWit}(C, E, x) \iff \bigcup_{i \in C} U_i = \text{Att}(x).$$

Moreover, $\text{CollDef}(E, x)$ iff S_x is a witness. Individual defense holds iff some singleton $\{i\} \subseteq S_x$ is a witness.

Proof. Every U_i contains only attackers. An attacker belongs to their union exactly when an eligible i supplies through $E \cap A_i$ a counterattack. Substitute $C = S_x$ for the collective clause; for a singleton this is fixed-agent defense, then choose i existentially. $\square$

Remark 2.5. A singleton witness need not be a singleton *minimal* witness. If x is unattacked, the empty coalition covers the empty attacker set and is uniquely minimal. Individual defense still holds when S_x is nonempty, and fails when S_x is empty. An attacked target with $S_x = \emptyset$ has no coalition witness.

Lemma 2.6 (Private-attacker test). *For finite $C \subseteq S_x$, a witness C is minimal iff for each $i \in C$ there is an attacker $b_i \in U_i \setminus \bigcup_{j \in C \setminus \{i\}} U_j$. For $C = \emptyset$ the condition is vacuous; it is minimal exactly when it covers the empty attacker set.*

Proof. Deleting i loses coverage precisely when such b_i exists. If every one-agent deletion fails, any proper subset $D \subsetneq C$ lies inside some deletion $C \setminus \{i\}$ and cannot recover its missing attacker, since coverage only decreases when members are removed. Conversely a minimal witness cannot retain coverage after any deletion. $\square$

Example 2.7 (Minimal does not mean smallest). For attackers $\{b_1, b_2\}$ let $U_{\text{Ann}} = \{b_1, b_2\}$, $U_{\text{Bob}} = \{b_1\}$ and $U_{\text{Cara}} = \{b_2\}$. Both $\{\text{Ann}\}$ and $\{\text{Bob}, \text{Cara}\}$ are inclusion-minimal; only the singleton minimizes cardinality. These coverage sets occur in the textbook’s complete Transfer input.

Theorem 2.8 (Finite minimal-witness enumeration). *Assume S_x is explicitly listed and finite. Compute the U_i , consider all $C \subseteq S_x$ in increasing cardinality, skip C if an accepted $D \subsetneq C$ exists, and otherwise append C exactly when it covers $\text{Att}(x)$. The procedure terminates and returns exactly the inclusion-minimal witnesses, regardless of the order among equal-size candidates.*

Proof. There are $2^{|S_x|}$ candidates. Every coverage test inspects finite A, E , so the procedure terminates. Every appended C covers. If a proper subset D covered, it was considered earlier. If accepted, C would be skipped. If skipped, D contained an accepted witness $D' \subseteq D \subsetneq C$, again forcing C to be skipped. It cannot have failed coverage and also be a witness. Thus appended C is minimal.

Conversely, a minimal witness C has no accepted proper subset, because accepted sets cover. Therefore it is not skipped and passes coverage. Proper inclusion implies smaller cardinality, so tied-size candidates cannot change this argument. All and only minimal witnesses are returned. $\square$

This theorem integrates the mathematics of textbook Study 2.S1 and its collected proof without reproducing the exercise or forming another answer collection. It makes no efficiency claim for large source sets.

2.3 Aggregating graphs and aggregating answers

Source-indexed defense is a property of one fixed graph. A different operation combines attack graphs. Let $A = \{a, b, c\}$ and

$$R_1 = \{(a, b), (b, c)\}, \quad R_2 = \{(b, c), (c, a)\}, \quad R_3 = \{(c, a), (a, b)\}.$$

Every unlisted edge, including loops, is absent. Strict edge majority gives

$$R_M = \{(a, b), (b, c), (c, a)\}.$$

A stable $E \subseteq A$ is conflict-free and attacks every outsider.

Counterexample 2.9 (Aggregation and stable evaluation). Each R_i is acyclic with a unique stable set: $E_1 = \{a, c\}$, $E_2 = \{a, b\}$, $E_3 = \{b, c\}$. The majority graph is a cycle with no stable set. Membership majority over E_1, E_2, E_3 gives A , which contains internal attacks in the majority graph.

Proof. Each displayed edge has two supporters; reverse edges and loops have none. In R_1 , unattacked a must belong to a stable set, excluding b ; then c must belong because no selected node attacks it. $\{a, c\}$ is conflict-free and attacks outsider b . Rotate names for R_2, R_3 . In R_M , the empty set attacks no outsider, each singleton misses one outsider, and every pair and the full set contain an attack. This exhausts 2^A . Every argument belongs to two E_i , so membership majority is A , containing $a \rightarrow b$, for instance. $\square$

This is a finite failure of acyclicity preservation for the stated rule and profile, and of the displayed order of aggregation and evaluation. It is not an impossibility theorem for every collective method.

2.4 Grounded construction and output types

For finite $F = (A, R)$ define the defense operator

$$D_F(X) = \{x \in A : \forall b \in \text{Att}(x) \exists d \in X (d, b) \in R\}.$$

If $X \subseteq Y$, a defender available in X is available in Y , so $D_F(X) \subseteq D_F(Y)$. Chapter 0's theorem therefore proves that bottom iteration reaches the least fixed point, the grounded set G_F . It is not globally inflationary: for $x \leftrightarrow y$ and $y \rightarrow z$, $\{z\}$ attacks nothing, so $D_F(\{z\}) = \emptyset$.

For R_1, R_2, R_3 the bottom traces are respectively

$$\emptyset, \{a\}, \{a, c\}, \{a, c\}; \quad \emptyset, \{b\}, \{a, b\}, \{a, b\}; \quad \emptyset, \{c\}, \{b, c\}, \{b, c\}.$$

In each path the initially accepted node is unattacked and counterattacks the attacker of the other endpoint. In the majority cycle no node is unattacked, so $D_{F_M}(\emptyset) = \emptyset$ and $G_{F_M} = \emptyset$. Membership majority over the three grounded sets is A , while evaluation of the majority graph gives $\emptyset$. Its grounded-output family is $\{\emptyset\}$, one member; its stable-output family is $\emptyset$, no member.

The supplied Study 2.S2 also changes a separate graph. On $A = \{x, y, z\}$ with $x \leftrightarrow y$ and $y \rightarrow z$, every node has an attacker, hence bottom iteration stops at $G = \emptyset$. Add a new unattacked t with the sole new edge $t \rightarrow y$. The stages now are

$$\emptyset \mapsto \{t\} \mapsto \{t, x, z\} \mapsto \{t, x, z\}.$$

The first stage admits only unattacked t ; it attacks y , defending both x and z at the next stage. No member attacks t , so y cannot be defended against that new attacker. The output families are respectively $\{\emptyset\}$ and $\{\{t, x, z\}\}$, each with one member.

A status ballot $\beta_i(x) \in \{\text{in}, \text{out}, \text{undec}\}$ can define a strict-majority Boolean predicate $\text{MajIn}(x)$. Failure of that predicate leaves the two other statuses open. Without semantic constraints on reports and a complete aggregation rule, no three-valued collective labelling theorem follows. Association also does not prove a public act.

Related textbook exercises

2.1–2.4 AAF data and defense certificates; **2.5–2.8** minimal coalitions, explicit defender sets, empty boundaries and transfer inputs; **2.9–2.10** ballot indicator and source-relation change; **2.11(a–d)** majority graph and stable outputs; **2.S1** enumeration proof; **2.S2(a–d)** grounded trace, output family and non-inflationarity.

3 Reconstructing arguments

Textbook correspondence: Chapter 3 and Study S3.1. Reconstruction chooses formal objects before a graph query is run. An exact query establishes a fact about its input representation; whether that input records the intended exchange is a separate evidential question.

3.1 Typed records and warranted edges

Definition 3.1 (Reconstruction record). *Let O, C, U, S, I, Ω be, respectively, sets of text occurrences, normalized claim contents, argument-unit identifiers, speakers, issues and options. Keep their roles disjoint even if their printed labels coincide. An occurrence map $c : O \rightarrow C$ records attributed content. An argument unit $u \in U$ has a finite set $P_u \subseteq O$ of premise occurrences, a conclusion occurrence $q_u \in O$, and a stated or analyst-supplied warrant. Its content projection is $(\{c(o) : o \in P_u\}, c(q_u))$. An act record is a tuple (s, t, κ, τ, o) of speaker, time, act kind, typed target and, where applicable, utterance occurrence. A target may be a claim, unit, earlier act, issue or option; its type is part of the record. The provenance map marks each occurrence and warrant as supplied, reported from another source or analyst-added. A reconstruction also declares its purpose and the queries that its graph is intended to answer.*

An assertion of p and a request for grounds for p can share content p without being the same act. An act withdrawing an earlier assertion targets that act and changes sponsorship; it does not create $\neg p$ or delete the historical occurrence. The projected content unit $(\{b\}, a)$ records an offered inference, not a truth-functional entailment $b \vdash a$. The warrant that makes the inference relevant may be an explicit rule, an attributed scheme, or an unresolved item.

Example 3.2 (Stipulated smoking reconstruction). Let a propose an indoor public smoking ban, b report harm to bystanders, c concede a liberty interest and d limit liberty where others are harmed. These four contents occur in the textbook’s short paragraph. The admission claim e , business-cost claim f , and mitigation claim g are added debate material. Neither expert testimony for e nor an actor’s practical goal occurs in that paragraph. An expert route would require, as separate premises, a source’s testimony, relevant expertise and domain fit; a practical route would require an actor, a goal and a means claim. A question about testimony requests a missing premise. It is not by itself a contrary assertion or an attack edge. An answer concerning adverse consequences may instead supply a competing reason, but that answer has to be recorded.

One possible expert warrant has linked premises

$$\text{Expert}(E, D), \text{InDomain}(e, D), \text{Says}(E, e) \Rightarrow \text{Plausible}(e);$$

removing any one of these premises breaks that instance’s route. A distinct practical warrant has the shape

$$\text{Goal}(s, H), \text{Means}(B, H) \Rightarrow \text{PracticallyOught}(s, B).$$

Its output concerns an action B , whereas the expert route assesses a proposition e . Separately warranted units from b and e can converge on a ; this is not the same structure as one linked unit with both b, e as necessary premises. A goal such as reduced harm states a desired outcome; a value such as public health explains why it matters, and neither alone establishes that the proposed means works.

For the graph fragment, take $V = \{a, b, c, d, e, f, g\}$ and $G = (V, E^+, E^-)$, where E^+ and E^- are typed support and attack relations on these claim-labelled vertices. This graph is a projection of a reconstruction: it retains neither who performed an act nor whether an edge was attributed to a participant or supplied by an analyst. A reply relation alone does not determine support or attack.

Proposition 3.3 (Information lost under a content projection). *There is no function of the content-labelled vertex set alone that returns every reconstruction’s local attackers of a , or its public sponsorship record.*

Proof. Keep all seven contents fixed. In one reconstruction put (c, a) in the attack relation; in another omit it. The content-labelled vertex sets are identical, but c belongs to the first local attacker set and not the second. A function on the shared projection would have to give both answers. For sponsorship, keep the same words and their content while recording one occurrence as a speaker’s assertion in one act record and as an interlocutor’s question in another. The first record sponsors the claim under the declared local convention, while the second does not. Again the proposed function has identical input and distinct required outputs. $\square$

This is an information-loss result, not a proof that either annotation is faithful. The issue *Should the ban be adopted?* and the option *adopt the ban* likewise have distinct types: a question with an acceptance standard is not an executable alternative.

3.2 Bounded graph questions

Definition 3.4 (Local graph queries). *For $x \in V$ and $k \in \mathbb{N}$, let $\text{Supporters}_k(x)$ contain vertices z for which an all-support walk from z to x has length between 1 and k , inclusive. Repeated vertices are allowed and length zero is excluded. Put*

$$\text{Attackers}(x) = \{y : (y, x) \in E^-\}, \quad \text{Defenders}(x) = \{z : \exists y ((z, y) \in E^- \wedge (y, x) \in E^-)\}.$$

The latter is a two-attack witness query, not the set-based defense operator used for abstract semantics in textbook Chapter 6.

Define $P^+(X) = \{z : \exists y \in X (z, y) \in E^+\}$. An enumeration may attach a predecessor z to a stored witness walk beginning at y ; it must retain the walk if witnesses are requested, even when a vertex has appeared before.

Theorem 3.5 (Finite bounded-walk enumeration). *Set $T_0(x) = \{x\}$ and $T_{j+1}(x) = P^+(T_j(x))$. Then*

$$\text{Supporters}_k(x) = \bigcup_{j=1}^k T_j(x).$$

For finite V, E^+ and finite k , a procedure that extends and stores each witness walk for $j = 0, \dots, k-1$ terminates and returns exactly the required vertices and witnesses. In particular, $\text{Supporters}_0(x) = \emptyset$.

Proof. By induction on j , $z \in T_j(x)$ exactly when a length- j support walk from z to x exists. The length-zero case has only the walk (x) . For the step, an edge $z \rightarrow y$ followed by a length- j walk from y is a length- $j+1$ walk; conversely, remove the first edge of any length- $j+1$ walk to obtain such a witness y . Prepending every available edge therefore generates all and only the next witnesses. Each level has finitely many walks, as there are finitely many edges and the depth is bounded. Unioning levels 1 through k proves the formula; for $k=0$ the union is empty. $\square$

Example 3.6 (A cycle is a witness). If $E^+ = \{(x, y), (y, x)\}$, then $T_1(x) = \{y\}$ and $T_2(x) = \{x\}$; hence $\text{Supporters}_2(x) = \{x, y\}$. The length-two witness $x \rightarrow y \rightarrow x$ repeats x . Replacing walks by simple paths changes the declared query.

The textbook fixes the following two annotated graphs:

$$\begin{aligned} E_0^+ &= \{(b, a), (e, a), (g, a)\}, & E_0^- &= \{(c, a), (f, a), (d, c), (g, f)\}, \\ E_1^+ &= E_0^+ \cup \{(d, a)\}, & E_1^- &= E_0^- \setminus \{(d, c)\}. \end{aligned}$$

The graph R_1 changes the target of d : it is direct support for a instead of a recorded response to the liberty objection. In R_0 , the coarse attack $d \rightarrow c$ answers the *use* of the conceded liberty interest against the ban; it does not assert that liberty interests are false. A fictional prediction P instead takes R_0 and *relabels* (d, c) as support, without adding (d, a) .

Proposition 3.7 (Three query signatures). *For $x = a, k = 2$, the three query answers are*

	R_0	R_1	P
Supporters ₂ (a)	$\{b, e, g\}$	$\{b, d, e, g\}$	$\{b, e, g\}$
Attackers(a)	$\{c, f\}$	$\{c, f\}$	$\{c, f\}$
Defenders(a)	$\{d, g\}$	$\{g\}$	$\{g\}$

Proof. The only support edges ending at a in R_0 start at b, e, g ; no support edge enters any of these vertices. In R_1 the new support edge starts at d , again with no support predecessor. In P , (d, c) cannot be followed by a support edge from c to a , because (c, a) remains an attack. The two attacks on a start at c, f in all three graphs. In R_0 , the two attack–attack walks are $d \rightarrow c \rightarrow a$ and $g \rightarrow f \rightarrow a$. Removing the first attack in R_1 or relabelling it in P leaves only the second. $\square$

If a requirement independently fixed R_0 as the reference, the computed answer

$$\text{Defenders}_P(a) = \{g\}$$

is exact for P . It does not meet the specified need to expose the response d . The proposition checks the graph query, not the interpretive choice of R_0 over another defensible reconstruction.

3.3 Status, countermodels and evaluation

For a generated text u , a local transparency condition is

$$\forall q \in \text{Claims}(u) \quad \text{Grounded}(q) \vee \text{Referenced}(q) \vee \text{UnsupportedLabel}(q).$$

It checks whether a factual claim’s support status is exposed. A label $\text{UnsupportedLabel}(q)$ can make the disjunction true without making q true or well supported. For a finite map of critical-question identifiers to $\{\text{open}, \text{answered}\}$, the open-item query scans the entries and returns just the identifiers whose stored value is **open**. This is exact for the recorded status by inspection of each entry, including the empty map; it does not test the adequacy of an answer marked **answered**.

Counterexample 3.8 (A missing source is not a contrary claim). Let $u = \text{Says}(E, e)$. A request for the source of u that receives no answer supplies no classical premise $\neg u$, still less $\neg e$. Even in a changed theory with r and strict $r \rightarrow \neg u$, the valuation $r = \text{true}, u = \text{false}, e = \text{true}$ is a model; thus $\neg e$ is not a consequence. Likewise a side-effects fact f alone has a valuation $f = a = \text{true}$, refuting $f \vdash \neg a$. A declared strict bridge $f \rightarrow \neg a$ changes the theory; a defeasible route $f \Rightarrow \neg a$ instead creates a contestable reason, which can face the mitigation consideration g .

Study S3.1 declares *exactly eight* candidate ordered pairs, with R_0 as the reference and P as a fictional prediction. One relation label differs, giving error $1/8$ and accuracy $7/8$. The nonempty defender reference answer $\{d, g\}$ has one retrieved member, giving recall $1/2$. These fractions use the declared denominator and query; adding unlisted negative pairs changes the benchmark. A local five-component quality score in Exercise 3.6 is a stipulated rubric, not a validated measure or truth probability. In that convention

$$Q(u) = \frac{Rel + Evid + CQ + Coh + Dial}{10};$$

the assigned vector $(2, 0, 1, 2, 0)$ gives $Q = 5/10$, and changing just evidence to 2 gives $Q = 7/10$. The arithmetic proves no validity of the component assignments.

Proposition 3.9 (Relation accuracy cannot certify a query). *For every integer $N \geq 2$, a fixed N -pair relation-label test can have accuracy $(N - 1)/N$ while recall of a nonempty local defender answer is zero.*

Proof. Make the sole defender of a target a the attack–attack witness $d \rightarrow c \rightarrow a$. Include (d, c) and (c, a) among the N candidate ordered pairs. For $N > 2$, add isolated vertices as needed to supply distinct extra pairs, labelled **none** in both graphs. Mislabel (d, c) as **none** in the prediction and label the other $N - 1$ pairs correctly. Exactly one pair is wrong, so accuracy is $(N - 1)/N$; the predicted defender answer misses the only reference member, so recall is $0/1 = 0$. $\square$

This construction is about a chosen formal benchmark, not observed mining performance. Annotation reliability, stakeholder requirements, and a user’s ability to use an explanation call for their own evidence.

Related textbook exercises

3.1 occurrences, additions and public acts; **3.2(a–e)**, **3.3** scheme warrants and critical-question roles; **3.4–3.5** R_0 , R_1 , P , bounded walks and witnesses; **3.6–3.8** local rubric, transparent status and finite open-item query; **3.9–3.10** missing-source and cost countermodels; **3.11**, **Study S3.1** fixed-candidate extraction and defender recall; **3.12** external literature comparison, not a theorem prerequisite.

4 Choosing the question

Textbook correspondence: Chapter 4. The same custody case supports an inferential question, a public responsibility question and a practical comparison. Their outputs have different types and need different inputs.

4.1 Local inference and public state

The fictional facts $w_1, \dots, w_5$ describe a child’s wish and understanding, the mother’s lesser wealth, the father’s work absence, and a stipulated code premise that the child’s opinion must be considered. Let m be the proposition recommending residence with the mother. Let M be the option of that arrangement and F the alternative option. In particular, $m \notin \{M, F\}$: a proposition about a proposal is not the arrangement itself.

An inquiry into the child’s understanding seeks evidence about w_2 ; persuasion seeks to defend a position against an interlocutor; deliberation compares what to do with M and F . The institution’s stipulated purpose is the child’s welfare, while a participant can seek her own proposed outcome. A goal describes a desired state, and a value explains why it matters. Greater resources and work absence are factual considerations whose relevance to care and comparative force need separate assessments. None of these purposes turns a permitted public move into the best next move automatically.

For this fragment the following are supplied defeasible inference permissions, not material implications:

$$\begin{aligned} r_1 : w_1, w_2 \Rightarrow m, \quad r_2 : w_3 \Rightarrow \neg m, \\ r_3 : w_4 \Rightarrow m, \quad r_4 : w_5, w_1 \Rightarrow \text{legallyRelevantPreference.} \end{aligned}$$

Let A_1 use r_1 and A_2 use r_2 . In this *local* comparison the contradictory conclusions attack each other. Stipulate $A_1 \succ A_2$ and that an attack by X on Y survives exactly if $Y \not\succeq X$.

Proposition 4.1 (Scope of the custody priority). *Under that local rule the attack $A_1 \rightarrow A_2$ is a defeat and $A_2 \rightarrow A_1$ is not. Neither the status of m under an acceptance semantics nor permission of M follows from this calculation alone.*

Proof. For $A_1 \rightarrow A_2$, its target A_2 is not preferred to A_1 , so the stipulated survival condition holds. For the reverse attack, its target A_1 is preferred to A_2 , so the condition fails. The definition has no acceptance operator and no map from an argument’s conclusion to an option permission. To see non-entailment formally, retain the same two arguments and priority while assigning either truth value to a fresh predicate $\text{Perm}(M)$; both expansions satisfy every premise of the local calculation. An acceptance semantics likewise requires additional selection clauses. The conclusion of r_4 is relevance of the child’s preference, not m . $\square$

Definition 4.2 (Limited public fragment). *Give each assertion act its own identifier and an active flag. For a speaker s and proposition p , derive $\sigma(s, p) = 1$ exactly when some active assertion act by s has content p , and 0 otherwise. A demand record $(d, s, \tau, \kappa, \text{status})$ identifies demander, requested speaker, typed target, requested kind of reason, and whether that request is open or answered. Keep an append-only act history. In the textbook’s local convention, assertion sets sponsorship and responsibility for grounds; a challenge undertakes opposition and requests an answer from the first speaker; a question requests grounds without undertaking opposition; an answer to the specified target closes that request but leaves responsibility for offered material. Withdrawal targets and deactivates one identified assertion act without deleting it or independently offered grounds. Other active assertions of the same content, if any, still sponsor it.*

The convention is deliberately narrower than a Chapter 9 dialogue protocol. It does not decide legality of arbitrary turns, who can challenge an individual premise, or whether offered material satisfies an issue's evidential standard.

Proposition 4.3 (The bounded public trace). *Starting with no recorded demands, let t_0 be the mother's assertion of m ; t_1 the father's challenge to it; t_2 the mother's answer with linked w_1, w_2 ; t_3 her request for his opposing grounds; and t_4 his answer with w_3 . The two requests in this fragment are answered after t_4 . Her responsibility for w_1, w_2 and his for w_3 remain. An ensuing withdrawal of her assertion removes only her current sponsorship of m .*

Proof. At t_0 , the assertion sets $\sigma(\text{mother}, m) = 1$. At t_1 , the challenge opens a demand for her answer and records the father's opposition. By the answer clause, t_2 closes that demand and records her offered pair, without certifying it. At t_3 , the question opens a demand on the father for the grounds of his challenge without creating a new contrary assertion. By the same clause t_4 closes that demand and records his offered ground. No other demand is introduced in this five-act fragment. The withdrawal clause then deactivates the mother's unique assertion of m in this fragment, so her derived $\sigma(\text{mother}, m)$ becomes 0. It has no operation deleting either answer, its premises or an earlier act. $\square$

A judge's possible t_5 request for the legal relevance of w_1 is intelligible. The fragment has no target-specific legality or discharge clause for that further move; its status cannot be certified by Proposition 4.3. Nor does answering the two recorded requests establish w_2 or w_3 to a decision maker's standard. The allocation of a burden identifies who must supply material; an issue standard identifies what showing is sufficient. For example, an issue record asking whether w_2 is established under a stated standard is distinct from the option M and from a public act asking for evidence of w_2 . A request may be answered while the issue remains unestablished.

4.2 Comparative permission and selection

Definition 4.4 (Two-option record model). *Fix distinct feasible options o, f and a complete, finite list D of records (g, x, y, j, r) , where $(x, y) \in \{(o, f), (f, o)\}$, g is a ground, and j, r are stipulated nonnegative exact weights. A record favours its second field against its third. Write $J_D(o, f)$ and $R_D(o, f)$ for the sums of its justifying and requiring weights in this direction, counting each record once. Empty sums are zero. Define*

$$\begin{aligned} \text{Perm}_D(o \mid f) &\iff J_D(o, f) \geq R_D(f, o), \\ P_D &= \{x \in \{o, f\} : \text{Perm}_D(x \mid \text{the other option})\}. \end{aligned}$$

A selector is an additional policy on permitted options; permission itself returns a set, possibly empty or plural.

The ground, favoured option and foil are three separate fields. The weights represent different practical forces. A missing weight means this numeric function's input is incomplete; it is not an empty list and cannot be silently treated as zero.

Theorem 4.5 (Pairwise computation and a one-coordinate change). *Scanning a valid list once, accumulating J_D and R_D for each ordered pair, and applying the two displayed inequalities returns exactly P_D . If only justifying weights on records favouring o against f are increased, while all other fields and weights are fixed, then permission of f is unchanged and permission of o can change only from false to true.*

Proof. Partition the finite records by their ordered option pairs. The four accumulators are initially zero; after any prefix, each is the corresponding sum over exactly that prefix, by

induction on the number of scanned records. At the end they equal the four sums in the definition, so the two tests give exactly P_D . Under the specified change only $J_D(o, f)$ may increase. The other test $J_D(f, o) \geq R_D(o, f)$ has neither side changed. In the test for o , the left side can only increase and the right side is fixed; a true inequality cannot become false. $\square$

Example 4.6 (Two distinct assessments). The supplied records give the child's preference for M against F weights $(JW, RW) = (8, 4)$, the father's absence for M weights $(5, 2)$, and the father's wealth for F weights $(3, 1)$. Thus $J(M, F) = 13, R(M, F) = 6$ and $J(F, M) = 3, R(F, M) = 1$. The two tests are $13 \geq 1$ and $3 < 6$, so $P_D = \{M\}$. Changing only wealth's justifying weight from 3 to 7 makes $7 \geq 6$ while the test for M remains $13 \geq 1$. Now $P_{D'} = \{M, F\}$. This is a change of stipulated practical force, not a change of $w_1, \dots, w_5$.

Two $(JW, RW) = (1, 1)$ records, one in each direction, give a nonzero double tie and permit both options. For equality in just one test, let the o -record have $(1, 2)$ and the f -record $(1, 1)$. Then $J(o, f) = R(f, o) = 1$, so o passes; $J(f, o) = 1 < R(o, f) = 2$, so f fails. The comparison is non-strict at its boundary, but permission need not be symmetric.

Proposition 4.7 (A fact-only permission function is impossible). *On a domain permitting both of the preceding completed assessments, no function of $w_1, \dots, w_5$ alone returns the two-option permission set for every completion.*

Proof. The two cases share all five facts and differ only in one authorized justifying weight. Their permission sets are $\{M\}$ and $\{M, F\}$. A function applied to the same five-fact input must return the same value, a contradiction. $\square$

The model therefore asks for the missing assessments and the authority to supply them. By contrast, an *explicitly complete empty* record list gives four zero totals and both tests $0 \geq 0$; it permits both options. A request for a unique option with $P_D = \{M, F\}$ requires a selection policy.

Proposition 4.8 (Finite selection after permission). *If $P_D \neq \emptyset$ and a supplied strict total priority orders the two options, the highest-priority member of P_D is unique. If $P_D = \emptyset$, this rule returns no selection.*

Proof. A nonempty subset of two options has either one member or both. In the singleton case that member is unique. In the two-member case strict totality compares the members in exactly one direction, giving a unique higher one. The empty set has no member to select. $\square$

Permission and selection are thus separate outputs; neither is execution or achievement. A public answer that might revise a weight needs an interface stating who may authorize the revision, which record it changes and when the calculation reruns. No utterance alone changes D . Similarly a rule result, an issue standard and a value assessment may be placed in a combined service only after their interfaces are stated.

Related textbook exercises

4.1 local conflict and missing acceptance rule; **4.2** t_0 – t_5 , sponsorship and response scope; **4.3**, **4.7** exact two-option sums, empty and tied cases; **4.4**–**4.6** type preservation and scoped hybrid interfaces; **4.8** missing weights, explicit empty records and missing selector; **4.9** external comparison of requirements and methods.

5 Formalising argumentation

Textbook correspondence: Chapter 5. The construction in this chapter produces arguments and a preference-resolved relation. Its graph is an input to the acceptance semantics of Chapter 6, not an acceptance result.

5.1 Canonical signature and recursive arguments

Definition 5.1 (System and knowledge base). *An ASPIC+ argumentation system has the signature*

$$AS = (\mathcal{L}, \bar{\cdot}, \mathcal{R}, n), \quad \mathcal{R} = \mathcal{R}_s \dot{\cup} \mathcal{R}_d.$$

The language $\mathcal{L}$ has a contrary map $\bar{\cdot} : \mathcal{L} \rightarrow 2^{\mathcal{L}}$. Here ψ is a contradictory of φ if $\psi \in \bar{\varphi}$ and $\varphi \in \bar{\psi}$; it is an asymmetric contrary if only the first membership holds. The canonical signature requires at least one contradictory for every formula. The strict and defeasible rule sets are disjoint; $n : \mathcal{R}_d \rightarrow \mathcal{L}$ names defeasible rules. A knowledge base is $\mathcal{K} = \mathcal{K}_n \dot{\cup} \mathcal{K}_p$, with axioms $\mathcal{K}_n$ and ordinary premises $\mathcal{K}_p$. These are model roles, not certifications of real-world truth.

A strict rule $\varphi_1, \dots, \varphi_m \rightarrow \psi$ is an inference permission, as is a defeasible rule with $\Rightarrow$ in place of $\rightarrow$. Unless inserted as a formula separately, neither rule is a material conditional in $\mathcal{K}$. The definitions in this section follow the 2013 ASPIC+ Definitions 2, 4–5 and 8–9, pp. 365–368; the strict preference correction follows its 2018 corrigendum, Eq. (3), p. 109 and Section 4, p. 110. Our finite calculations use a declared firm rule-only fragment, not the full ordinary-premise preference treatment.

Definition 5.2 (Argument tree and structural functions). *For each $\varphi \in \mathcal{K}$, form a distinct premise argument $\langle \varphi \rangle$ with final conclusion $\text{Conc} = \varphi$. If $A_1, \dots, A_m$ have conclusions matching a rule r 's antecedents, form the tree $[A_1, \dots, A_m \rightarrow \psi]_r$ or $[A_1, \dots, A_m \Rightarrow \psi]_r$. Its final conclusion is ψ . Different derivation trees remain distinct even when their conclusions coincide.*

For a premise A , put $\text{Prem}(A) = \{\text{Conc}(A)\}$ and $\text{Sub}(A) = \{A\}$. Also $\text{DefRules}(A) = \text{LastDefRules}(A) = \emptyset$, while $\text{TopRule}(A)$ is undefined. For a composite tree A with children A_i , set

$$\text{Prem}(A) = \bigcup_i \text{Prem}(A_i), \quad \text{Sub}(A) = \{A\} \cup \bigcup_i \text{Sub}(A_i), \quad \text{TopRule}(A) = r.$$

If $r \in \mathcal{R}_s$, set $\text{DefRules}(A) = \bigcup_i \text{DefRules}(A_i)$ and $\text{LastDefRules}(A) = \bigcup_i \text{LastDefRules}(A_i)$; if $r \in \mathcal{R}_d$, set $\text{DefRules}(A) = \{r\} \cup \bigcup_i \text{DefRules}(A_i)$ and $\text{LastDefRules}(A) = \{r\}$. These are sets of rule identities, not counts of their occurrences. The argument is strict if $\text{DefRules}(A) = \emptyset$ and firm if $\text{Prem}(A) \subseteq \mathcal{K}_n$.

The textbook displays only the contrary entries relevant to its finite arguments. Such a table is not asserted to be the entire canonical map. One possible ambient completion adds an unused fresh contradictory partner to each formula lacking one, preserving already declared asymmetric entries, and gives the fresh partner its reciprocal entry. With no premise or rule producing a fresh partner, the completion creates no additional attack among the displayed arguments. This completion is possible, not uniquely intended.

Lemma 5.3 (Subarguments and defeasible ancestry). *If $Z \in \text{Sub}(Y)$ and $Y \in \text{Sub}(W)$, then $Z \in \text{Sub}(W)$. If $Y \in \text{Sub}(W)$, then $\text{DefRules}(Y) \subseteq \text{DefRules}(W)$. The inclusion need not hold if both occurrences of DefRules are replaced by LastDefRules .*

Proof. Structural induction on W proves both positive statements. For a premise W , its only subargument is itself and its rule set is empty. For a composite W , every proper subargument lies in a child's subargument set. Applying the induction hypothesis inside that child and then the definition of $\text{Sub}(W)$ gives transitivity. Every child's defeasible rule set is included in $\text{DefRules}(W)$, whether the top rule is strict or defeasible, yielding the inclusion. For failure with LastDefRules , take $Y = [\langle p \rangle \Rightarrow a]_r$ and $W = [Y \Rightarrow b]_s$ with $r \neq s$. Then $\text{LastDefRules}(Y) = \{r\}$ but $\text{LastDefRules}(W) = \{s\}$. $\square$

A strict top step can thus have defeasible ancestry and need not make its whole argument strict. The universe of recursively formed finite trees need not be finite, even for a finite input.

Proposition 5.4 (Finite rules can yield infinitely many arguments). *With axiom p and the sole rule $r : p \Rightarrow p$, there are infinitely many distinct finite argument trees. All positive-depth trees have $\text{DefRules} = \text{LastDefRules} = \{r\}$.*

Proof. Let $A_0 = \langle p \rangle$ and $A_{i+1} = [A_i \Rightarrow p]_r$. By induction, A_i has depth i , so the trees are pairwise distinct. The first application has both selected sets $\{r\}$; every later application unions the same identity into DefRules and replaces LastDefRules by $\{r\}$. $\square$

5.2 The location of attack and the defeat test

Definition 5.5 (Attacks on an exact subargument). *Let $Y' \in \text{Sub}(Y)$. An attacker X uses only its final conclusion in the following clauses:*

$$\begin{aligned} X \text{ undermines } Y \text{ on } Y' &\iff Y' = \langle \varphi \rangle, \varphi \in \mathcal{K}_p, \text{Conc}(X) \in \overline{\varphi}, \\ X \text{ rebuts } Y \text{ on } Y' &\iff Y' = [Y_1, \dots, Y_m \Rightarrow \varphi]_r, \text{Conc}(X) \in \overline{\varphi}, \\ X \text{ undercuts } Y \text{ on } Y' &\iff \text{TopRule}(Y') = r \in \mathcal{R}_d, \text{Conc}(X) \in \overline{n(r)}. \end{aligned}$$

Rebut is restricted to a defeasible top at the attacked subargument. Neither an axiom premise nor a strict top is an eligible target of the corresponding premise/rebut clause.

Definition 5.6 (Defeat from a declared preference). *Write $X \prec Y'$ when the policy says the attacker is strictly less preferred than the exact attacked subargument. An undercut is preference-independent. So is an undermine or rebut initiated by an asymmetric contrary of the target formula. For a symmetric contradictory undermine or rebut, the attack succeeds precisely if $X \not\prec Y'$. An argument X defeats Y when some attack by X on Y at a subargument Y' succeeds.*

Theorem 5.7 (Incoming inheritance, with the same target). *If X attacks Y on Z and $Y \in \text{Sub}(W)$, then X attacks W on that same Z . If the original attack is a defeat, so is the inherited attack under the same preference policy.*

Proof. The first premise entails $Z \in \text{Sub}(Y)$. By Lemma 5.3, $Z \in \text{Sub}(W)$. Every attack clause tests X 's final conclusion and the features of Z , all unchanged when the enclosing target becomes W . An undercut or asymmetric contrary remains preference-independent; for a symmetric attack the only preference operand is still Z , so the truth of $X \not\prec Z$ is unchanged. The same attack witness therefore defeats W . $\square$

Counterexample 5.8 (Outgoing attacks do not inherit). Take axioms p, q , rules $r : p \Rightarrow a$, $s : a \Rightarrow b$, $t : q \Rightarrow \neg a$, and no relevant contrary entries among constructed arguments except the mutual $a, \neg a$ entries. Put $A = [\langle p \rangle \Rightarrow a]_r$, $T = [A \Rightarrow b]_s$, and $U = [\langle q \rangle \Rightarrow \neg a]_t$. Then U rebuts T on A and A rebuts U . But T 's final conclusion is b , not a , so T has no attack on U from the declared entries. Replacing the top of T by a strict rule concluding b would also prevent a rebut on T 's top, while leaving an eligible defeasible subargument below it.

5.3 Corrected link preferences

The corrigendum defines the *strict elitist* lifting directly from a strict ingredient order $<$:

$$\Gamma \triangleleft_{\text{Eli}} \Delta \iff \exists r \in \Gamma \forall s \in \Delta (r < s). \quad (5.1)$$

It is not obtained by taking the strict part of the older non-strict lifting. For the following calculation assume all premises are axioms, compare only defeasible rules, and assign distinct integer ranks to rule identities, with lower rank strictly weaker. The selected sets are finite, although the argument universe need not be. Put

$$\begin{aligned} X \prec_{WL} Y &\iff \text{DefRules}(X) \triangleleft_{\text{Eli}} \text{DefRules}(Y), \\ X \prec_{LL} Y &\iff \text{LastDefRules}(X) \triangleleft_{\text{Eli}} \text{LastDefRules}(Y). \end{aligned}$$

This is a firm, rule-only fragment. A symmetric attack on an ordinary premise may require a premise preference that it does not supply.

Theorem 5.9 (The finite rank test, including empty sets). *For any finite selected sets Γ, Δ in this strictly ranked fragment, define $\rho(\emptyset) = \infty$ and otherwise let $\rho(\Gamma)$ be the minimum rank in Γ . Then*

$$\Gamma \triangleleft_{\text{Eli}} \Delta \iff \rho(\Gamma) < \rho(\Delta).$$

In particular, an empty left set is never strictly weaker; a nonempty set is strictly weaker than an empty right set; and equal minima, including two empty sets, give neither strict comparison.

Proof. If $\Gamma \neq \emptyset$, a witness $r \in \Gamma$ to (5.1) has rank below every rank in Δ . When $\Delta \neq \emptyset$, the minimum rank of Γ is at most that of r , hence strictly below the minimum in Δ . Conversely if those minima are strictly ordered, choose r attaining the minimum of Γ ; it is below every $s \in \Delta$. If $\Gamma = \emptyset$, the existential in (5.1) is false and $\infty < \rho(\Delta)$ is false. If $\Gamma \neq \emptyset, \Delta = \emptyset$, any $r \in \Gamma$ witnesses the vacuous universal, while $\rho(\Gamma) < \infty$. These cases cover two empty sets as well. Equality makes a strict inequality false. $\square$

Thus when symmetric contradictory attacks occur in both directions and the compared selected strengths are equal, neither attacker is strictly weaker than its attacked subargument; both attacks survive. This says nothing about eventual acceptance.

Proposition 5.10 (Scot construction and reversal). *Let $\mathcal{K}_n = \{\top\}$, with no ordinary premises or strict rules, and take*

$$d_1 : \top \Rightarrow a \text{ (1)}, \quad d_3 : a \Rightarrow b \text{ (3)}, \quad d_2 : \top \Rightarrow \neg b \text{ (2)}.$$

Assume the only contrary-map entries that produce attacks among constructed arguments are the mutual $b, \neg b$ entries, and that no constructed conclusion conflicts with a rule name. The complete argument set is

$$P = \langle \top \rangle, \quad A = [P \Rightarrow a]_{d_1}, \quad B = [A \Rightarrow b]_{d_3}, \quad C = [P \Rightarrow \neg b]_{d_2}.$$

The raw attack set is $\{(B, C, C), (C, B, B)\}$, where the third coordinate is the attacked subargument. Weakest link has only $C \rightarrow B$ as defeat; last link has only $B \rightarrow C$. If the rank of d_1 alone becomes 4, both policies have only $B \rightarrow C$.

Proof. From $\top$, only d_1, d_2 apply. Only d_3 can follow the resulting conclusion a , and no rule has antecedent b or $\neg b$. This exhausts the four trees. Their defeasible-rule sets are, in order, $\emptyset, \{d_1\}, \{d_1, d_3\}, \{d_2\}$, and their last-rule sets are $\emptyset, \{d_1\}, \{d_3\}, \{d_2\}$. The only final conclusions contrary to an eligible defeasible-top conclusion are b and $\neg b$; therefore B, C rebut each other on the full opposite target and no other raw attack exists.

Originally weakest ranks for B, C are 1, 2. By Theorem 5.9, $B \prec_{WL} C$, so B 's attack fails and C 's survives. Last ranks are 3, 2, so $C \prec_{LL} B$, reversing the surviving direction. When only d_1 's rank changes to 4, construction and raw attacks do not change. The weakest ranks are now 3, 2 and last ranks remain 3, 2; in both policies only B 's attack survives. $\square$

For a five-argument transfer, take axioms *snores*, *professor*, no other premises or rules, and only *accessDenied*, $\neg$ *accessDenied* as relevant contrary entries among the constructed arguments, with no rule-name conflicts. The rules are

$$\begin{aligned} r_1 : \textit{snores} &\Rightarrow \textit{misbehaves} & (1), \\ r_2 : \textit{misbehaves} &\Rightarrow \textit{accessDenied} & (3), \\ r_3 : \textit{professor} &\Rightarrow \neg \textit{accessDenied} & (2). \end{aligned}$$

The two premise trees S_0, P_0 , the first-chain trees

$$S_1 = [S_0 \Rightarrow \textit{misbehaves}]_{r_1}, \quad S_2 = [S_1 \Rightarrow \textit{accessDenied}]_{r_2}, \quad P_1 = [P_0 \Rightarrow \neg \textit{accessDenied}]_{r_3}$$

exhaust the enabled rules. Only S_2, P_1 mutually rebut. Their weakest ranks are 1, 2, hence $P_1 \rightarrow S_2$ is the only weakest-link defeat; their last ranks are 3, 2, hence $S_2 \rightarrow P_1$ is the only last-link defeat. Two roots and five arguments make this a different graph from Scot's shared-root four-argument construction. Ranks alone do not select which policy an application should use.

Proposition 5.11 (Why the whole target is the wrong operand). *In the earlier A, T, U construction, assign ranks $\text{rank}(r) = 1, \text{rank}(s) = 5, \text{rank}(t) = 2$. Under last link U defeats T on A ; a comparison to the whole T would reject it. After changing just r 's rank to 3 and s 's to 1, the correct local attack fails, while whole-target comparison would admit it.*

Proof. The raw rebut $U \rightarrow T$ has the unique attacked subargument A , whose defeasible top concludes a . Initially $\text{LastDefRules}(U) = \{t\}$ of rank 2 and $\text{LastDefRules}(A) = \{r\}$ of rank 1, so $U \not\prec_{LL} A$ and the attack succeeds. The whole T instead has last set $\{s\}$ of rank 5, making $U \prec_{LL} T$ and falsely rejecting it. After the stated change, $2 < 3$ makes $U \prec_{LL} A$, so the attack fails, but $2 \not< 1$ would make a whole-target test accept it. In both cases A 's outgoing rebut on U does not become T 's outgoing attack, because $\text{Conc}(T) = b$. $\square$

Example 5.12 (Preference-independent routes and a boundary). Let $t : p \Rightarrow q$ have rank 9 and $u : z \Rightarrow \neg ok_t$ rank 1, with axioms p, z , $n(t) = ok_t$, and $\neg ok_t \in \overline{ok}_t$. The argument for $\neg ok_t$ undercuts $[\langle p \rangle \Rightarrow q]_t$ on that defeasible top and defeats it under either link policy: no comparison of 1 with 9 is made. For instance, if $\eta \in \mathcal{K}_p$, $p \in \bar{\eta}$ but $\eta \notin \bar{p}$, an argument concluding p undermines $\langle \eta \rangle$ by asymmetric contrary and succeeds without a rank comparison. For a symmetric undermine of an ordinary premise, the rule-only fragment has no premise ordering and does not determine every defeat. A strict top cannot be rebutted at that top, though a defeasible subargument below it can be; an axiom premise cannot be undermined.

Example 5.13 (Strict and firm attackers). Take axioms $\neg b, z, p$, a strict rule $z \rightarrow \neg b$, a defeasible rule $d : p \Rightarrow b$ of rank 9, and only the mutual $b, \neg b$ entries relevant to these arguments. Both $\langle \neg b \rangle$ and $[\langle z \rangle \rightarrow \neg b]$ rebut $[\langle p \rangle \Rightarrow b]_d$. Their selected sets are empty under both link policies, so Theorem 5.9 makes neither attacker strictly weaker than its target: both defeats succeed. The reverse conclusion b cannot rebut an axiom or a strict top, and neither is an ordinary premise to undermine. The one-way relation is therefore structural as well as preferential.

5.4 Other retained reasons and the handoff to semantics

Different constructions keep different grounds of attack. The following two bounded cases make that distinction without importing their complete evaluation procedures.

Definition 5.14 (Minimal classical support, local case). *For a finite classical formula base K , a deductive argument (Γ, φ) has $\Gamma \subseteq K$ consistent, $\Gamma \vdash \varphi$, and no proper subset of Γ entailing φ . This is the minimal-support variant in Besnard and Hunter, Handbook I.9, Definition 4.8, p. 446; it is not the most general deductive pair definition.*

Proposition 5.15 (Two supports are arguments, the full base is not). *Let $K = \{p, p \rightarrow q, \neg q\}$. Then $(\{p, p \rightarrow q\}, q)$ and $(\{\neg q\}, \neg q)$ meet the definition, but (K, q) does not.*

Proof. The first support has a model $p = q = \text{true}$ and entails q by modus ponens. Removing p admits $p = q = \text{false}$; removing $p \rightarrow q$ admits $p = \text{true}, q = \text{false}$. Thus neither proper subset entails q . The second singleton has a model with $q = \text{false}$ and entails its own member. The empty set does not entail $\neg q$, as $q = \text{true}$ witnesses. Finally K has no model and contains the smaller support already entailing q , so it fails consistency and minimality. $\square$

The formula $p \rightarrow q$ here is a *member* of the classical support. Substituting an uninterpreted name for an inference rule would invalidate the first entailment: set the name and p true but q false. No attack or selection policy has been supplied for these deductive pairs.

Proposition 5.16 (A minimal DeLP support pair). *Take the strict fact $\Pi = \{\text{bird}(t)\}$ and two defeasible rules with heads $\text{flies}(t)$ and $\sim \text{flies}(t)$, each with body $\text{bird}(t)$. In the local DeLP construction, each singleton rule set is an inclusion-minimal consistent defeasible support for its respective head. The two conclusions counterargue at their contrary conclusions. Their derivations alone do not decide warrant.*

Proof. Together with Π , either singleton derives its own head and not the opposite head. The empty defeasible support derives neither, proving minimality. The two-rule set derives contradictory literals and fails the stipulated consistency test. The opposing conclusions therefore give the stated counterargument relation. No warrant predicate, defeat comparison or dialectical evaluation rule is defined in these local construction clauses, so derivation alone supplies no warrant conclusion. $\square$

This is the bounded construction from García and Simari, Handbook I.8, Definitions 4.5–4.6, pp. 416–417. It does not assert the technical DeLP dialectical procedure. Flat assumption-based argumentation locates attacks at designated assumptions; the textbook’s further-reading route introduces it, while its full developments belong to their own definitions and evaluation contracts.

For an input $T = (AS, \mathcal{K}, \text{preference policy})$, the structured route passes

$$F_T = (\text{Arg}(T), \text{Defeat}(T))$$

to a later abstract semantics. A raw attack is not passed as if it had already survived preference, and a relation already named attack in a source may itself be preference-resolved. To compare this route with a source method, one must then specify a semantics and an output extraction. Scot’s reversal alone is neither an acceptance result nor a PDL representation theorem; Exercise 5.6 makes that comparison after textbook Chapter 7.

5.5 Proof and source status

The ASPIC+ signature, recursive functions and attack clauses are imported *definitions* from Modgil and Prakken (2013), Definitions 2, 4–5, 8–9, pp. 365–368. The strict elitist clause is imported from their 2018 corrigendum, Eq. (3), p. 109 and Section 4, p. 110; the minimum-rank theorem here is proved only for the declared firm, rule-only, finite ranked fragment. The textbook’s sparse tables are relevant computational displays within the canonical ambient signature. No rationality theorem, full premise preference construction, general DeLP warrant theorem or source/representation equivalence has been silently imported into these results. Chapter 3’s graph queries and Study S3.1 use the approved textbook’s stipulated reconstructions; the countermodels use the opening group’s classical consequence convention. Chapter 4’s trace and weights are local stipulations; they do not describe a jurisdiction’s law or a deployed service.

Related textbook exercises

5.1 Scot’s trees, raw attacks and changed rank; **5.2** exact local target versus whole-target comparison; **5.3(a–d)** ordinary-premise, strict-top, asymmetric-contrary and strict/firm boundaries; **5.4** corrected strict lifting, empty sets and repeated-rule infinity; **5.5** five-argument transfer; **5.6** later Chapter 7 source-output comparison; **5.7–5.8** optional minimal deductive and DeLP constructions; **5.9** preference-independent undercut.

6 Finite abstract semantics

Textbook correspondence: Chapter 6. An abstract framework receives arguments and its evaluated arrow relation from a construction such as Chapter 5's. Once the input is fixed, defense and selection can be studied independently of the argument language. That mathematical reuse does not certify that the construction retained the right content or acts.

6.1 Defense, admissibility and the characteristic map

Definition 6.1 (Finite framework and defense). *Fix $F = (A, R)$ with A finite and $R \subseteq A^2$; self-attacks are allowed. For $x \in A$ and $S \subseteq A$, put*

$$x^- = \{y : yRx\}, \quad S^+ = \{y : \exists s \in S (sRy)\}.$$

The set S is conflict-free if $R \cap S^2 = \emptyset$. It defends x if

$$\forall y \in A (yRx \implies \exists s \in S (sRy)). \quad (6.1)$$

The witness s may depend on y . Write $\text{Def}_F(S) = \{x : S \text{ defends } x\}$. An admissible set is conflict-free and satisfies $S \subseteq \text{Def}_F(S)$.

Unattacked arguments are defended even by $\emptyset$. These are Dung's Definitions 2, 5–6 and 16 (1995, pp. 326, 328–329). If R came from Chapter 5's defeat relation, abstract semantics calls it attack but does not reapply preferences.

Lemma 6.2 (Defense invariants). *For any $S \subseteq T \subseteq A$, $\text{Def}_F(S) \subseteq \text{Def}_F(T)$. If S is conflict-free, so is $\text{Def}_F(S)$. Consequently, if S is admissible, then $\text{Def}_F(S)$ is admissible and $S \subseteq \text{Def}_F(S) \subseteq \text{Def}_F^2(S)$.*

Proof. Every counterattacker in S remains in T , proving monotonicity. For the second assertion, suppose $x, y \in \text{Def}_F(S)$ and xRy , including the case $x = y$. Since S defends y , some $s \in S$ attacks x . Since S defends x , some $t \in S$ attacks s . This is an internal attack tRs , contrary to conflict-freeness of S . Thus $\text{Def}_F(S)$ is conflict-free. If S is admissible, its defining inclusion and monotonicity give $\text{Def}_F(S) \subseteq \text{Def}_F(\text{Def}_F(S))$, so the conflict-free image is admissible. $\square$

Lemma 6.3 (Fundamental extension lemma). *If S is admissible and $x, y \in \text{Def}_F(S)$, then $S \cup \{x\}$ is admissible and still defends y .*

Proof. If $x \in S$, there is nothing to prove. Otherwise suppose sRx for some $s \in S$. Defense of x supplies $t \in S$ with tRs , an internal attack. Thus S does not attack x . If xRs for some $s \in S$, admissibility of S supplies $t \in S$ with tRx , just excluded. Finally if xRx , defense of x again supplies a member of S attacking x , also excluded. These cases establish conflict-freeness of $S \cup \{x\}$. Every one of its members is defended by S , either by admissibility or the hypothesis on x , and remains defended by the larger set using Lemma 6.2. The same monotonicity preserves defense of y . $\square$

Admissibility in this lemma is essential. In the chain $x \rightarrow y \rightarrow z$, $S = \{y\}$ is conflict-free and defends the unattacked x , but $S \cup \{x\}$ is not conflict-free. The opening group's general monotone fixed-point lemma alone supplies none of these admissibility claims.

6.2 Four selection standards

Definition 6.4 (Extension families). *A complete extension is admissible and contains all arguments it defends; equivalently it is a conflict-free fixed point of Def_F . A preferred extension is inclusion-maximal admissible. A stable extension is conflict-free and attacks every outsider, equivalently $E \cup E^+ = A$. The grounded extension G_F is the least fixed point of Def_F . Write $\text{Ext}_{\text{co}}(F)$, $\text{Pref}(F)$, $\text{Stable}(F)$ for families of complete, preferred and stable sets. Grounded semantics returns the singleton family $\{G_F\}$, even if $G_F = \emptyset$.*

The complete characterization requires conflict-freeness: an arbitrary fixed point need not be complete. The one-node self-attacker has $\text{Def}_F(\{u\}) = \{u\}$ but $\{u\}$ is not conflict-free.

Theorem 6.5 (Finite semantic dependencies). *Every admissible S is contained in a preferred extension. At least one preferred extension exists, and*

$$\text{Stable}(F) \subseteq \text{Pref}(F) \subseteq \text{Ext}_{\text{co}}(F). \quad (6.2)$$

For $q \in A$, preferred credulous acceptance of q is equivalent to the existence of an admissible set containing q .

Proof. The finite family of admissible supersets of S is nonempty because it contains S . Choose one of largest cardinality *within that family*. Any admissible proper superset would be a still larger member, so the choice is inclusion-maximal among all admissible sets and is preferred. Taking $S = \emptyset$ gives existence. This finite maximization finds one maximal element; it does not identify preferredness with largest cardinality across incomparable sets.

For preferred P , admissibility gives $P \subseteq \text{Def}_F(P)$. If x were in the difference, Lemma 6.3 would extend P admissibly, a contradiction. Hence $P = \text{Def}_F(P)$ and P is complete. For stable E , an attacker y of $x \in E$ lies outside E by conflict-freeness, and stability supplies a member of E attacking y . Thus E defends all its members. Any proper superset adds an outsider already attacked by E , so it is not conflict-free. Hence E is preferred and therefore complete. A preferred witness containing q is admissible; conversely extend any admissible witness containing q by the first argument. $\square$

The directed mutual pair has complete empty set and two preferred singletons, disproving $\text{complete} \Rightarrow \text{preferred}$. The self-attacker has preferred $\emptyset$ and no stable extension, disproving $\text{preferred} \Rightarrow \text{stable}$. Repeatedly adding defended outsiders does not in general find a preferred set: on the mutual pair it stops at complete $\emptyset$.

Theorem 6.6 (Grounded iteration and its invariant). *For $n = |A|$, put $S_0 = \emptyset$ and $S_{i+1} = \text{Def}_F(S_i)$. Every S_i is admissible and $S_i \subseteq S_{i+1}$. Equality is detected within $n + 1$ evaluations of Def_F , including the final equality check and the $n = 0$ case. The resulting G_F is the unique least fixed point and least complete extension. It lies in every complete, preferred and existing stable extension.*

Proof. The empty set is admissible. Lemma 6.2 inductively makes each image admissible and includes the previous one. Each strict inclusion adds at least one node, so at most n strict increases occur. The next evaluation confirms equality, within $n + 1$ evaluations; if $A = \emptyset$, the first gives $\emptyset$. Let equality occur at G_F . It is a fixed point and, by the separate invariant, conflict-free and complete.

For any fixed point T , with no conflict-freeness assumed, induction gives $S_i \subseteq T$: the base is empty inclusion, and monotonicity gives $\text{Def}_F(S_i) \subseteq \text{Def}_F(T) = T$. Thus $G_F \subseteq T$. Two least fixed points are equal by inclusion in both directions. All complete extensions are fixed points; the other semantic inclusions follow from Theorem 6.5. The evaluation count does not bound the cost of one evaluation. $\square$

An arbitrary starting set need not grow: on the chain $\text{Def}_F(\{y\}) = \{x\}$. Nor does fixed-point equality alone imply completeness, as the self-attacker shows. These are the two failed shortcuts repaired by the proof.

6.3 Two five-node graphs and boundary frameworks

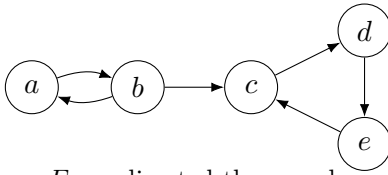
The main semantic discriminator is

$$\begin{aligned} F_{\text{link}} &= (\{a, b, c, d, e\}, R_{\text{link}}), \\ R_{\text{link}} &= \{a \rightarrow b, b \rightarrow a, b \rightarrow c, c \rightarrow d, d \rightarrow e, e \rightarrow c\}. \end{aligned}$$

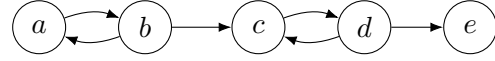
The two-cycle and three-cycle are linked by $b \rightarrow c$. The separate computational-reuse graph is

$$R_{\text{reuse}} = \{a \rightarrow b, b \rightarrow a, b \rightarrow c, c \rightarrow d, d \rightarrow c, d \rightarrow e\}.$$

In particular $e \rightarrow c$ belongs only to the first graph; $d \rightarrow c$ belongs only to the second.



F_{link} : directed three-cycle



F_{reuse} : second mutual pair

Proposition 6.7 (Linked-cycle families). *For F_{link} , the admissible sets are $\emptyset, \{a\}, \{b\}, \{b, d\}$, and*

$$\begin{aligned} \text{Ext}_{\text{co}} &= \{\emptyset, \{a\}, \{b, d\}\}, \\ \text{Pref} &= \{\{a\}, \{b, d\}\}, \quad \text{Stable} = \{\{b, d\}\}, \quad G_F = \emptyset. \end{aligned}$$

Proof. Any admissible set containing c must answer $e \rightarrow c$ with d , creating the internal attack $c \rightarrow d$. A set containing e must answer $d \rightarrow e$ with c , creating $e \rightarrow c$. Thus c, e are absent. Defending d against c now requires b ; conflict-freeness excludes a, b together. This leaves the four listed candidates, all of which defend their members. Their characteristic images, in order, are $\emptyset, \{a\}, \{b, d\}, \{b, d\}$. The fixed admissible sets are complete; maximality gives the two preferred sets. The set $\{a\}$ leaves c, d, e unattacked, while $\{b, d\}^+ = \{a, c, e\}$, proving the stable claim. Every node is attacked, so $\text{Def}_F(\emptyset) = \emptyset$, giving the grounded set. The preferred singleton $\{a\}$ is incomparable with the larger $\{b, d\}$; cardinality is not the criterion. $\square$

Proposition 6.8 (The separate reuse graph). *For F_{reuse} , the admissible family is*

$$\{\emptyset, \{a\}, \{b\}, \{d\}, \{a, c\}, \{a, d\}, \{b, d\}, \{a, c, e\}\}.$$

Its complete family is $\{\emptyset, \{a\}, \{d\}, \{a, d\}, \{b, d\}, \{a, c, e\}\}$; preferred and stable coincide as $\{\{a, d\}, \{b, d\}, \{a, c, e\}\}$. The grounded set is $\emptyset$. After adding only $e \rightarrow a$, preferred and stable both become $\{\{a, d\}, \{b, d\}\}$.

Proof. Conflict-freeness forbids the pairs a, b, b, c, c, d, d, e . The resulting 13 sets consist of the empty set, five singletons, six allowed pairs and $\{a, c, e\}$. Of these, $\{c\}, \{c, e\}$ do not answer $b \rightarrow c$; and $\{e\}, \{a, e\}, \{b, e\}$ do not answer $d \rightarrow e$. Each of the other eight sets defends its members, giving the displayed admissible family. Among them $\{b\}$ also defends d , and $\{a, c\}$ also defends e ; the other six are fixed. The three maximal admissible sets attack every outsider, so are both preferred and stable. Every node is attacked, making the first grounded iterate empty.

After $e \rightarrow a$, the former $\{a, c, e\}$ has an internal attack. Any admissible set containing a must now answer $e \rightarrow a$ with d ; this prevents c and e from joining it. The surviving admissible sets are $\emptyset, \{b\}, \{d\}, \{a, d\}, \{b, d\}$. Their maximal members $\{a, d\}, \{b, d\}$ attack every outsider, as claimed. $\square$

Example 6.9 (No stable output versus an empty output). For the empty AF, the preferred and stable families are both $\{\emptyset\}$. For a sole self-attacker or a directed odd three-cycle, the only admissible set is $\emptyset$; it is preferred but not stable, because it attacks no outsider. Their stable family is $\emptyset$, with *no* member. On the chain $w \rightarrow x \rightarrow y \rightarrow z$, the grounded iterates are $\emptyset, \{w\}, \{w, y\}, \{w, y\}$. The first attack excludes x and then defends y ; z remains opposed by y . These are different boundary inputs.

6.4 Labellings, queries and extraction

Definition 6.10 (Complete and stable labels). *A total map $\lambda : A \rightarrow \{\text{in}, \text{out}, \text{undec}\}$ is a complete labelling exactly when, for each node: it is in iff all attackers are out; it is out iff some attacker is in; and it is undec iff no attacker is in but at least one is not out. A stable labelling is complete with no undec arguments. These labelling clauses are from the later Handbook account (Baroni, Caminada and Giacomin 2018, Definitions 3.1, 3.19–3.20, 3.36, pp. 170, 178, 184), not Dung’s 1995 wording.*

Theorem 6.11 (Complete labelling correspondence). *For every complete extension E , the partition*

$$\text{in} = E, \quad \text{out} = E^+, \quad \text{undec} = A \setminus (E \cup E^+) \quad (6.3)$$

is a complete labelling. Conversely every complete labelling is obtained this way from its in-set. It is stable exactly when its undec set is empty.

Proof. Conflict-freeness makes $E \cap E^+ = \emptyset$. Every attacker of an in-node is out: if yRx with $x \in E$, defense by E puts $y \in E^+$. Every out-node has an in-attacker by the definition of E^+ . If an undec-node had an in-attacker it would be out; if all its attackers were out, the complete extension would defend and contain it. Thus the displayed partition satisfies all clauses.

Conversely, in-nodes cannot attack one another: an in-node’s attackers are all out. Every attacker of an in-node is out, and each out-node has an in-attacker, so the in-set defends its members. If the in-set defends x , each attacker y of x has an attacker in the in-set. The legal clauses force every node attacked by an in-node to be out: it cannot be in, since in-nodes have only out-attackers, or undec, since undec-nodes have no in-attacker. Hence every attacker of x is out, and the legal clauses force x to be in. Thus the in-set is complete. Conversely out means an in-attacker, so the out-set is exactly what the in-set attacks, giving (6.3). Finally no undec nodes means $E \cup E^+ = A$, precisely stability. $\square$

For F_{link} , the legal triples (in, out, undec) are

$\emptyset$	$\emptyset$	$\{a, b, c, d, e\}$
$\{a\}$	$\{b\}$	$\{c, d, e\}$
$\{b, d\}$	$\{a, c, e\}$	$\emptyset$

Only the last is stable. For the reuse graph, $\{a\}$ has labels $(\{a\}, \{b\}, \{c, d, e\})$; $\{b, d\}$ has $(\{b, d\}, \{a, c, e\}, \emptyset)$. Conflict-freeness alone would make the three sets a partition but would not make their labels legal.

Definition 6.12 (Guarded acceptance and conclusions). *For an extension family $\mathcal{E} = \text{Ext}_\sigma(F)$,*

$$\begin{aligned} \text{Cred}_\sigma(x) &\iff \exists E \in \mathcal{E} (x \in E), \\ \text{Skept}_\sigma(x) &\iff \mathcal{E} \neq \emptyset \wedge \forall E \in \mathcal{E} (x \in E), \\ \text{NoOutput}_\sigma(F) &\iff \mathcal{E} = \emptyset. \end{aligned}$$

The guard makes both Boolean acceptance answers false when the semantic family has no member. If arguments have conclusions, define $\text{Conc}(E) = \{\text{Conc}(a) : a \in E\}$ and the output

family $\mathcal{O}_\sigma(F) = \{\text{Conc}(E) : E \in \mathcal{E}\}$. This is an additional extraction, not the argument extension family.

In the linked graph, preferred-credulous nodes are a, b, d , with no preferred-skeptical node. Stable-credulous and stable-skeptical nodes are both b, d ; grounded accepts none. In the reuse graph every node is stable-credulous but none is stable-skeptical. An empty stable family makes both queries false and the no-output diagnostic true, without manufacturing an existing counterexample extension.

Two distinct arguments can conclude the same formula. Thus

$$\forall E \in \mathcal{E} \exists a \in E (\text{Conc}(a) = \varphi)$$

need not imply $\exists a \forall E \in \mathcal{E} (a \in E \wedge \text{Conc}(a) = \varphi)$. On two alternative singleton extensions with different arguments of conclusion φ , the first holds and the second fails. Chapter 7 will use this distinction in guarded source-output queries.

6.5 Bounded alternatives

The optional textbook cases retain mathematical objects without developing a second general semantics. In a flat ABA fragment with assumptions p, q , rules $x \leftarrow q, y \leftarrow p$, and contraries $\bar{p} = x, \bar{q} = y$, assumption set $\{p\}$ attacks sets containing q , and $\{q\}$ symmetrically attacks sets containing p . The empty set and the two singletons are admissible; the full set attacks itself. Thus the two singletons are preferred and stable assumption sets. Deleting only $x \leftarrow q$ leaves only $\{p\}$ stable: it still attacks omitted q , whereas $\{q\}$ no longer attacks omitted p . This finite enumeration is not a general ABA correspondence theorem.

In an ADF two-valued model of $\varphi_a = \top, \varphi_b = b, \varphi_c = a \wedge b, \varphi_d = \neg b$, a is true and b is free; the true sets are $\{a, d\}$ and $\{a, b, c\}$. Changing only φ_c to $a \vee b$ keeps c true even when b is false, giving $\{a, c, d\}$ and $\{a, b, c\}$. The two cases for b prove exhaustiveness; untyped parent arrows into c would not record conjunction versus disjunction.

Finally, for raw attack $a \rightarrow b$ and preference $b \succ a$, a deletion policy removes the arrow, yielding grounded $\{a, b\}$ and stable family $\{\{a, b\}\}$. A reversal policy instead uses $b \rightarrow a$, yielding grounded $\{b\}$ and stable family $\{\{b\}\}$. These follow from the respective unattacked nodes and outsider coverage. The first output conflicts with the *raw* arrow; neither policy is Chapter 5's local ASPIC+ defeat clause.

Related textbook exercises

6.1 linked-cycle defense, families, labels and acceptance; **6.2(a–d)** distinct computational-reuse graph, witnesses, labels and added edge; **6.3** empty, self-attacking, odd-cycle and chain boundaries; **6.4–6.6** defense invariants, Fundamental Lemma, semantic dependencies and grounded proof repair; **6.7–6.9** optional bounded ABA, ADF and abstract preference-policy comparisons.

7 Representation and source solutions

Textbook correspondence: Chapter 7 and its normal-program mathematical supplement. A source procedure determines its solutions before an attack graph is proposed. This order of definition matters: semantic accuracy about the proposed graph does not establish fidelity to the source procedure.

7.1 A representation contract and its extraction

Definition 7.1 (Exact representation). *For each input T in a declared domain $\mathcal{D}$, let the independent source procedure return a family $\text{Sol}(T)$. A construction $T \mapsto F_T$, semantics σ , and output map Out_T represent that procedure exactly if*

$$\text{Sol}(T) = \{\text{Out}_T(E) : E \in \text{Ext}_\sigma(F_T)\} \quad (T \in \mathcal{D}). \quad (7.1)$$

The domain, construction, semantics and extraction are all part of the claim. If different extensions have the same output, (7.1) is weaker than a bijective correspondence between extensions and source solutions.

For arguments bearing conclusions, $\text{Conc}(E) = \{\text{Conc}(a) : a \in E\}$. With $\mathcal{E} = \text{Ext}_\sigma(F_T)$, a conclusion q is credulous when $\exists E \in \mathcal{E} \exists a \in E (\text{Conc}(a) = q)$, and skeptical when $\mathcal{E} \neq \emptyset$ and $\forall E \in \mathcal{E} \exists a \in E (\text{Conc}(a) = q)$. The latter permits different supporting arguments in different extensions. Both queries fail for an empty extension *family*; the family $\{\emptyset\}$ is a distinct, existing output. We compare full families before collapsing them to queries.

Proposition 7.2 (Extraction after finite repairs). *Let K be finite, P consistent hard premises, and $\mathcal{M} = \text{MCS}(K, P)$ the inclusion-maximal $S \subseteq K$ for which $P \cup S$ is classically consistent. Then $\mathcal{M} \neq \emptyset$, and*

$$\text{Cn}\left(P \cup \bigcap \mathcal{M}\right) \subseteq \bigcap_{S \in \mathcal{M}} \text{Cn}(P \cup S). \quad (7.2)$$

The inclusion may be strict, even when an argumentation construction represents the repair family exactly.

Proof. The empty subset is feasible, and finiteness gives a maximal feasible subset, so $\mathcal{M}$ is nonempty. For each $S \in \mathcal{M}$, $P \cup \bigcap \mathcal{M} \subseteq P \cup S$. Monotonicity of classical consequence gives the indicated inclusion after intersecting over S .

For strictness take $P = \emptyset$, $K = \{u, v\}$, $u = p \wedge q$, $v = \neg p \wedge q$. The repairs are $\{u\}, \{v\}$; their common premise set is empty. Both repairs imply q . Conversely, a valuation of q satisfies u when p is true and v when p is false. Thus $\text{Cn}(\{u\}) \cap \text{Cn}(\{v\}) = \text{Cn}(\{q\})$, while $\text{Cn}(\{u\} \cap \{v\}) = \text{Cn}(\emptyset)$; q is not a tautology. $\square$

The opening group's base-argument representation has arguments $\emptyset, \{u\}, \{v\}$ on this input. The nonempty arguments attack each other by inconsistent premises; its stable extensions are $\{\emptyset, \{u\}\}$ and $\{\emptyset, \{v\}\}$, whose union-of-premises readback recovers the two repairs. It does not equate the two expressions in (7.2). Adding a fresh premise r to K gives repairs $\{u, r\}, \{v, r\}$: their intersection closes to $\text{Cn}(r)$ while the intersection of their closures is $\text{Cn}(q, r)$. This is the packed-premise distinction in the Handbook of Formal Argumentation II, Chapter 12, Definition 44 and Example 45. The set $\Gamma = \{u, r\}$ is irredundant for its entire closure: deleting u loses p , and deleting r loses r . Yet (Γ, q) is not a conclusion-minimal support–conclusion argument, since $\{u\}$ already entails q . These are distinct notions of minimality.

7.2 Application priorities and local defeat

For the bounded normal-default comparisons below, a rule $d : \theta \Rightarrow \phi$ has a numerical rank, with larger ranks stronger. Start with facts W , repeatedly choose a highest-ranked available rule whose prerequisite is obtained and whose addition is consistent, add its head and take the declared strict closure, and stop when no permitted rule remains. An already established head or contrary prevents reapplication. Ties permit alternative sequences. The resulting *PDL domain-output family* below lists derived literals, suppressing the common $\top$, without silently adding full classical closure. This is the cited normal-default procedure of Pardo et al. (2025, Remark 1 and Definition 15), on the displayed fragment.

Consider

$$d_1 : \top \Rightarrow a \ (1), \quad d_2 : \top \Rightarrow \neg b \ (2), \quad d_3 : a \Rightarrow b \ (3). \quad (7.3)$$

The only initial competitors are d_1, d_2 . PDL chooses d_2 , then d_1 ; d_3 is now blocked by $\neg b$. Its family is $\{\{a, \neg b\}\}$. Let $P = \langle \top \rangle$, $A = [P \Rightarrow a]_{d_1}$, $B = [A \Rightarrow b]_{d_3}$, and $C = [P \Rightarrow \neg b]_{d_2}$. The structural rebuts are B against C and C against the defeasible top of B ; no other pair rebuts. Preference is compared at the *attacked subargument*. Simple weakest link uses $\rho(\text{DefRules}(X))$, last link uses $\rho(\text{LastDefRules}(X))$, where $\rho(D)$ is the minimum numerical rank among rules in D , with $\rho(\emptyset) = \infty$. For this rebut, disjoint weakest link uses

$$\rho(\text{DefRules}(X) \setminus \text{DefRules}(Y')) \quad \text{versus} \quad \rho(\text{DefRules}(Y') \setminus \text{DefRules}(X)), \quad (7.4)$$

where Y' is the exact attacked subargument. The rebut by X survives as a defeat if Y' is not strictly preferred to X ; equivalently, in this total-rank fragment, the left strength is at least the right strength. These are the corrected strict comparison and the disjoint *rule identity* subtraction of Chapter 5; no second preference filter follows Dung evaluation.

Here weakest and disjoint weakest link retain only $C \rightarrow B$. Last link retains only $B \rightarrow C$. An unattacked P, A belong to every stable extension; hence the respective unique extensions are $\{P, A, C\}$ and $\{P, A, B\}$, extracting $\{a, \neg b\}$ and $\{a, b\}$. Only the first output matches the displayed PDL source. If the task instead explicitly prefers the complete outcome using d_1, d_3 to the one using d_1, d_2 , $\{a, b\}$ serves that supplied objective. A rank alone does not decide whether it orders rule application or desired contributions. Raising d_1 's rank to 4 makes it initially selected, then d_3 , so both source and all three displayed comparison policies give $\{a, b\}$. These are instance calculations, not a representation theorem.

Counterexample 7.3 (A shared weakest rule). Replace d_2 by $d'_2 : a \Rightarrow \neg b \ (2)$. PDL first selects d_1 , then the higher ranked d_3 , giving only $\{a, b\}$. Replace C by $\hat{C} = [A \Rightarrow \neg b]_{d'_2}$. The two structural rebuts are $B \leftrightarrow \hat{C}$. Simple weakest link compares 1 with 1, preserving both defeats. Its stable family is $\{\{P, A, B\}, \{P, A, \hat{C}\}\}$, extracting both $\{a, b\}$ and $\{a, \neg b\}$. Disjoint comparison cancels the shared *rule* d_1 : the remaining strengths are 3 and 2. Only $B \rightarrow \hat{C}$ survives, giving the sole extension $\{P, A, B\}$ and the PDL output. Distinct rules with the same numerical rank would not cancel.

Even distinct ranks do not make raw disjoint weakest link a general PDL representation. With $W = \{\top\}$, the *only* strict rule $s : a \rightarrow \neg b$, and $d_1 : \top \Rightarrow a \ (1)$, $d_2 : \top \Rightarrow b \ (2)$, PDL selects d_2 ; d_1 would strictly entail $\neg b$ and is inconsistent with the selection. The output is $\{b\}$. The arguments are $P, A = [P \Rightarrow a]_{d_1}$, $B = [P \Rightarrow b]_{d_2}$, $C = [A \rightarrow \neg b]_s$. Only C structurally rebuts B : B cannot rebut a strict top, and A concludes a . Disjoint weakest link on the original ranks rejects C 's rebut because $1 < 2$. The empty defeat graph has the unique stable extension containing all four, whose extracted domain output is $\{a, b, \neg b\}$. This directional-strict example alone does not test the full classical/SP construction. The distinct obstruction in §7.3 does.

7.3 The classical/SP source claim and an exact-domain obstruction

The reported PDL claim concerns a different construction from the domain-literal illustrations above. Let W be a consistent, possibly infinite set of closed first-order formulas. Let D be a finite set of closed normal defaults $d : \theta : \phi / \phi$, with a strict total priority. Keep distinct default identities when translating them to defeasible rules $r_d : \theta \Rightarrow \phi$. Write R_d for this rule set, and $h(r_d) = \phi$ for a rule’s head. Use W as axioms, no ordinary premises, all classically valid finitary strict inferences, syntactic negation as contrary, and no undercuts. Arguments are all finite trees licensed by these rules, including trees with inconsistent intermediate conclusions. Their universe need not be finite. For $R \subseteq R_d$, let

$$\text{Arg}(R) = \{A : \text{DefRules}(A) \subseteq R\}.$$

Assume every rule in R_d occurs in at least one freely constructed argument. This is the free-rule qualification in Young–Modgil–Rodrigues, *On the Interaction between Logic and Preference in Structured Argumentation* (2017, §3.2, author manuscript p. 6). It does not impose consistency on free construction. The earlier arXiv:1609.05224v1, §3.4.3, p. 32, expressly allows inconsistent arguments.

For this source domain Th denotes full first-order consequence. The PDL iteration begins at $E_0 = \text{Th}(W)$. A default $\theta : \phi / \phi$ is active at E_i when $\theta \in E_i$ and $\phi, \neg\phi \notin E_i$. Choose the highest-priority active default and put $E_{i+1} = \text{Th}(E_i \cup \{\phi\})$, or stop if none is active. Call the final full theory E_{PDL} . Unlike the preceding domain-literal projections, this output includes every classical consequence.

Definition 7.4 (Structure-preference enumeration). *Beginning with no selected defeasible rules, repeatedly choose the highest originally ranked unused rule whose prerequisite has an argument in $\text{Arg}(R)$, where R is the set already selected. Earlier choices receive stronger SP priority. Free constructibility does not require consistency with accepted PDL conclusions. If unused rules remain, take a finite free argument containing one and choose its first unused rule occurrence in construction order. All defeasible rules below that occurrence have already been selected, so an eligible unused rule exists. Thus the free-rule qualification makes the enumeration a strict total ordering of R_d .*

Write $r <_{\text{SP}} s$ when r is weaker. For arguments X, Y' , put

$$X <_{\text{SP}} Y' \iff \begin{aligned} &\exists r \in \text{DefRules}(X) \setminus \text{DefRules}(Y') \\ &\forall s \in \text{DefRules}(Y') \setminus \text{DefRules}(X) \ (r <_{\text{SP}} s). \end{aligned}$$

An attacker X defeats Y exactly when it rebuts a subargument $Y' \in \text{Sub}(Y)$ with defeasible top and $X \not<_{\text{SP}} Y'$. The comparison is with that exact Y' , not with Y . The existential clause retains the empty-difference convention of Chapter 5. A stable extension is a conflict-free set defeating every argument outside it, also for an infinite universe.

Reported claim 7.5—refuted under the recorded construction. On the preceding source domain and construction, the SP-disjoint local-defeat graph has a unique stable extension S , and its full conclusion set equals the unique classically closed PDL extension: $\text{Conc}(S) = E_{\text{PDL}}$.

Young, Modgil and Rodrigues state this correspondence in AAMAS 2016, Theorem 5.3, pp. 632–633, following the unique-stability claim in Theorem 5.2. Its extended argument is arXiv:1609.05224v1, Theorems 3.13–3.14, pp. 24–27; the TAFA-17 author manuscript, Theorem 1, p. 10, imports the correspondence. Under the construction recorded in Definition 7.4, however, Counterexample 7.5d supplies an input for which no stable extension has the PDL full conclusion set. The conjunction is therefore refuted on this domain and is not used as a theorem. This result does not separately disprove stable existence or uniqueness.

Lemma 7.5a (The PDL output is defined). *The stated PDL iteration has a unique final consistent, classically closed theory after at most $|D|$ applications, for finite or infinite W .*

Proof. The initial theory is consistent and closed. If a chosen head ϕ made $E_i \cup \{\phi\}$ inconsistent, classical consequence would give $\neg\phi \in E_i$, contrary to activity. Closure therefore preserves consistency. The sequence increases. Once a default has been applied, its head remains present and it cannot be active again. There are only $|D|$ defaults. At each nonterminal stage the strict total order gives one greatest active default; hence induction gives one sequence and one final theory. This is a mathematical construction, not a decidability claim for arbitrary first-order consequence. $\square$

Lemma 7.5b (Two structural facts on the full argument universe). *Every stable extension S of the stated local-defeat graph is subargument closed. For every argument A , independently of acceptance,*

$$W \cup \{h(r) : r \in \text{DefRules}(A)\} \models \text{Conc}(A).$$

Proof. If $B \in \text{Sub}(A)$, $A \in S$ and $B \notin S$, stability supplies $X \in S$ defeating B at some $B' \in \text{Sub}(B)$. The same rebut and the same comparison with B' defeat A , contradicting conflict-freeness. This proves subargument closure without any finiteness assumption on S or the argument universe.

For the entailment, use induction on the finite tree A . An axiom is in W . At a strict node, the induction hypotheses and the validity of its inference entail the conclusion from the union of the children's head sets and W . At a defeasible node, its conclusion is the head of its top rule and is itself in the displayed premise set. Monotonicity completes each step. $\square$

Lemma 7.5c (An inconsistent free prefix changes availability). *Suppose a selected free prefix permits arguments concluding x and $\neg x$. Every remaining closed prerequisite is then freely constructible. From that point the SP enumeration lists the remaining rules in descending original priority, regardless of their prerequisites.*

Proof. For any closed θ , the strict inference $x, \neg x \rightarrow \theta$ is classically valid. Combining the two arguments with it gives a finite prerequisite argument using only rules already selected. Every unused rule is thus eligible; each subsequent choice is the greatest one left. No consistency test is part of this free enumeration. $\square$

Counterexample 7.5d (Failure of the recorded full-conclusion claim). *Let x, a, b, c be independent closed atomic formulas, take $W = \{\neg x\}$, and assign the following ranks, larger being stronger:*

$$\begin{aligned} q & : \top \Rightarrow x \quad (5), \\ r_4 & : a \Rightarrow \neg(b \wedge c) \quad (4), \\ r_3 & : a \Rightarrow c \quad (3), \\ r_2 & : \top \Rightarrow b \quad (2), \\ r_1 & : \top \Rightarrow a \quad (1). \end{aligned}$$

This input satisfies every domain condition above, but no stable extension of its SP-defeat graph has conclusion set E_{PDL} .

Proof. The premise set is consistent, the five defaults are normal and closed, and their ranks are strict and total. Tautological prerequisites have strict arguments. The rule r_1 freely supplies a , so both r_3 and r_4 also occur in free finite arguments. Thus every rule is freely available even without appealing to explosion.

In PDL, q is blocked from the start by $\neg x$. The successive selected defaults are r_2, r_1, r_4 . After r_2 only r_1 is active; after r_1 , r_4 outranks r_3 . Its head together with b entails $\neg c$, blocking r_3 . Hence

$$E_{\text{PDL}} = \text{Th}(\neg x, a, b, \neg c).$$

The equivalent head presentation is $\text{Th}(\neg x, b, a, \neg(b \wedge c))$. These theories are consistent.

Free enumeration, by contrast, first selects q : unlike PDL, it does not reject a head inconsistent with W . The free prefix now constructs both x and $\neg x$, so Lemma 7.5c gives the SP list

$$q, r_4, r_3, r_2, r_1 \quad (\text{strongest to weakest}).$$

In particular $r_2 <_{SP} r_3$.

Suppose, for a contradiction, that a stable extension S has $\text{Conc}(S) = E_{\text{PDL}}$. By Lemma 7.5b it contains every subargument of each of its members. No member can therefore use q or r_3 : the corresponding subarguments would conclude x or c , neither belonging to the consistent theory E_{PDL} . Thus every $X \in S$ satisfies $\text{DefRules}(X) \subseteq \{r_1, r_2, r_4\}$.

If $X \in S$ concludes $\neg c$, it must use r_2 and r_4 . Indeed, without r_2 , the valuation $\neg x, a, \neg b, c$ satisfies W and every possible remaining head but falsifies $\neg c$; without r_4 , use $\neg x, a, b, c$. Both exclusions follow from the head-entailment part of Lemma 7.5b. An occurrence of r_4 requires a proper subargument concluding a . If r_1 were absent, that subargument's heads could only be b and $\neg(b \wedge c)$; the valuation $\neg x, \neg a, b, \neg c$ refutes the required entailment of a . Consequently

$$\text{DefRules}(X) = \{r_1, r_2, r_4\} \quad (X \in S, \text{Conc}(X) = \neg c).$$

These valuations may be realized by interpretations of four independent closed predicates, so the argument is also a first-order countermodel argument, not a replacement of the first-order source domain.

Let T be a strict argument for the tautology $\top$, and take the actual finite arguments $A = [T \Rightarrow a]_{r_1}$ and $C = [A \Rightarrow c]_{r_3}$. Since $c \notin E_{\text{PDL}}$, $C \notin S$, so stability requires some member of S to defeat C . Its only defeasible-top subarguments are A and C . No member concludes $\neg a$, so none rebuts A . For a rebut on C , the preceding calculation gives the exact differences

$$\text{DefRules}(X) \setminus \text{DefRules}(C) = \{r_2, r_4\}, \quad \text{DefRules}(C) \setminus \text{DefRules}(X) = \{r_3\}.$$

The witness $r_2 <_{SP} r_3$ proves $X \prec_{SP} C$, so this rebut does not survive as defeat. There are no ordinary-premise attacks or undercuts. Thus no member of S defeats C , a contradiction. The argument considers *every* potential attacker in the full possibly infinite universe, not only the three displayed trees. $\square$

The counterexample refutes reported claim 7.5 through its full-conclusion requirement; it does not assert that existence or uniqueness separately fails. The author accepts this as a developed obstruction on the recorded domain. No request to prove the same conjunction remains open, and no replacement theorem, changed construction or new domain is adopted. The source's proof route has additional unproved steps: its Algorithm 1 can discard a rule whose prerequisite becomes available only later, and Lemma 3.12 transports defeat between occurrences of a rule without preserving their different prerequisite supports in the local comparison. Exact witnesses and source locators are retained in the companion's completion review evidence. These proof-route issues are distinct from the counterexample to the recorded statement itself.

On the earlier examples, free enumeration still selects d_2, d_1, d_3 in (7.3), and d_1, d_3, d'_2 in the shared-rule variant. Those successful calculations remain valid, but neither is a general correspondence proof. The raw-priority disjoint relation also remains unsupported as a general correspondence.

7.4 Positive closure and stable models of normal programs

Definition 7.6 (Finite programs and reduct). *Fix a finite propositional atom universe U . A normal rule is $h \leftarrow B, \text{not } C$, with $h \in U$ and finite positive and default-negative body sets*

$B, C \subseteq U$. It is a fact if both bodies are empty, and positive if C is empty. A set $J \subseteq U$ models a positive rule $h \leftarrow B$ when $B \subseteq J$ implies $h \in J$. For a candidate $M \subseteq U$, the reduct P^M deletes rules for which $C \cap M \neq \emptyset$ and erases not C from the rest. It is positive. A stable model satisfies $M = \text{LM}(P^M)$.

not c tests candidate absence. It is not classical negation; an atom named neg_p has no contradictory meaning merely because of its spelling. The candidate decides the reduct and does not seed the positive derivation.

Proposition 7.7 (Least positive model). *For a finite positive Q , define*

$$C_Q(S) = S \cup \{h : (h \leftarrow B) \in Q, B \subseteq S\}, \quad S_0 = \emptyset, \quad S_{i+1} = C_Q(S_i).$$

After at most $|U|$ strict increases and a final equality check, the iteration returns the unique least model $\text{LM}(Q)$.

Proof. Each step retains its input; a strict step adds an atom, so finiteness gives the bound, including $U = \emptyset$. At equality, any rule whose body is contained in the returned set has its head there, so that set is a model. If J is any model, induction gives $S_i \subseteq J$: true at $i = 0$, and a new head in S_{i+1} has its body in $S_i \subseteq J$, so belongs to J . Thus the returned model is contained in every model; two least models coincide. $\square$

For $Q = \{p \leftarrow p\}$, both $\emptyset$ and $\{p\}$ are models but the least is empty. For $\{p \leftarrow \text{not } p\}$, neither of the two candidates equals the least model of its own reduct; there is no stable model.

The Nixon program makes the general calculation concrete:

$$\begin{aligned} q &\leftarrow, & r &\leftarrow, \\ p &\leftarrow q, \text{not } neg_p, & neg_p &\leftarrow r, \text{not } p. \end{aligned} \tag{7.5}$$

Every reduct keeps the two facts, so every stable model has q, r . The four choices for p, neg_p give the following *complete* reduct audit:

M	$\text{LM}(P^M)$
$\{q, r, p\}$	$\{q, r, p\}$
$\{q, r, neg_p\}$	$\{q, r, neg_p\}$
$\{q, r\}$	$\{q, r, p, neg_p\}$
$\{q, r, p, neg_p\}$	$\{q, r\}$

For the first row the surviving nonfact is $p \leftarrow q$, and the positive rounds are $\emptyset, \{q, r\}, \{q, r, p\}$, then equality. The second row is symmetric; the other two fail equality. Adding $u \leftarrow p, v \leftarrow u, r$ makes the first model $\{q, r, p, u, v\}$, with rounds $\emptyset, \{q, r\}, \{q, r, p\}, \{q, r, p, u\}, \{q, r, p, u, v\}$, then equality. The other model remains $\{q, r, neg_p\}$. The candidate $N = \{q, r, neg_p, u, v\}$ has $\text{LM}(P^N) = \{q, r, neg_p\}$: its u, v lack a derivation from the empty start. Starting closure at N would wrongly accept unsupported atoms. If the r fact is omitted from (7.5), only $\{q, p\}$ is stable; adding that fact can therefore remove skeptical p . In a separate comparison, construct $\alpha = [q; q \Rightarrow p]$ when factual q is present, and $\beta = [r; r \Rightarrow \neg p]$ only when factual r is also present. Stipulate mutual rebut between the derived arguments and read the ordinary atom neg_p as the contested output $\neg p$. Initially only α is available, extracting $\{p\}$; after the r fact is added, $\{\alpha\}$ and $\{\beta\}$ are the two stable extensions, extracting $\{p\}$ and $\{\neg p\}$. This agrees with the program's contested outputs on these two inputs; it is not a general translation of arbitrary normal programs into AFs. These positive-body programs illustrate the study's reduct machinery. The following exact translation has a more restricted syntax.

7.5 AA2LP is an exact finite graph encoding

Definition 7.8 (AA2LP). *For any finite AF $F = (A, R)$, self-attacks allowed, choose injective atom names $\iota : A \rightarrow U$ and take exactly $U = \iota[A]$. Put $\text{Att}(a) = \{b \in A : bRa\}$. For each $a \in A$ place one rule*

$$\iota(a) \leftarrow \text{not } \iota(b_1), \dots, \text{not } \iota(b_t), \quad \{b_1, \dots, b_t\} = \text{Att}(a) \quad (7.6)$$

in P_F . All attackers occur conjunctively. An unattacked argument has a fact. The construction is the Handbook’s AA2LP (Sá-Dvořák–Caminada 2024, Section 3.2, Definition 11).

Theorem 7.9 (Stable AA2LP bijection). *The maps $E \mapsto \iota[E]$ and $M \mapsto \{a \in A : \iota(a) \in M\}$ are inverse bijections between $\text{Stable}(F)$ and the stable models of P_F , including the empty AF and frameworks with self-attacks.*

Proof. For every $E \subseteq A$, injectivity makes the rule for a survive the reduct at $\iota[E]$ precisely if $\text{Att}(a) \cap E = \emptyset$. All surviving rules are facts, and there are no other rules, hence

$$\text{LM}(P_F^{\iota[E]}) = \{\iota(a) : \text{Att}(a) \cap E = \emptyset\}. \quad (7.7)$$

If E is stable, its members have no attacker in E by conflict-freeness, and each outsider does have one by stable coverage. Equation (7.7) is therefore $\iota[E]$, making this a stable model.

Conversely, for a stable model $M \subseteq U$, set $E = \{a : \iota(a) \in M\}$. Exactness of $U = \iota[A]$ gives $M = \iota[E]$. Stability and (7.7), followed by injectivity, give

$$E = \{a : \text{Att}(a) \cap E = \emptyset\}. \quad (7.8)$$

An internal attack would put its target on the left but not the right, so E is conflict-free. Every omitted a has $\text{Att}(a) \cap E \neq \emptyset$, proving stable outsider coverage. The two maps are inverse on the entire power sets by injective naming and exact image universe, and thus on the indicated subfamilies. No step assumes $A \neq \emptyset$ or irreflexivity. At $A = \emptyset$ both families have the sole member $\emptyset$; a sole self-attacker has neither a stable extension nor a stable model. $\square$

This is a complete finite proof of the special two-valued correspondence reported in the Handbook, Section 3.2, Corollary 2. It is not a reverse translation from arbitrary normal programs. For the simple-WL shared-rule graph in Counterexample 7.3, (7.6) is

$$x_P \leftarrow, \quad x_A \leftarrow, \quad x_B \leftarrow \text{not } x_{\widehat{C}}, \quad x_{\widehat{C}} \leftarrow \text{not } x_B.$$

The models $\{x_P, x_A, x_B\}$ and $\{x_P, x_A, x_{\widehat{C}}\}$ precisely recover its two extensions. AA2LP preserves the graph’s *extra* $\{a, \neg b\}$ output as well as its PDL output. Graph-to-program exactness cannot repair a graph-to-PDL mismatch.

For a separate edge sensitivity calculation, take $a \rightarrow b$, $b \rightarrow c$, $c \rightarrow b$, $c \rightarrow d$. Its sole stable set is $\{a, c\}$. Omitting not a from the AA2LP rule for b is exactly deletion of $a \rightarrow b$; the altered graph has stable sets $\{a, c\}$ and $\{a, b, d\}$. The latter conflicts with the *original* edge. Adding $d \rightarrow a$ instead to the original graph gives stable sets $\{a, c\}$ and $\{b, d\}$. Each claim follows by forcing an unattacked node where available, then choosing one member of the b, c mutual pair and checking outsider coverage; the second change removes the premise that a is unattacked.

Example 7.10 (Two output boundaries). The mutual pair $u \leftrightarrow v$, both concluding p , has two stable extensions and two differently named AA2LP models, but its conclusion-output family is only $\{\{p\}\}$. Neither argument is skeptical although p is. An empty AF has the family $\{\emptyset\}$; a lone self-attacker has an empty stable family. Their guarded acceptance queries agree in being false, but their solution families are different. Yet the graph $s \rightarrow t$, $t \rightarrow t$ has stable extension $\{s\}$: a self-attacker elsewhere need not prevent a stable output.

7.6 Objects other than belief sets

The type of Out_T need not be a formula set. Dung’s 1995 Sections 3–4 construct frameworks also for stable sets of payoffs, stable matchings, defaults and programs; their separate general representation theorems are imported source results, not consequences of Theorem 7.9. For a bounded two-by-two instance, let both men prefer their opposite-indexed women and both women their opposite-indexed men, with strict complete preferences. Of the two complete matchings $M_1 = \{(m_1, w_1), (m_2, w_2)\}$ and $M_2 = \{(m_1, w_2), (m_2, w_1)\}$, both missing pairs block M_1 ; neither missing pair blocks M_2 . At the *whole-matching* graph level, assign an arrow from one matching node to another when it contains a pair blocking the latter. The sole arrow is $A_{M_2} \rightarrow A_{M_1}$, so the sole stable extension $\{A_{M_2}\}$ extracts M_2 . This calculation is an instance, distinct from Dung’s general *individual-pair* construction and proof (Theorem 39).

Related textbook exercises

Exercises 7.1 (changed rank and the source-policy question); 7.2 (repair construction, both extraction orders, general inclusion and support minimality); 7.3 (both AA2LP directions, inverse maps and empty/self-attack boundaries); 7.4 (omitted attacker condition, deleted edge and added edge); 7.5 (argument/conclusion quantifiers and empty-family boundaries); 7.6 (blocking pairs, whole-matching extraction and its limited scope); 7.7 (directional-strict raw-DWL counterexample); and 7.8* (both SP enumerations and source scope). Supplementary S7.1 (positive closure rounds and unsupported candidate) and S7.2 (fact addition and skeptical loss) connect to the normal-program subsection. The textbook’s statements and collected solutions remain the canonical exercise record.

8 Principles, closure and a representation limit

Textbook correspondence: Chapter 8. A correct extension calculation and a faithful representation answer different questions. Postulates constrain accepted arguments or conclusions; construction principles constrain relations across inputs. Their domains and output types must be fixed before an example can test them.

8.1 Four output tests and finite strict closure

Definition 8.1 (Output postulates). *For a structured extension E , let $C_E = \text{Conc}(E)$. Let $\text{Cl}_s(C)$ be the least superset of C closed under the declared strict rules. A formula set is directly consistent if it contains no pair φ, ψ with $\psi \in \bar{\varphi}$, using the declared contrary map (which may include self-contraries). Four tests are*

$$\begin{array}{ll}
 \text{subargument closure} & X \in E \implies \text{Sub}(X) \subseteq E, \\
 \text{strict closure} & \text{Cl}_s(C_E) = C_E, \\
 \text{direct consistency} & C_E \text{ has no contrary pair,} \\
 \text{indirect consistency} & \text{Cl}_s(C_E) \text{ has no contrary pair.}
 \end{array} \tag{8.1}$$

They are universal claims over relevant extensions on a specified input domain. If a semantics gives no extension, their vacuous satisfaction says nothing about the existence of a usable output.

An argument $X = [Y \rightarrow q]_r$ can be accepted while Y is omitted even if another accepted argument concludes $\text{Conc}(Y)$: identity, not merely conclusion, matters to subargument closure. Strict closure concerns formulas, not derivation identity. A directional strict rule is not silently supplemented by all classical inferences.

Proposition 8.2 (Finite strict closure). *Fix finite formula universe U , $C \subseteq U$, and finitely many strict rules $B \rightarrow h$ with $B \subseteq U$ and $h \in U$. Starting with $C_0 = C$, set*

$$C_{i+1} = C_i \cup \{h : (B \rightarrow h) \in \mathcal{R}_s, B \subseteq C_i\}. \tag{8.2}$$

The first equality occurs after at most $|U \setminus C|$ strict increases and one equality check. It returns $\text{Cl}_s(C)$.

Proof. The sequence is increasing and each strict increase adds an element of $U \setminus C$. At equality, the result contains the head of each rule whose body it contains, hence is closed. If D is any closed superset of C , induction gives $C_i \subseteq D$: initially true, and each newly added head belongs to D because its body does. The fixed set is therefore contained in every closed superset and is the least. Empty bodies are covered by the same induction. $\square$

Proposition 8.3 (Consistency implications and boundaries). *Indirect consistency implies direct consistency. Strict closure and direct consistency together imply indirect consistency. Neither direct nor indirect consistency alone implies strict closure, and strict closure alone implies neither consistency condition.*

Proof. Since $C \subseteq \text{Cl}_s(C)$, any contrary pair in C persists in its closure. If $C = \text{Cl}_s(C)$, direct and indirect consistency are exactly the same test.

For the converse boundaries, with sole rule $p \rightarrow q$, $C = \{p\}$ is indirectly (and directly) consistent but not closed. With rule $p, r \rightarrow \perp$, self-contrary $\perp$, and $C = \{p, r\}$, direct consistency holds, but closure adds $\perp$, so indirect consistency fails. With no applicable strict rule and $p, \neg p$ declared contrary, $C = \{p, \neg p\}$ is closed but fails both consistency tests. These examples use only the displayed rule and contrary systems. $\square$

Positive ASPIC+ rationality theorems for complete extensions import additional hypotheses on well-defined systems, contraries, strict inference and reasonable argument orderings (Modgil–Prakken 2013, Definitions 12 and 18, Theorems 12–15 and Corollary 17). Their proofs are imported results, not proved by Proposition 8.3; the corrected lifting of Chapter 5 and the 2018 corrigendum are required for preference claims. The small diagnostics here do not assert that an unrestricted ASPIC+ system under those hypotheses generates a violating output.

8.2 Permissive tests and sensitivity to a new route

In the textbook’s snoring-professor construction, the weakest-link graph has only $P_1 \rightarrow S_2$, and the last-link graph only $S_2 \rightarrow P_1$. Their unique stable extensions are

$$E_{WL} = \{S_0, S_1, P_0, P_1\}, \quad E_{LL} = \{S_0, S_1, S_2, P_0\}. \quad (8.3)$$

The unchallenged arguments must occur; the displayed edge excludes and attacks the sole outsider in either case, proving uniqueness. Each selected chain retains its subarguments; there are no strict rules; and each output includes exactly one of *accessDenied*, $\neg$ *accessDenied*. Hence all four tests pass both different conclusions. The postulates leave the application’s choice of priority purpose open.

For a second, expressly *stipulated* diagnostic, let A_1, A_2, N_1, N_2 conclude $i_1, i_2, \neg i_2, \neg i_1$, and put

$$R_0 = \{N_2 \rightarrow A_1, N_2 \rightarrow N_1\}.$$

Adding a defeasible route to an already given fact i_s permits a new argument N'_1 concluding $\neg i_2$. Keep the old arguments and evidence. Compare

$$R_{\text{good}} = R_0 \cup \{N_2 \rightarrow N'_1\}, \quad R_{\text{bad}} = R_{\text{good}} \cup \{N'_1 \rightarrow N_2\}. \quad (8.4)$$

In the original and good graphs the unique stable extension is $\{A_2, N_2\}$: A_2, N_2 are unattacked and N_2 covers every outsider. In the bad graph, if N_2 is selected that extension remains. Otherwise its only attacker N'_1 must be selected; A_1, N_1 must be selected because their only attacker N_2 is absent, while A_2 is unattacked. Thus the other and only other extension is $\{A_1, A_2, N_1, N'_1\}$. Its conclusions contain both i_2 and $\neg i_2$. The query $\neg i_1$ remains credulous but loses skeptical status; the full family of extracted stable outputs changes from $\{\{i_2, \neg i_1\}\}$ to $\{\{i_2, \neg i_1\}, \{i_1, i_2, \neg i_2\}\}$. These arrows are assigned for diagnosis. They are not claimed to be the full graphs of corrected ASPIC+.

Dung’s irrelevance-of-redundant-defaults condition demands equality of both stable and complete *belief-set families* when a default concluding an already given fact is added (Dung 2016, Definition 4.6). The stipulated calculation tests the stable part on one input, not that general result. Its attack-monotonicity clauses concern a relation of *strengthening of arguments*: if X strengthens X' , outgoing attacks from X' must persist from X , and attacks on X must persist on X' . For the fact-based N_1 strengthening the new defeasible N'_1 , these respectively read

$$N'_1 \rightarrow B \implies N_1 \rightarrow B, \quad B \rightarrow N_1 \implies B \rightarrow N'_1. \quad (8.5)$$

One failed edge implication alone does not compute the belief-set families. The conditional source theorem connecting attack monotonicity, context independence and redundant-default irrelevance on a sensible domain closed under those additions is imported from Dung 2016, Theorem 4.1; it is not a consequence of (8.4).

8.3 Disjoint components and an existence guard

Proposition 8.4 (Stable product and projection). *For finite AFs $F = (A, R)$, $H = (B, S)$ with $A \cap B = \emptyset$ and no cross edges, write $F \sqcup H = (A \cup B, R \cup S)$. Then*

$$\text{Stable}(F \sqcup H) = \{E_F \cup E_H : E_F \in \text{Stable}(F), E_H \in \text{Stable}(H)\}. \quad (8.6)$$

If $\text{Stable}(H) \neq \emptyset$, the projection of this family to A is exactly $\text{Stable}(F)$, even when the latter is empty. Existing-output skeptical comparison further requires $\text{Stable}(F) \neq \emptyset$.

Proof. If E is stable in the union, an internal conflict in $E \cap A$ or $E \cap B$ would conflict in E . Every outsider in A must be attacked from $E \cap A$, as no edge crosses components, and similarly for B . Both projections are therefore stable. Conversely, the union of stable component sets is conflict-free, and each outsider is covered within its own component. This proves (8.6). Every combined projection lies in $\text{Stable}(F)$; if $E_H \in \text{Stable}(H)$, pair it with each $E_F \in \text{Stable}(F)$ for the reverse inclusion. $\square$

The attack-free chain from fact p to q and an independent attack-free chain from r to s retain the old $\{p, q\}$ projection. Replacing the second component by a singleton self-attacker makes its stable family empty, and hence the combined family empty even though the first graph has not changed. Both guarded Boolean statuses of q become false with a *no-output* diagnosis. Separate source vocabularies alone do not ensure disconnected constructed graphs: classical strict inference from a contradiction may construct cross-vocabulary arguments. Proposition 8.4 does not prove general noninterference or crash resistance.

8.4 Exact PDL capture, context independence and attack closure

Now fix the finite normal-default fragment of Chapter 7, allowing tied ranks. An input $T = (\mathcal{R}_s, \mathcal{R}_d, \preceq, \mathcal{K}_n)$ has finite defaults, total-preorder priority, no ordinary premises, and consistent strict closure of its evidence. $\text{PDL}(T)$ is the *entire* source output family. Let $\text{Arg}(T)$ consist of exactly the finite induced arguments under the declared rules. One attack assignment att on domain $\mathcal{D}$ chooses a relation on $\text{Arg}(T)$ for *every* $T \in \mathcal{D}$, with no attack on a strict argument ($\text{DefRules}(X) = \emptyset$). Its relation is already evaluated, with no further preference filter. Exact capture is

$$\{\text{Conc}(E) : E \in \text{Stable}(\text{Arg}(T), \text{att}(T))\} = \text{PDL}(T) \quad (T \in \mathcal{D}). \quad (8.7)$$

A *sensible* domain is nonempty and contains every consistent evidence change of an included theory with its rules and priorities held fixed. Sensibility says nothing about deleting rules; inclusion of the three inputs below is an explicit additional premise.

Definition 8.5 (P2 and P6 on this domain). *For shared arguments $X, Y \in \text{Arg}(T) \cap \text{Arg}(T')$, put $D(X) = \text{DefRules}(X)$. Context independence (P2) requires*

$$\preceq_T \upharpoonright_{D(X) \cup D(Y)} = \preceq_{T'} \upharpoonright_{D(X) \cup D(Y)} \implies (X \text{ att}(T) Y \iff X \text{ att}(T') Y). \quad (8.8)$$

Attack closure (P6) requires every assigned attack to be an undercut or a contradiction of the conclusion of some subargument of its target; every undercut must be assigned. In the witness there are no undercuts, so P6 restricts eligible arrows to subargument contradictions, without requiring all such arrows.

Take $\mathcal{K}_n = \{\top\}$, no strict rules, and

$$\begin{aligned} d_1 : \top \Rightarrow a \ (1), \quad d_2 : \top \Rightarrow b \ (1), \\ d_3 : a \Rightarrow \neg b \ (2), \quad d_4 : b \Rightarrow \neg a \ (2). \end{aligned} \quad (8.9)$$

Let T contain all four, T_1 delete only d_4 , and T_2 delete only d_3 ; priorities restrict to each fragment. Define

$$S_1 = \{\top, a, \neg b\}, \quad S_2 = \{\top, b, \neg a\}, \quad S_3 = \{\top, a, b\}.$$

Lemma 8.6 (Source and argument families of the witness). *The exhaustive PDL calculation is*

	permitted sequences	PDL
T	$(d_1, d_3), (d_2, d_4)$	$\{S_1, S_2\}$
T_1	$(d_1, d_3), (d_2, d_1)$	$\{S_1, S_3\}$
T_2	$(d_2, d_4), (d_1, d_2)$	$\{S_2, S_3\}$.

(8.10)

With $P = \langle \top \rangle$, $A_1 = [P \Rightarrow a]_{d_1}$, $A_2 = [P \Rightarrow b]_{d_2}$, $A_3 = [A_1 \Rightarrow \neg b]_{d_3}$ and $A_4 = [A_2 \Rightarrow \neg a]_{d_4}$, the argument sets are respectively $\{P, A_1, A_2, A_3, A_4\}$, $\{P, A_1, A_2, A_3\}$ and $\{P, A_1, A_2, A_4\}$. Each S_i has exactly one possible argument-set preimage:

$$E_1 = \{P, A_1, A_3\}, \quad E_2 = \{P, A_2, A_4\}, \quad E_3 = \{P, A_1, A_2\}. \quad (8.11)$$

Proof. Initially only the tied d_1, d_2 are active in T . After d_1 , the newly enabled rank-2 d_3 must precede d_2 , and its $\neg b$ blocks d_2 . After d_2 , the symmetric d_4 blocks d_1 . In T_1 , the second branch cannot enable d_4 and continues with d_1 ; its existing b then blocks d_3 . T_2 is symmetric. The two initial choices exhaust each input and the continuation is forced, proving (8.10). The only rules construct the five displayed arguments and the deleted rules remove the corresponding A_4, A_3 . Each of $\top, a, b, \neg a, \neg b$ has the unique indicated argument on these inputs, so the preimages in (8.11) are forced. Such uniqueness need not hold in other structured theories. $\square$

P6 permits, in T , only

$$A_1 \rightarrow A_4, \quad A_2 \rightarrow A_3, \quad A_3 \rightarrow A_2, \quad A_3 \rightarrow A_4, \quad A_4 \rightarrow A_1, \quad A_4 \rightarrow A_3. \quad (8.12)$$

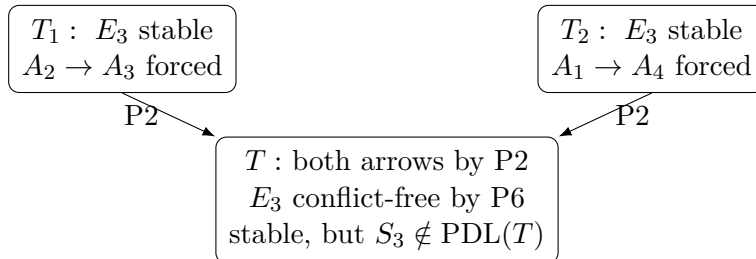
For example A_3 contradicts A_4 's subargument A_2 ; P contradicts none and cannot be targeted. This is an eligibility list, not an instruction to draw every arrow. In T_1 , among members of E_3 only A_2 can attack outsider A_3 ; in T_2 , only A_1 can attack outsider A_4 .

Theorem 8.7 (Bounded PDL/P2/P6 obstruction). *On every sensible domain $\mathcal{D}$ that contains T, T_1, T_2 , no attack assignment on their prescribed induced arguments jointly satisfies exact family capture (8.7), P2 and P6 under stable semantics.*

Proof. Assume an assignment with all three properties. Since $S_3 \in \text{PDL}(T_1)$, the source-to-graph direction of (8.7) requires a stable preimage. Lemma 8.6 makes it E_3 . Its sole outsider in T_1 is A_3 ; stability requires an arrow from E_3 to A_3 . P6 leaves only $A_2 \rightarrow A_3$. Similarly S_3 in T_2 forces E_3 stable, and P6 leaves only $A_1 \rightarrow A_4$ to cover its sole outsider. Thus

$$A_2 \text{ att}(T_1) A_3, \quad A_1 \text{ att}(T_2) A_4. \quad (8.13)$$

The first compared pair uses exactly $D(A_2) \cup D(A_3) = \{d_1, d_2, d_3\}$; its relative priority is identical in T_1, T . The second uses $D(A_1) \cup D(A_4) = \{d_1, d_2, d_4\}$, identically ordered in T_2, T . P2 transports both arrows in (8.13) to T . No two members of $E_3 = \{P, A_1, A_2\}$ have an undercut or subargument contradiction, so P6 forbids every internal assigned arrow: E_3 is conflict-free in T . Its two outsiders A_3, A_4 are covered by the transported arrows. Hence E_3 is stable in the full graph and extracts S_3 . But (8.10) excludes S_3 from $\text{PDL}(T)$; the graph-to-source direction of (8.7) fails. $\square$



The arrows between boxes depict transfer of forced attacks, not argument attacks. This proves the source’s Theorem 6.1 (Pardo et al. 2025, pp. 26–27) on the declared witness-containing domain, using its distinct source outputs and both inclusions of exactness. It does not follow from failure of a selected link policy.

The distinction between full families and single-literal queries is consequential here. The families $\{S_1, S_2\}$ and $\{S_1, S_2, S_3\}$ have identical unions and intersections: every one of $a, b, \neg a, \neg b$ is credulous and none skeptical in either family. An extra S_3 would escape those tests while violating (8.7).

One can assign, on *these three inputs*,

$$\begin{aligned} Q_0 &= \{A_3 \rightarrow A_2, A_3 \rightarrow A_4, A_4 \rightarrow A_1, A_4 \rightarrow A_3\} && \text{to } T, \\ Q_1 &= \{A_2 \rightarrow A_3, A_3 \rightarrow A_2\} && \text{to } T_1, \\ Q_2 &= \{A_1 \rightarrow A_4, A_4 \rightarrow A_1\} && \text{to } T_2. \end{aligned} \tag{8.14}$$

Every arrow is P6 eligible. In Q_0 , isolated P is included and exactly one of A_3, A_4 is selected: selection of A_3 excludes A_2, A_4 and forces A_1 ; selection of A_4 symmetrically forces A_2 . The stable family is $\{E_1, E_2\}$. The mutual pairs in Q_1, Q_2 , with their other unattacked nodes, give $\{E_1, E_3\}$ and $\{E_2, E_3\}$. These graphs recover (8.10) and violate P2 at precisely the two transported pairs. Three calculations do not define an assignment on a sensible domain or prove general context-dependent capture. Restricting to total priorities excludes this tied witness but supplies no new positive theorem, and does not revive raw-DWL correspondence. Changing arguments, semantics, extraction or the output requirement changes the contract and requires a new proof.

Related textbook exercises

Exercises 8.1 (three closure/consistency cases, implications, finite iteration and subargument identity); 8.2 (both professor policies and all four tests); 8.3 (original, good and bad Sherlock graphs, exhaustive stable families, query and consistency); 8.4 (both attack-monotonicity directions and the fact/route distinction); 8.5 (both product directions, projection, existence and independent blocks); 8.6(a–d) (the entire PDL source table, both exactness directions, P2 sets, P6 and domain boundary); 8.7 (literal-query insufficiency and all three context-dependent graphs); and 8.8(a–d) (changed representation contracts and their remaining proof obligations). The textbook’s collected solutions remain the canonical exercise collection.

9 Dialogue: transitions, reasons and semantic games

A dialogue can be evaluated as a public record, as a purposeful continuation, or as a game characterizing a semantics. These questions require different state and outcome functions. We first make public responsibilities explicit, then retain the occurrence-sensitive demands that even this state can omit.

9.1 An exact local public protocol

Definition 9.1 (Typed state and moves). *Let Ag, C be finite sets of agents and opaque claim labels, let $J = 2^C \setminus \{\emptyset\}$, and let $U = C \dot{\cup} J$. The disjoint union tags a claim P differently from the support unit $\{P\}$. A state is $s = (A_s, K_s, H_s, D_s, p_s)$, where*

$$\begin{aligned} A_s &\subseteq Ag \times C, \\ K_s &\subseteq Ag \times J \times \{+, -\} \times U, \\ H_s &\subseteq Ag \times Ag \times U, \quad D_s \subseteq Ag \times U \times \{+, -\}, \\ p_s &\in \{\perp\} \cup (Ag \times U \times \{+, -\}). \end{aligned}$$

These fields record active assertions, active justifications, recorded challenges (challenger, addressee, target), standing obligations, and a pending immediate reply. Initially the four sets are empty and $p_s = \perp$. Public commitments are derived:

$$\text{Com}_i(s) = \{c : (i, c) \in A_s\} \cup \bigcup \{\Phi : (i, \Phi, \sigma, u) \in K_s\}.$$

Write $\text{DOS}(s) = D_s$. The well-typed moves are

$$\text{assert}(i, c), \quad \text{question}(j, i, u), \quad \text{challenge}(j, i, u), \quad \text{justify}(i, \Phi \vdash^\sigma u), \quad \text{retract}(i, u).$$

A justification records supplied support, without asserting that Φ is a valid inference to or against u .

The state fields can be read as follows; commitments are recomputed from the first two fields.

Field	Public information retained
A_s	Agent–claim pairs for active assertions.
K_s	Speaker, nonempty support unit, sign and tagged target for each active justification.
H_s	Challenger, addressee and tagged target for recorded challenges.
D_s	Agent, tagged target and sign for standing obligations.
p_s	The required immediate reply, or $\perp$ if none is pending.

Example 9.3 follows these fields through a public record.

An *active witness* for (i, u) is $(i, u) \in A_s$ when u is a claim, or a record $(i, u, \sigma, v) \in K_s$ when u is a unit. A *challenge witness* is $\exists k \in Ag \ (i, k, u) \in H_s$, independently of the current requester. The following clauses define $\text{Legal}(s)$ and the transition $\delta(s, m)$; all fields not mentioned remain unchanged.

If $p_s = (i, u, \sigma)$, the only legal moves are $\text{justify}(i, \Phi \vdash^\sigma u)$, $\Phi \in J$. They add (i, Φ, σ, u) to K_s , set $p' = \perp$, and put

$$D' = (D_s \setminus \{(i, u, \sigma)\}) \cup \{(i, \Phi, \sigma)\}. \quad (9.1)$$

This is CR3 for $\sigma = +$, CR5 for $\sigma = -$. When $p_s = \perp$, the other clauses are:

- (i) CR1 permits $\text{assert}(i, c)$, adding (i, c) to A_s and $(i, c, +)$ to D_s .

- (ii) A question or challenge (j, i, u) requires $j \neq i$. An active witness gives CR2 and $p' = (i, u, +)$. Otherwise a challenge witness gives CR4 and $p' = (i, u, -)$. Without either witness the move is illegal. Thus CR2 has priority. A question changes only p_s ; a challenge also adds (j, i, u) to H_s and $(j, u, -)$ to D_s .
- (iii) CR6 permits $\text{retract}(i, u)$ exactly when an active witness exists. For a claim, remove its assertion record; for a unit, remove *every* $(i, u, \sigma, v) \in K_s$, of either sign and any target. Remove $(i, u, +)$ and $(i, u, -)$ from D_s .

Commitments are always recomputed from the remaining records. In particular, retraction neither removes recorded challenges nor recreates an obligation previously discharged by (9.1). A premise is not independently retractable merely because it occurs in a unit.

Proposition 9.2 (Finite public checking). *This protocol has a finite state space and a single successor for each legal state–move pair. A finite proposed record has either a first malformed move, a first illegal transition after a well-typed legal prefix, or a unique state sequence with rule certificates. A legal record need not end with $p = \perp$, and finiteness of the state space does not imply termination.*

Proof. Every state field ranges over a finite powerset or finite union, so their product is finite. First check move constructors, tags, agent membership and nonempty units. On a well-typed move the pending field either fixes the sole permitted justification form, or is empty. In the latter case the three move classes are disjoint; the only overlapping request witnesses are resolved by CR2 priority. Each update uses determined set operations. Induction on the record length therefore yields either the first failed test or one successor at every position, proving the certificate claim. An assertion followed by a legal challenge ends with a pending reply. For any agent and claim, alternating assertion and its retraction is an arbitrarily long legal record. Thus neither remaining assertion holds generally. $\square$

For implementation, a serialized claim and a serialized unit must retain different constructors; a string such as $Q\&S$ remains one opaque claim. The proposition checks a proposed formal record. It does not validate the interpretation of an utterance as that move.

Source boundary. Fatio supplies the five locutions and six combination rules (McBurney and Parsons, ULCS-04-002, §3, PDF pp. 4–5). Its positive justification retains the claim obligation *and* adds the argument obligation (§4.1, pp. 6–7). Equation (9.1), typed whole units, active witnesses, CR2 priority and symmetric unit withdrawal are local textbook conventions. The printed CR6 has a positive branch; the semantic discussion treats negative justification and argument retraction by analogy. The local rule is not evidence that the source prohibits negative withdrawal.

The source also gives mental-state conditions: the mental component of assertion by i 's precondition is $\bigwedge_{j \neq i} D_i B_j B_i \phi$, and that of its postcondition is $\bigwedge_{k \neq i} \bigwedge_{j \neq i} B_k D_i B_j B_i \phi$. These are not its full conditions, which also constrain and update the dialogue-obligation store. Here B_i, D_i are the source's belief and desire operators; we do not supply a modal semantics for them. Its operational account also uses decision mechanisms D1–D5 (§4.2). Our construction is therefore a *local public transition system*, not a proof of the full source's operational or BDI conditions. The supplied mechanization discussed in the textbook omits **pre-con** from its **Done** condition; that omission cannot establish those conditions either.

Example 9.3 (Responsibilities survive an immediate answer). Use the restaurant claim χ , meal claim P , and opposing claims Q, S , with agents A, B, C . The legal record

$$\begin{aligned} &\text{assert}(A, \chi), \text{ challenge}(B, A, \chi), \text{ justify}(A, \{P\} \vdash^+ \chi), \\ &\text{question}(C, B, \chi), \text{ justify}(B, \{Q, S\} \vdash^- \chi), \text{ retract}(A, \chi) \end{aligned}$$

has rule certificate CR1, CR2, CR3, CR4, CR5, CR6. Write $K_A = (A, \chi, +)$, $K_B = (B, \chi, -)$, $D_A = (A, \{P\}, +)$, $D_B = (B, \{Q, S\}, -)$. The ledger after its successive moves is

$$\{K_A\}, \{K_A, K_B\}, \{D_A, K_B\}, \{D_A, K_B\}, \{D_A, D_B\}, \{D_A, D_B\}.$$

Only turns 2 and 4 leave a pending reply, respectively $(A, \chi, +)$ and $(B, \chi, -)$. The final commitments are $\text{Com}_A = \{P\}$, $\text{Com}_B = \{Q, S\}$, $\text{Com}_C = \emptyset$. The challenge witness, not an assertion by B , licenses turn 4.

After turn 3, retracting the unit $\{P\}$ instead would leave $\text{Com}_A = \{\chi\}$ and ledger $\{K_B\}$; it would not restore K_A . After turn 5, withdrawing B 's negative unit would clear Com_B and D_B , but retain B 's recorded challenge. If s occurs only in the unit $\{s, f\}$, a challenge to atom s has no CR2 witness. A separate assertion of s can supply one; once challenged, its retraction cannot evade the pending reply.

9.2 Purpose and unfinished reasons

A protocol can be packaged as $\Pi = (Ag, M, S, s_0, \text{Legal}, \delta, W)$, with a separately supplied outcome condition W . A bounded continuation problem additionally specifies offered moves, a response policy, a budget, an observation boundary and an objective. These are not implicit in Legal.

After turn 3 of Example 9.3, offer A withdrawal or inquiry into B 's challenge. Fix B 's response to inquiry as $\{Q, S\}$. Stop after at least one further move, with no pending reply, once A 's assertion is withdrawn or B 's countergrounds are disclosed. For a two-move budget, the completed observations are

	retained	exposed	n	(exposed, retained, $-n$)
withdraw	0	0	1	(0, 0, -1)
inquire, answer	1	1	2	(1, 1, -2).

Lexicographic maximization chooses inquiry. Under a one-move budget only withdrawal reaches the stopping boundary; the single question leaves a reply pending. This calculation certifies the choice among the offered completed continuations under the supplied policy. It proves neither that all legal choices were offered nor that a participant will supply those grounds.

Definition 9.4 (A documentary frontier). *A reason ledger retains primary occurrence identifiers, their speaker, time, content, act, target and context, together with documentary answers and source/version references. A demand is a typed request with a target occurrence or comparison, context and required documentary source type. An answer closes it only when all these fields match. Let $F(L)$ be the set of open demands after a finite ledger L . Opening adds a demand; a matching answer removes it; a documentary answer in the following construction opens no new demand.*

Counterexample 9.5 (Forgetting a demand's target). Let q_R be the general claim that the restaurant suits this dinner, distinct from both option R and excellence claim χ . Let o_g be a primary reason targeting $\text{General}(q_R)$, and o_{RC} a comparative occurrence favouring R over café option C . Both use company as content. Distinguish a request d_0 for a general reason, d_1 for the grounds of o_g , d_2 for the R/C comparison, and d_3 for its warrant. In one fixed context

D_1 , use the following sequence:

	event	$F(L)$
0	start	$\emptyset$
1	request general reason	$\{d_0\}$
2	o_g answers and opens a grounds request	$\{d_1\}$
3	matching invitation evidence	$\emptyset$
4	request the R/C comparison	$\{d_2\}$
5	o_{RC} answers and requests its warrant	$\{d_3\}$
6	matching policy evidence, weights (4, 2)	$\emptyset$

Every documentary entry retains its source version. Define π_{occ} to retain just the primary occurrences and their attacks, and π_{con} additionally to identify equal content. Here $\pi_{\text{occ}}(L_5) = \pi_{\text{occ}}(L_6)$: both contain o_g, o_{RC} and no attacks. Yet $F(L_5) \neq F(L_6)$. If $F = f \circ \pi_{\text{occ}}$, equality of the projections would force equality of the frontiers, a contradiction. The same pair proves nonfactorization through π_{con} .

Changing the foil to takeaway T creates a different comparison target. Evidence for R/C with (4, 2) cannot close its warrant request; the stipulated R/T row uses (7, 5). This is loss under the declared projections, not an impossibility theorem about every enriched graph encoding.

9.3 Grounded acceptance as a finite dialogue game

Let $F = (A, R)$ be a finite AF. We use Chapter 6’s defense operator Def_F , its admissibility theorem, and the grounded iterates $G_0 = \emptyset$, $G_{k+1} = \text{Def}_F(G_k)$, whose stable limit is G_F . These supply semantic facts beyond the general fixed-point theorem.

Definition 9.6 (Standard grounded game). *A play begins with P ’s query argument x . Players alternate, each move attacking the immediately preceding argument. P may not repeat an argument at its own positions on the same play; O is unrestricted. A play ends when the next player has no legal move, and the last player wins.*

A winning strategy tree is rooted at $P : x$. At each P -node there is exactly one O -child for every attacker of its label; at each O -node there is exactly one chosen legal P -response. Every leaf is a P -node at which no O -move is available. Nonrepetition is path-local, so separate branches may reuse a label.

Theorem 9.7 (Finite grounded-game correspondence). *P has a winning strategy tree for x if and only if $x \in G_F$.*

Proof. For $x \in G_F$, let $\rho(x)$ be its first positive iteration index. If $\rho(x) = 1$, it has no attacker. If $\rho(x) = r > 1$, every attacker yRx is attacked by some $z \in G_{r-1}$. At the $O : y$ child choose such a z , and repeat. The ranks at successive P -positions strictly decrease. Hence P never repeats, the choices terminate, and leaves have rank one and no attacker. All O -attacks have been included. This constructs a winning tree.

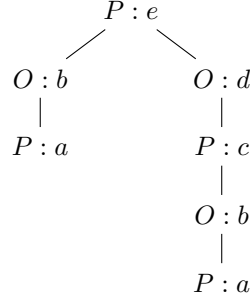
Conversely, on any play there are at most $|A|$ P -moves, so its length is at most $2|A|$. A strategy tree is finitely branching with this finite depth bound, hence finite. At a P -node let h be the maximum number of P -nodes on a path starting there. We prove that its label belongs to G_h by induction on h . For $h = 1$, any O -child would require a further P -child, since O -leaves are excluded. Thus the label is unattacked and belongs to G_1 . For $h > 1$, every attacker occurs as an O -child and has a chosen response whose subtree has height at most $h - 1$. By induction and the increasing grounded sequence, that response belongs to G_{h-1} . Consequently G_{h-1} attacks every attacker of the node’s label; the label belongs to G_h . Apply this to the root. $\square$

This is a complete proof for finite AFs of the convention in *Handbook of Formal Argumentation*, vol. 1, §7, Definitions 17–18, printed pp. 512–513. The source states the correspondence there with references; its earlier Theorem 6 concerns a different game. No infinite-framework or stronger game theorem is imported here.

Example 9.8 (One winning play is insufficient). Let $A = \{a, b, c, d, e, f\}$ and

$$R = \{(a, b), (b, c), (c, d), (b, e), (d, e), (a, f), (b, f)\}.$$

The grounded sequence is $\emptyset, \{a\}, \{a, c\}, \{a, c, e\}$, then constant. The winning tree for e has two branches:



For f , the play $[f, b, a]$ wins but $[f, a]$ loses, since a is unattacked. Deleting only (a, f) removes this losing branch and yields grounded extension $\{a, c, e, f\}$. In the separate mutual pair gRh, hRg , the play $[g, h]$ ends with O : repeating g as a P -move is forbidden. Its grounded extension is empty.

Thus a public certificate, an open-demand frontier, a winning play, a winning strategy and an extension are distinct outputs. The majority-aggregation example of Chapter 2 remains another boundary: the individual grounded sets $\{a, c\}, \{a, b\}, \{b, c\}$ accept each argument by majority, whereas the majority attack graph is a three-cycle, with empty grounded extension and no stable extension. Aggregating semantic answers need not commute with aggregating inputs.

9.4 Owned reasons and Boolean dependencies

Study S9.1 supplies a different, Libra-based public model. Its tokens include options $\text{Opt}(a)$, grounds $G_{\text{exo}} \dot{\cup} G_{\text{qry}}$ and weight atoms. Exogenous grounds cannot be questioned or retracted; queryable grounds can. Ownership by an agent, membership in the live option set Ω , current relevance $\text{Rel}(i, a) \subseteq O \setminus \{a\}$, outstanding targets, and truth values are separate fields. Weight atoms use $low \prec mid \prec high$ but are Boolean variables, not the numerical weights of Chapter 10.

For each relevant ordered pair there is a permission variable $\text{Perm}_i^{(a,b)}$. A disclosed contrast element contains a ground and justifying/requiring weight atoms for that *same* pair; its body is their conjunction. Comparison-package adequacy is the supplied abstract predicate $\text{Bal}(a, b, \Gamma)$. An element answer instead uses a finite ground package and installs $\chi := \bigwedge \Gamma$. A broad target demands value one for all current relevant pairs; a focal target demands it for one pair; an element target demands a ground or weight atom's value one. The bounded study admits only finite acyclic equation fragments with defined parents, one target per justification, and no unanswered broad target across an enlargement of relevance.

The operations have the following distinct effects. Assertion, permitted only for a token not already owned by its speaker, introduces ownership and a live option, initializes relevance, and installs its pair-permission constants at one; it opens no target. Questioning another agent's owned option, queryable ground or weight token opens a broad or element target without

changing equations. Challenging a against b , with distinct agents and options, introduces the challenger j 's ownership of b , adds b to $\text{Rel}(i, a)$, opens the owner's focal target and sets its pair constant to one. A new owner of b gets $\text{InitRel}(j, b) \cup \{a\}$ and all its corresponding permission constants at one; an existing owner adds a and resets that pair. Adequate justification adds ownership of each disclosed queryable ground or weight atom and sets it to one, replaces equations with the same left-hand side, and closes its target. Option withdrawal removes that ownership, sets the owner's corresponding permission variables to zero and removes the source-specified option/focal targets; the option ceases to be live only when its last owner withdraws. It does not prune relevance. Ground/weight withdrawal sets its atom to zero and removes that ownership and target, retaining dependent equations and without reopening a discharged burden.

Lemma 9.9 (Evaluation of a declared acyclic fragment). *Let a finite dependency graph have an edge from every variable used on a right-hand side to its left-hand variable. Suppose it is acyclic, every variable has one equation, and every equation is a total function of its displayed parents. There is a unique valuation satisfying the equations. If some equations are replaced and the resulting graph is still acyclic and fully defined, a variable can change only if it is a changed left-hand variable or is reachable from one in the union of the old and new dependency graphs.*

Proof. Every nonempty finite DAG has a vertex without incoming edges: otherwise following predecessors would repeat a vertex and give a cycle. Removing such vertices successively gives a topological order. Evaluate each equation when its parents have been evaluated. Totality gives existence at every step. Any satisfying valuation must agree at the first vertex and then at each subsequent vertex, which proves uniqueness. Outside the stated reachability set, the equation is unchanged and all its parents also lie outside that set. Induction along the induced topological order there shows that their values, and hence its value, agree in both valuations. $\square$

Example 9.10 (Closed targets need not preserve values). Abbreviate the permission variables by

$$p = \text{Perm}_A^{(\text{Save1}, \text{Bystander})}, \quad q = \text{Perm}_A^{(\text{Save1}, \text{Cafe})}, \quad z = \text{Perm}_B^{(\text{Cafe}, \text{Save1})}.$$

Initially all public fields are empty. Fix the policy entries

$$\text{InitRel}(A, \text{Save1}) = \{\text{Bystander}\}, \quad \text{InitRel}(B, \text{Cafe}) = \{\text{Save1}\}.$$

There are no exogenous grounds. Let j_1, j_2 be high-justifying atoms and r_1, r_2 low-requiring atoms for their respective pairs, with the queryable ground g . Stipulate exactly the two comparisons

$$\begin{aligned} \text{Bal}(\text{Save1}, \text{Bystander}, \{\langle g, j_1, r_1 \rangle\}), \\ \text{Bal}(\text{Save1}, \text{Cafe}, \{\langle g, j_2, r_2 \rangle\}). \end{aligned}$$

The six-move study first asserts Save1 , questions it, and justifies its original comparison, replacing $p := 1$ by $p := g \wedge j_1 \wedge r_1$. All three disclosed atoms are one. A challenge using Cafe adds the new relevance pair, sets $q := 1, z := 1$, and opens the focal q -target. An adequate second package replaces q by $g \wedge j_2 \wedge r_2$, with its disclosed atoms one. Withdrawal of Cafe by its sole owner finally sets $z := 0$ and removes it from Ω . The old relevance entries remain: $p = q = 1$ still constitute A 's two tests, although only Save1 is live.

Continue by questioning j_1 , answering with $j_1 := g$, and withdrawing the owned ground g , which sets $g := 0$. Lemma 9.9 yields

$$(p, q, z, j_1, j_2, r_1, r_2, g) = (0, 0, 0, 0, 1, 1, 1, 0).$$

Ownership of j_1 , the live option and the closed targets can all remain while these values change. No collective action or general persistence theorem follows from the six-move calculation.

Applications and proof status. The supplied Libra study is used within its declared fragment and abstract adequacy test: Pasetto, Skafte, van der Torre and Yu, *The Libra Dialogue Protocol for Comparative Deliberation* (2026), twelve-page author-supplied version, Definitions 4.4–4.13 and §5, pp. 9–10. This treatment follows approved Study S9.1 and its solutions. No fresh direct primary-paper comparison was completed in the recorded companion review. No general Libra soundness result is assumed. Study S9.2 distinguishes the formal EQR account, an implementation’s selected locutions, and evidence about people using CONSULT/EQRbot (Xydis, Castagna, Sklar and Parsons, *Applications of Argumentation-based Dialogues*, Handbook vol. 3, 2024, §4.2, pp. 251–252). An encoding/evaluator/dialogue interface can return acceptable arguments for a recommendation while leaving usability unmeasured. A reported admissibility soundness/completeness result requires its own formal protocol and proof; we do not import it for the smaller implemented move set. Neither an exact trace nor a semantic theorem establishes human benefit.

Related textbook exercises

- **9.1–9.4:** public stores, source/local distinctions, forced questions and challenge obligations.
- **9.5(a,b), 9.6–9.8:** tagged-target and pending-reply countertraces, malformed versus illegal records, whole-unit withdrawal and CR2 priority.
- **9.9–9.10:** budgeted continuation and occurrence-sensitive documentary frontiers, including the changed foil.
- **9.11–9.12, 9.13(a–e):** strategy quantifiers, grounded-game transfers, distinct output sorts and aggregation.
- **9.14–9.15** (Study S9.1): Libra’s six public states and the subsequent ground withdrawal; **9.16** (S9.2): formal, implementation and empirical evidence.
- **9.17:** external investigation of dialogue sources, including abduction and mechanization; no further theorem is required by its literature mention.

10 Issue standards, comparative permission and valuations

A statement can meet an issue standard, an option can be permitted, and a policy can select among permitted options. None of these outputs by itself records execution. This chapter connects the formal evaluations without identifying their objects.

10.1 Premise occurrences and fixed-input standards

Definition 10.1 (Issue-evaluation kernel). *A finite argument graph has statement nodes Q and argument nodes B . Each argument has finitely many premise occurrences, each with a statement label and one role: ordinary, assumption or exception. It has a conclusion issue and pro or con orientation. Edges run from premise statements to arguments and from arguments to their issues; the graph is acyclic. A context assigns each statement a status, each evaluated issue a proof standard, and a strict partial priority order $\succ$ on arguments.*

Let $A(q)$ denote issue q 's Boolean acceptability result. The holding test for an occurrence of q is:

<i>status</i>	<i>ordinary</i>	<i>assumption</i>	<i>exception</i>
<i>stated</i>	$A(q)$	1	$\neg A(q)$
<i>questioned</i>	$A(q)$	$A(q)$	$\neg A(q)$
<i>accepted</i>	1	1	0
<i>rejected</i>	0	0	1

(10.1)

An argument is defensible when all its premises hold. With currently defensible pro and con sets $\mathcal{P}(q), \mathcal{C}(q)$ supplied, define

$$\begin{aligned}
 SE(q) &\iff \mathcal{P}(q) \neq \emptyset, \\
 BA(q) &\iff \exists p \in \mathcal{P}(q) \forall c \in \mathcal{C}(q) (p \succ c), \\
 DV(q) &\iff \mathcal{P}(q) \neq \emptyset \wedge \mathcal{C}(q) = \emptyset.
 \end{aligned}
 \tag{10.2}$$

The assigned standard determines $A(q)$.

Premise role belongs to an *occurrence*; context status belongs to a statement. In particular accepted status is an input to (10.1), not another name for the output $A(q)$. An exception in stated or questioned status holds precisely when the blocking circumstance fails its standard. Holding that premise permits the argument; it does not establish the circumstance.

The standards and table are from Carneades 2007, Definitions 5–10, pp. 884–886 (the graph definition is Definition 4, p. 882). The textbook selects the fixed-input kernel (10.2), not the entire source evaluator, whose context additionally constrains complements and their standards. No general source theorem is transferred to a model lacking those conditions. A fully declared acyclic network of the displayed Boolean tests is evaluable by Lemma 9.9; that fact proves totality of such a *local* network only. The 2012 ASPIC+ translation concerns a later stage-specific CAES and additional representation structure, not an automatic correspondence for every instance of Definition 10.1.

Proposition 10.2 (Hierarchy and quantifier boundary). *For any fixed defensible sets and priority order, $DV(q) \Rightarrow BA(q) \Rightarrow SE(q)$. Neither converse holds in general. All three standards fail when $\mathcal{P}(q) = \emptyset$. Replacing BA 's $\exists p \forall c$ by $\forall c \exists p$ changes the test.*

Proof. DV supplies a pro witness; with no cons, it exceeds every con vacuously, proving BA. A BA witness supplies a pro, proving SE. If there is no pro, neither existential test can hold, nor can DV. For BA without DV take $\mathcal{P} = \{p\}$, $\mathcal{C} = \{c\}$, and $p \succ c$. For SE without BA reverse the sole priority. For the quantifier distinction take pros p_1, p_2 , cons c_1, c_2 , with exactly $p_1 \succ c_1$ and $p_2 \succ c_2$. This relation is irreflexive and transitive: it has no two-edge chain. Each con has a superior pro, but neither pro exceeds both. $\square$

Example 10.3 (Questioning an assumption). For the custody *issue* m , take $a_1 : w_1, w_2, w_5 \Rightarrow m$, $a_2 : w_3 \Rightarrow \neg m$, and $a_3 : w_4 \Rightarrow m$. Here a_2 is con m . Let w_1, w_3, w_4, w_5 be accepted ordinary premises and w_2 a stated assumption. All arguments are defensible, so $\mathcal{P}(m) = \{a_1, a_3\}$, $\mathcal{C}(m) = \{a_2\}$. With $a_1 \succ a_2$, SE and BA hold, but DV fails. All three linked premises of a_1 must hold; a_3 is a separate, convergent route.

Questioning w_2 replaces its unconditional holding test by $A(w_2)$. If this is false, a_1 drops out and BA requires $a_3 \succ a_2$. In a completely specified partial order with a_3 incomparable to a_2 , BA is false. For a separate priority input described only by $a_2 \succ a_1$, both $a_3 \succ a_2 \succ a_1$ and $a_2 \succ a_3 \succ a_1$ are possible completions, with opposite BA answers. The uncertainty then lies in the supplied order, not in (10.2). Failure to establish m establishes neither $\neg m$ nor a different custody option.

Example 10.4 (Production, an exception and a dependent issue). Let s be a blocking safety circumstance. Use SE as its production test and DV as its acceptability standard. The sole pro argument A_M for issue M has accepted ordinary premises and s as a stated exception. There are no con arguments for M ; assign BA to M . Write $d(A_M)$ for its defensibility. Then

$$A(s) = DV(s), \quad d(A_M) = \neg A(s), \quad \mathcal{P}(M) = \begin{cases} \{A_M\} & d(A_M) = 1, \\ \emptyset & d(A_M) = 0. \end{cases}$$

With $\mathcal{C}(M) = \emptyset$, (10.2) gives the exact trace

	$\mathcal{P}(s)$	$\mathcal{C}(s)$	$SE(s)$	$A(s)$	$d(A_M)$	$BA(M)$
no evidence	$\emptyset$	$\emptyset$	0	0	1	1
bare allegation	$\emptyset$	$\emptyset$	0	0	1	1
one report, rebuttal	$\{r_1\}$	$\{c\}$	1	0	1	1
two reports, rebuttal	$\{r_1, r_2\}$	$\{c\}$	1	0	1	1
rebuttal loses support	$\{r_1, r_2\}$	$\emptyset$	1	1	0	0

The last change establishes the exception, thereby disabling A_M ; it does not add a direct con argument for M . Counting reports cannot replace DV's no-con condition. These are fictional policy inputs, with no argument establishing a competing option F and no legal or factual verdict.

10.2 Comparative reasons and all-foil permission

Definition 10.5 (Weighted comparative instance). Let G, O be finite sets of grounds and feasible options. A reason is a triple

$$r = (g, o, f) \in R_{cr} \subseteq \{(g, o, f) \in G \times O \times O : o \neq f\}.$$

Its favoured option is o and its foil is f . A weight assignment is $w : R_{cr} \rightarrow \mathbb{R}_{\geq 0}^2 \setminus \{(0, 0)\}$, written $w(r) = (JW(r), RW(r))$. For unequal o, f , put

$$J(o, f) = \sum_{(g, o, f) \in R_{cr}} JW(g, o, f), \quad W(o, f) = \sum_{(g, o, f) \in R_{cr}} RW(g, o, f),$$

with empty sums zero. On this complete supplied instance,

$$\text{Perm}(o \mid f) \iff J(o, f) \geq W(f, o), \quad P_O = \{o \in O : \forall f \in O \setminus \{o\}, \text{Perm}(o \mid f)\}. \quad (10.3)$$

The justifying coordinate supports permission for o ; the requiring coordinate places a demand on the foil. Thus the opposing W in (10.3) has reversed indices. Equality passes. Reasons are set elements: a repeated utterance does not create a second copy. Empty sums

represent absent reasons in the complete instance, not an unelicited assessment. For $O = \emptyset$, $P_O = \emptyset$; for a singleton O , its option passes the vacuous all-foil test.

These definitions are the selected kernel of Yu, Cai, Markovich and van der Torre, AAMAS 2026 author version, §3.1, Definitions 3.1–3.4, PDF p. 3. The source’s further representation and selection results are not assumed.

Example 10.6 (The foil changes the reason). Let R, C, T denote a restaurant meal, a café meal and separate takeaway. The six comparative inputs and their resulting tests are:

g	(o, f)	(JW, RW)	$J(o, f) \geq W(f, o)$	$J(o, f) - W(f, o)$
company	(R, C)	$(4, 2)$	$4 \geq 3$	1
budget	(C, R)	$(3, 3)$	$3 \geq 2$	1
company	(R, T)	$(7, 5)$	$7 \geq 3$	4
speed	(T, R)	$(4, 3)$	$4 \geq 5$	−1
budget	(C, T)	$(2, 1)$	$2 \geq 3$	−1
speed	(T, C)	$(3, 3)$	$3 \geq 1$	2

Only R passes every foil, so $P_O = \{R\}$. The same company ground has different comparison triples and weights; content alone does not recover them.

Change only $RW(\text{speed}, T, C)$ from 3 to 2. The C/T margin becomes zero and the other five are unchanged. Hence $P'_O = \{R, C\}$. A least-cost policy with $C < T < R$ selects C ; a priority policy $R > C > T$ selects R . Both need a separate selection rule, and neither executes an action.

Across these two instances the justifying totals over all foils remain $R : 11, C : 5, T : 7$. Any function of just those totals has the same input in both instances and therefore the same output. Since permission changes, no such function computes (10.3) on both. This is a projection counterexample, not a claim that every ranking lacks every useful connection with permission.

On a distinct two-option input, rows for M/F with weights $(8, 4), (5, 2)$, and for F/M with weight $(4, 1)$, give $13 \geq 1$ and $4 < 6$: only M is permitted. Replacing the last row by $(7, 3)$ gives $13 \geq 3$ and $7 \geq 6$, permitting both. The favoured option’s two coordinates must not be confused with competing justifying totals.

10.3 Scores, ranks and what they forget

A bipolar graph is $B = (A, R^-, R^+)$, with attack and support relations on A . A deductive reading of xR^+y requires $x \in E \Rightarrow y \in E$; a necessary-support reading requires $y \in E \Rightarrow x \in E$. Either reading needs further clauses to evaluate arbitrary interactions. A support arrow by itself supplies neither a premise role nor a proof standard.

Consider instead declared numerical operations on a finite DAG: attenuation $\tau \prod_i (1 - s_i)$, support aggregation $1 - \prod_i (1 - s_i)$, and products of previously evaluated values. Every base is in $[0, 1]$.

Proposition 10.7 (Bounds and the categoriser). *Such an acyclic calculation has uniquely determined values in $[0, 1]$. On a finite acyclic AF the categoriser equations*

$$\text{Cat}(x) = \frac{1}{1 + \sum_{yRx} \text{Cat}(y)} \quad (10.4)$$

have a unique solution with all values in $(0, 1]$. The relation $x \succeq_{\text{Cat}} y \iff \text{Cat}(x) \geq \text{Cat}(y)$ is a total preorder.

Proof. Use the topological order from Lemma 9.9. Complements, finite products and one minus a finite product preserve $[0, 1]$, with empty product one. Induction proves the first

bound and the unique determined calculation. In (10.4), a sum of earlier positive values is finite and nonnegative; its denominator is at least one. Each next value is therefore uniquely in $(0, 1]$. Induction proves existence, and forces any other solution to agree at each vertex. Reflexivity, transitivity and totality of real comparison give the three preorder properties. $\square$

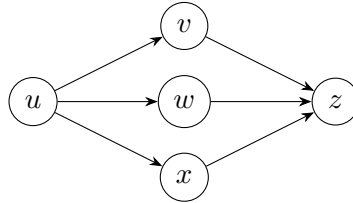
The named categoriser and its induced ranking follow Bonzon et al., Definitions 9–10, p. 916. Our proof is confined to DAGs; it establishes no convergence of iteration on a cyclic graph.

For the study's bipolar calculation, let $R^+ = \{(a_1, m), (a_3, m)\}$, $R^- = \{(a_2, m), (a_3, a_2)\}$. Bases .8, .6, .7 for a_1, a_2, a_3 give

$$s(a_2) = .6(1 - .7) = .18, \quad S^+(m) = 1 - (1 - .8)(1 - .7) = .94, \quad s(m) = .94(1 - .18) = .7708.$$

The same a_3 supports m directly and weakens its attacker. Whether both routes correctly represent dependent evidence is a modelling question left open by the interval bound.

Counterexample 10.8 (No acceptance threshold on these ranks). Take the five-node AF drawn below, with no other attacks:



Equation (10.4) gives $u : 1$, $v, w, x : 1/2$, $z : 2/5$. The attack product $s(a) = \prod_{bRa} (1 - s(b))$, with all bases one, gives $u, z : 1$ and $v, w, x : 0$.

No admissible set contains a middle vertex, because its attacker u cannot be attacked. A set containing z must contain u to answer its attackers. Thus the admissible sets are exactly $\emptyset, \{u\}, \{u, z\}$. The first two omit defended arguments; only $\{u, z\}$ is complete. It is maximal admissible and attacks every outsider, so all four semantics of Chapter 6 return that extension (grounded iteration is $\emptyset, \{u\}, \{u, z\}, \{u, z\}$). An upper threshold, strict or weak, that includes value $2/5$ must also include the larger $1/2$ values. It cannot recover exactly $\{u, z\}$.

Delete only (x, z) . Now $\text{Cat}(z) = 1/2$, but the admissible-set argument and product values are unchanged. The categorical agreement of the product on these two inputs is no general equivalence theorem.

Votes and factors. For finite nonnegative vote counts, the stipulated base

$$\tau(x) = \frac{V^+(x) + 1}{V^+(x) + V^-(x) + 2}$$

is strictly between zero and one: its numerator is positive and the denominator exceeds it by $V^-(x) + 1 > 0$. With no votes it is $1/2$. Counts $(8, 1), (3, 6), (5, 0)$ give bases $9/11, 4/11, 6/7$ in the preceding bipolar example. Consequently $s(a_2) = 4/77$, $S^+(m) = 75/77$, and $s(m) = 5475/5929$. Public endorsement, private belief, collective acceptance under an aggregation rule and this calculated score are different objects; the larger score is no measured gain in accuracy.

An ordinal factor representation retains another input:

$$\text{Case} = (F_M^+, F_M^-, F_F^+, F_F^-, \preceq, \text{outcome}).$$

The supplied case has $F_M^+ = \{\text{childPreference}, \text{fatherAway}\}$, $F_M^- = \{\text{motherLessWealthy}\}$, $F_F^+ = \{\text{fatherWealthier}\}$, and $F_F^- = \{\text{fatherAway}, \text{childPreference}\}$. The stipulated paired

comparisons $\text{childPreference} \succ \text{fatherWealthier}$ and $\text{fatherAway} \succ \text{motherLessWealthy}$ favour M locally. They give no general lifting rule for extra or unpaired factors. Presence, direction and priority are separate targets for challenge; mirrored descriptions of one circumstance are not independent corroboration.

10.4 Audience priorities and finite probability

Definition 10.9 (Audience defeat). *For an AF (A, R) , value assignment $\text{val} : A \rightarrow V$, and an audience's strict value order $>_v$, set*

$$R_v = \{(a, b) \in R : \neg(\text{val}(b) >_v \text{val}(a))\}.$$

Evaluate the resulting defeat graph with the selected semantics.

This is the selected value-based policy of Atkinson and Bench-Capon, Definitions 2.1–2.7, pp. 305–306. For mutual attackers a, b promoting privacy and convenience, a privacy-first audience retains only $a \rightarrow b$, with grounded, preferred and stable extension $\{a\}$. The opposite audience retains only $b \rightarrow a$, yielding $\{b\}$. Indeed a single-edge graph's source must be accepted and its target cannot be defended. No uniqueness theorem for every value-based framework, or moral authority for either audience, follows from these two inputs.

Definition 10.10 (Finite probability and two interpretations). *Let $\Omega \neq \emptyset$ be finite and $w : \Omega \rightarrow [0, 1]$ satisfy $\sum_{\omega} w(\omega) = 1$. For an event $E \subseteq \Omega$, put $P(E) = \sum_{\omega \in E} w(\omega)$. Then $P(\emptyset) = 0$, and disjoint partition into E and its complement gives $P(\Omega \setminus E) = 1 - P(E)$. For $P(H) > 0$, define $P(E \mid H) = P(E \cap H)/P(H)$; it is undefined for $P(H) = 0$.*

In a constellation model, worlds specify which graph exists before semantic evaluation. In an epistemic model on fixed $F = (A, R)$, worlds $Y \subseteq A$ are belief interpretations with mass $w(Y)$, and $P(a) = \sum_{Y \ni a} w(Y)$ is an argument marginal. Equal underlying powersets do not identify these interpretations.

Example 10.11 (Presence is not acceptance). Let a, b be independently present with probabilities $3/5, 1/2$; include attack $a \rightarrow b$ exactly when both endpoints are present. The product of presence/absence factors gives:

world	$\emptyset$	$\{a\}$	$\{b\}$	$\{a, b\}$
mass	$1/5$	$3/10$	$1/5$	$3/10$
grounded extension	$\emptyset$	$\{a\}$	$\{b\}$	$\{a\}$

The masses sum to one. Thus $P(a \text{ accepted}) = 3/5$, $P(b \text{ accepted}) = 1/5$, while $P(b \text{ present}) = 1/2$. Acceptance implies presence, so

$$P(b \text{ accepted} \mid b \text{ present}) = (1/5)/(1/2) = 2/5.$$

Absence is not an out label on a present argument. The graph does not entail the independence assumption.

Proposition 10.12 (Coherence, rationality and labels). *For an epistemic distribution on a finite AF define*

$$\begin{aligned} \text{COH}_F(P) &\iff \forall(a, b) \in R \quad P(a) \leq 1 - P(b), \\ \text{RAT}_F(P) &\iff \forall(a, b) \in R \quad P(a) > \frac{1}{2} \implies P(b) \leq \frac{1}{2}. \end{aligned}$$

COH implies RAT, but not conversely. Labelling an argument in, out, undec as its marginal is above, below or equal to $1/2$ need not give a complete Dung labelling, even under COH.

Proof. For any attack, COH gives $P(b) \leq 1 - P(a)$. If $P(a) > 1/2$, then $P(b) < 1/2$, which implies the required weak bound. This also covers self-attacks.

For the converse use fixed $A = \{a, b\}$, $R = \{(a, b)\}$ and masses $(1/5, 3/10, 1/10, 2/5)$ on $(\emptyset, \{a\}, \{b\}, \{a, b\})$. They are nonnegative and sum to one; the marginals are $P(a) = 7/10$, $P(b) = 1/2$. RAT holds, whereas COH would require $7/10 \leq 1/2$. The threshold labels are in, undec; Chapter 6’s complete conditions require in, out.

To refute sufficiency of COH, put all mass on $\emptyset$ on the same graph. Both marginals are zero, so COH and RAT hold. Thresholding labels both arguments out, but an unattacked a must be in in every complete labelling. $\square$

For comparison, masses $(1/5, 1/2, 1/10, 1/5)$ give marginals $7/10, 3/10$, satisfying COH at equality and matching the complete labels on this graph. The counterexample proves why that agreement cannot be generalized. The probability conventions and postulates follow Hunter, Polberg, Potyka, Rienstra and Thimm, Definitions 3.1–3.3, pp. 407–410, and Definitions 4.2–4.6, pp. 419–421. No cyclic scoring or epistemic-to-Dung correspondence theorem is imported.

10.5 Calculation and the authority of inputs

A weight audit separates its source, the representation and rule, the evaluated target and cases, and the basis for accepting or contesting the result. Repeatability of assignments and agreement with selected outcomes are different empirical measurements. Neither logically implies unique weights. For example, replacing every comparative weight by twice that weight doubles each side of (10.3), leaving every comparison and the permission set unchanged. Both assignments satisfy the nonnegative, nonzero-pair condition, and they differ whenever a reason exists. Even full agreement on finitely many outputs would leave other inputs and normative justification untested.

Study S10.4 reports Liga, Markovich and Yu’s distinction between fixed-graph assignments and jointly generated topology and weights (§§4.1–4.3, PDF pp. 4–5). Their one-step measurement differs from the recursive Euler-based evaluation applied later to averaged weights. We import neither as a replacement for (10.3). A Libra response may disclose a disputed source, weight or rule; its abstract adequacy predicate cannot certify the weights of a different model. The formal constructions here supply no new participant data or empirical result.

Related textbook exercises

- **10.1–10.3, 10.7–10.10:** priority completions, premise roles, linked/convergent structure, support readings, standard hierarchy (including the quantifier test), and the safety dependency.
- **10.6, 10.11–10.12:** distinct two-option and all-foil instances, the opposing requiring coordinate, and permission versus selection.
- **10.4–10.5, 10.13–10.15** (Study S10.1): attenuation, smoothed bases, interval bounds, finite-DAG categoriser, threshold impossibility and edge deletion.
- **10.16** (S10.2): audience defeat; **10.17–10.19** (S10.3): normalized possible-graph and belief distributions, conditioning, COH/RAT and complete labels.
- **10.20–10.22:** weight authority (S10.4), the local factor comparisons, and public votes versus belief and acceptance.
- **10.23:** external comparison of source objects and versions, including the 2012 stage-specific Carneades translation. The chapter’s exercises have no lettered subparts.

11 Comparing methods: locality, mechanisms and change

A principle quantifies over methods on a stated domain. A useful example can refute that universal statement; agreement on examples cannot establish it. We begin with ordinary semantics, then change the input and output types to compare support mechanisms, public transitions and practical permission.

11.1 Function properties and upstream restriction

An extension semantics on finite AFs assigns $\sigma(F) \subseteq 2^A$ to $F = (A, R)$. A principle has form

$$P(\sigma) \iff \forall F \in \mathcal{D} \ p(\sigma, F),$$

or quantifies over declared input changes or restrictions. Conflict-free and complete *outputs* constrain every $E \in \sigma(F)$; existence requires $\sigma(F) \neq \emptyset$; uniqueness here means at most one output. The empty AF has stable family $\{\emptyset\}$, while a self-attacker has stable family $\emptyset$. Chapter 6’s guarded skeptical query requires a nonempty family as well as membership in every returned extension.

For $U \subseteq A$, put $F|_U = (U, R \cap (U \times U))$ and $\mathcal{E}|_U = \{E \cap U : E \in \mathcal{E}\}$. In particular $\emptyset|_U = \emptyset$, not $\{\emptyset\}$. Say U is *unattacked* when no attack enters it from outside, and *isolated* when no cross attack exists in either direction. Directionality is

$$\sigma(F|_U) = \sigma(F)|_U \quad \text{for every unattacked } U. \quad (11.1)$$

Noninterference requires (11.1) only for isolated U . The definitions and their distinction follow van der Torre and Vesic, §4, Definitions 4.6–4.7, pp. 816–818. Directionality implies noninterference because every isolated set is unattacked.

Lemma 11.1 (Defense and admissibility under restriction). *If U is unattacked, then for every $S \subseteq A$,*

$$\text{Def}_F(S) \cap U = \text{Def}_{F|_U}(S \cap U). \quad (11.2)$$

A globally admissible set projects to a locally admissible set. A locally admissible $Q \subseteq U$ is globally admissible. If E is globally admissible, Q is locally admissible and $E \cap U \subseteq Q$, then $E \cup Q$ is globally admissible.

Proof. For $x \in U$, every attacker yRx belongs to U ; every possible counterattacker zRy also belongs to U . Thus the global and local tests have the same attackers and witnesses, with $z \in S$ equivalent to $z \in S \cap U$. This proves (11.2) in both directions, including no attackers. Restriction preserves conflict-freeness, and (11.2) preserves self-defense, proving the projection assertion. A local admissible Q has no external attackers, so its same defenses work globally; it is globally conflict-free.

For the union, only cross conflicts remain to exclude. An attack from $E \setminus U$ to Q would enter U . If $q \in Q$ attacked $e \in E \setminus U$, admissibility of E would supply $d \in E$ attacking q . Since no attack enters U , $d \in E \cap U \subseteq Q$, contradicting Q ’s conflict-freeness. Thus $E \cup Q$ is conflict-free. Each member of E retains its defense from E , and each member of Q its global defense from Q . The union is admissible. $\square$

Theorem 11.2 (Finite directionality). *Grounded, complete and preferred semantics satisfy directionality on finite AFs.*

Proof. For grounded semantics, start both defense iterations at $\emptyset$. Identity (11.2) and induction give $G_k^U = G_k \cap U$ for every k . Chapter 6 proves increasing admissible iteration and finite stabilization. Continue a stabilized sequence constantly while the other finishes. The limits obey $G_{F|_U} = G_F \cap U$; their singleton families obey (11.1).

If E is complete in F , it is admissible and $E = \text{Def}_F(E)$. The lemma makes $E \cap U$ admissible and, by (11.2), a fixed point of local defense, so it is complete locally. Conversely let Q be locally complete. It is globally admissible. Set $S_0 = Q$, $S_{k+1} = \text{Def}_F(S_k)$. Chapter 6's defense theorem for admissible sets makes all iterates admissible and increasing. Identity (11.2) and $\text{Def}_{F|U}(Q) = Q$ imply $S_k \cap U = Q$ at every stage. Finiteness yields an admissible fixed point S , hence a global complete extension with $S \cap U = Q$. Both family inclusions follow.

Let E be globally preferred. Its projection is locally admissible. Were it properly contained in a local admissible Q , the lemma would make $E \cup Q$ a proper global admissible superset, a contradiction. Thus its projection is locally preferred. Conversely let Q be locally preferred. It is globally admissible, and Chapter 6's finite maximal-extension theorem gives a global preferred $E \supseteq Q$. Its projection is locally admissible and contains Q , so local maximality forces $E \cap U = Q$. This gives the converse family inclusion. $\square$

The complete/preferred cells are literature results in the textbook (van der Torre and Vesic, p. 817); their finite versions have now been proved here. The argument uses admissibility preservation and finite maximal extension, not merely a general fixed-point theorem. It asserts no theorem for infinite AFs.

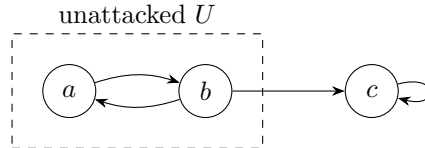
Proposition 11.3 (Stable projection). *If U is unattacked, then $\text{Stable}(F)|_U \subseteq \text{Stable}(F|_U)$; equality can fail even when both families are nonempty.*

Proof. For global stable E , $E \cap U$ is conflict-free. Each $x \in U \setminus E$ is attacked by an element of E ; the no-incoming condition puts that attacker in $E \cap U$. So the projection is locally stable.

For failure of equality take

$$A = \{a, b, c\}, \quad R = \{(a, b), (b, a), (b, c), (c, c)\}, \quad U = \{a, b\}.$$

Locally $\{a\}, \{b\}$ are exactly the stable extensions. Globally c 's self-attack excludes it and the mutual pair excludes $\{a, b\}$; only $\emptyset, \{a\}, \{b\}$ are conflict-free. The first covers nothing, the second misses c , and the third attacks both outsiders. Thus $\text{Stable}(F) = \{\{b\}\}$, proving strict inclusion. $\square$



Deleting only (c, c) admits the formerly blocked completion $\{a, c\}$; the whole stable family is then $\{\{a, c\}, \{b\}\}$. Indeed those sets cover all outsiders, while every other conflict-free candidate is empty or a singleton missing an outsider. This repairs this input, not stable directionality universally. For the chain $xRyRz$, $G_F = \{x, z\}$; restriction to unattacked $\{x, y\}$ gives $\{x\}$, but to attacked $\{y, z\}$ gives $\{y\}$, demonstrating the hypothesis.

Together with Chapter 6 this proves the comparison table:

	CF outputs	complete outputs	exists	at most one	directional
complete	yes	yes	yes	no	yes
grounded	yes	yes	yes	yes	yes
preferred	yes	yes	yes	no	yes
stable	yes	yes	no	no	no

The mutual pair has three complete and two preferred/stable extensions, refuting the corresponding uniqueness cells. A self-attacker refutes stable existence. Complete-output satisfaction, existence and directionality still do not characterize complete semantics: grounded semantics satisfies them and has different outputs on the mutual pair. The everywhere-empty-family function satisfies all per-output requirements vacuously and is directional, but fails existence.

11.2 Three support mechanisms

Use finite BAFs $B = (A, R^-, R^+)$; attack/support overlap is permitted. First evaluate the attack-only stable family $\mathcal{E}(B) = \text{Stable}(A, R^-)$, and define

$$\text{Score}_B(E) = |R^+ \cap (E \times E)|, \quad \text{Sel}(B) = \{E \in \mathcal{E}(B) : \forall T \in \mathcal{E}(B) \text{ Score}_B(E) \geq \text{Score}_B(T)\}. \quad (11.3)$$

This is internal directed-edge selection (Yu, Al Anaissy, Vesic, Li and van der Torre, *A Principle-Based Analysis of Bipolar Argumentation Semantics*, JELIA 2023 supplied manuscript, Definitions 7–8, p. 8), specialized to a stable base. Both directions of a mutual support count separately.

Proposition 11.4 (Selection and inactive support). *Selection inherits every property required of each base extension. For a finite base it returns a nonempty family exactly when the base is nonempty, and retains every maximizer. With attacks fixed, deleting any set of support edges internal to no base candidate leaves the selected family unchanged.*

Proof. Equation (11.3) returns a subfamily, proving inheritance. A finite nonempty set of real scores has a largest value, and (11.3) returns exactly its attaining candidates. An empty base has no candidate to return. Under the stated deletion every intersection $R^+ \cap (E \times E)$ for a base candidate is unchanged. Attacks, hence candidates, are unchanged too. Their scores and all comparisons in (11.3) are therefore unchanged. $\square$

These are per-output and preservation statements; they imply no locality or closure under the meaning assigned to support.

The second operation is a one-step attack reduction:

$$R_1^- = R^- \cup (R^+; R^-), \quad (x, z) \in (R^+; R^-) \iff \exists y (xR^+y \wedge yR^-z). \quad (11.4)$$

Both right-hand relations are the original inputs. The third is a defended-target query for supplied $S \subseteq A$:

$$D_B^+(S) = \{x : \forall y [yR^-x \Rightarrow \exists z \in S (zR^-y \vee \exists w (zR^+w \wedge wR^-y))]\}. \quad (11.5)$$

All quantified vertices lie in A ; the intermediary w need not lie in S . Equation (11.5) defines a subset of arguments, not a family of extensions.

Proposition 11.5 (Query monotonicity and iterated reduction). *The query D^+ is monotone in S , and, with attacks fixed, in support inclusion. Repeatedly applying the support-prefix rule to newly added attacks terminates at*

$$R_\infty^- = (R^+)^*; R^-,$$

where $(R^+)^*$ is reflexive transitive closure and R^- on the right is the original attack relation.

Proof. Increasing S preserves every old defending witness. Increasing support preserves each old direct attack and each old two-edge route while adding possible routes; the attackers tested are unchanged. Each old defended target therefore remains defended.

For the reduction put $R_0^- = R^-$ and $R_{n+1}^- = R_n^- \cup (R^+; R_n^-)$. Induction gives $R_n^- = \bigcup_{0 \leq k \leq n} (R^+)^k; R^-$: the base is $k = 0$, and prefixing one support edge adds precisely the next path length. The increasing sequence of subsets of $A \times A$ stabilizes finitely. Its union is exactly the displayed support-path composition, proving the assertion. $\square$

Even this iterated construction is not the source's full deductive reduction, which also includes super-mediated attacks (Definitions 9–10, pp. 8–9). The source's defenses 1–3 strengthen ordinary defense; none is the permissive (11.5) (Definitions 3–5, pp. 3–4). Their extension theorems are not imported.

Example 11.6 (A common input, distinct outputs). Take $A = \{a, b, c, d\}$,

$$R^- = \{(a, b), (b, a), (c, d), (d, c)\}, \quad R^+ = \{(a, c), (c, a), (b, d)\}.$$

A stable set chooses exactly one vertex from each mutual pair: two would conflict and none would leave an outsider uncovered. Thus the base family is $\{\{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}\}$. Their internal-support scores are 2, 0, 0, 1, giving $\text{Sel}(B) = \{\{a, c\}\}$. Delete only support (a, c) ; the scores become 1, 0, 0, 1, so both $\{a, c\}, \{b, d\}$ must be retained. Guarded skeptical acceptance then loses a, c .

The one-step reduction adds exactly $(a, d), (c, b), (b, c)$. It eliminates the two conflicting cross choices $\{a, d\}, \{b, c\}$. The other two base pairs remain stable. No singleton covers every outsider: a, b, c, d respectively miss c, d, a, a ; the empty set fails and the original mutual conflicts exclude triples. Hence the reduced stable family is exactly $\{\{a, c\}, \{b, d\}\}$.

For $S = \{a\}$, ordinary defense is $\{a\}$, but aR^+cR^-d also counters the attacker of c . No route from a counters a or c , the attackers of b and d , respectively. Therefore $D_B^+(\{a\}) = \{a, c\}$. The identical names in this set and in the selected family do not identify their output types.

11.3 Closure, locality and a supported target

Deductive closure requires every output E to satisfy $x \in E \wedge xR^+y \Rightarrow y \in E$. Inverse closure requires $y \in E \wedge xR^+y \Rightarrow x \in E$. On $A = \{a, b\}$ with sole attack aR^-b , the sole stable candidate is $\{a\}$, necessarily retained by selection. Support aR^+b refutes deductive closure; the alternative support bR^+a refutes inverse closure. The first example deliberately overlaps attack and support.

For support-aware locality, restrict both relations to U and require that no attack or support enters U . Equation (11.1) with Sel still can fail. On $A = \{a, b, c\}$, mutual attacks aR^-b, bR^-a , sole support aR^+c , and $U = \{a, b\}$, the whole stable candidates are $\{a, c\}, \{b, c\}$. They must contain unattacked c and choose one mutual opponent. Their scores 1, 0 select only $\{a, c\}$. Locally $\{a\}, \{b\}$ both score zero. Thus

$$\text{Sel}(B|_U) = \{\{a\}, \{b\}\} \not\supseteq \text{Sel}(B)|_U = \{\{a\}\}.$$

Nothing enters U , but outgoing support makes it nonisolated and affects the global score. This is the explicit bipolar analogue, not an ordinary-AF directionality theorem.

Counterexample 11.7 (Support need not preserve skeptical target acceptance). Apply preferred or stable semantics *after* (11.4) to $A = \{x, y, z, w\}$, initially no support, and attacks

$$R^- = \{(w, x), (x, y), (x, z), (y, w), (z, x)\}.$$

Add only support (x, y) . It composes with yR^-w to add only attack (x, w) . Before and after, x conflicts with all other vertices and y conflicts with w . Hence the complete conflict-free candidate list is

$$\emptyset, \{x\}, \{y\}, \{z\}, \{w\}, \{y, z\}, \{z, w\}.$$

There is no conflict-free triple. Their admissibility is:

	before	after	defense test
$\emptyset$	1	1	no members
$\{x\}$	0	1	new xR^-w answers the missing attacker
$\{y\}$	0	0	x unanswered
$\{z\}$	1	1	zR^-x answers its sole attacker
$\{w\}$	0	0	y unanswered
$\{y, z\}$	1	1	z answers x for both members
$\{z, w\}$	0	0	yR^-w unanswered

The unchanged attack xR^-z answers x 's other attacker. The unique old preferred extension is $\{y, z\}$, which is also stable since it attacks x, w . After the change the maximal admissible sets are $\{y, z\}, \{x\}$, both stable: x now attacks all outsiders. No other stable set can occur, because stable implies preferred by Chapter 6. Thus supported y loses guarded skeptical acceptance but remains credulously accepted. No vertex is unattacked before or after, so $\text{Def}(\emptyset) = \emptyset$ and both grounded extensions are empty. Overlap at (x, y) is part of this example.

A numerical principle instead quantifies over a valuation $s_B : A \rightarrow [0, 1]$: for each fixed A, R^- , weights and other parameters, adding a single absent support (x, y) must give $s_{B'}(y) \geq s_B(y)$. The Boolean counterexample above supplies no such valuation, hence does not refute that different principle. The meaning of support motivates a requirement; it does not prove it.

11.4 Preference policies and minimum edits

For a raw attack relation R and strict preference $\succ$ on arguments, define two distinct defeat policies:

$$\begin{aligned} R_1 &= \{(x, y) \in R : \neg(y \succ x)\}, \\ R_2 &= R_1 \cup \{(y, x) : (x, y) \in R, (y, x) \notin R, y \succ x\}. \end{aligned} \quad (11.6)$$

These are Reductions 1 and 2 of Kaci, van der Torre, Vesic and Villata, Definitions 2.4–2.5 and 2.7, pp. 216–217. They are not a revision of Chapter 5's exact-subargument defeat construction. With sole attack aRb and $b \succ a$, $R_1 = \emptyset$ and $R_2 = \{(b, a)\}$. The grounded and unique stable outputs are respectively $\{a, b\}$ and $\{b\}$. Conflict-freeness is tested in the chosen defeat graph; the first output conflicts under raw attack.

Definition 11.8 (Finite deletion enforcement). *Fix $F = (A, R)$, deletable $D \subseteq R$, and target $T \subseteq A$. For $\Delta \subseteq D$, write $F - \Delta = (A, R \setminus \Delta)$. Membership success means $T \subseteq G_{F-\Delta}$; exact-target success means $T = G_{F-\Delta}$. Cost is $|\Delta|$. The required output is the family of all minimum-cost successful deletions, or an infeasibility certificate.*

Theorem 11.9 (Contract of finite exhaustive enforcement). *List deletions by increasing cardinality. At the first successful level, finish that level and return all successes; if every level fails, report infeasibility. Using Chapter 6's grounded evaluator, this procedure terminates and meets the required output contract. Every minimum-cardinality success is inclusion-minimal.*

Proof. There are $2^{|D|}$ subsets, and each finite grounded calculation terminates, so the procedure terminates. Every returned edit passed the chosen success test. All smaller levels failed, proving minimum cost. Completing the first successful level includes every edit of that cost. If all levels fail, every permitted edit has been tested and none succeeds, proving infeasibility within this edit class. Finally a successful proper subset of a minimum edit would have smaller cardinality, a contradiction. $\square$

The converse minimality implication fails for general success conditions: a success family with inclusion-minimal members $\{r_1\}$ and $\{r_2, r_3\}$ has just the first as minimum. This order-theoretic example is not a claimed realization by a particular enforcement AF.

For $A = \{p, q\}$, $R = D = \{(p, q)\}$, $T = \{q\}$, the only permitted edits give grounded sets $\{p\}$ and $\{p, q\}$. Membership therefore has minimum cost one, while exact-target enforcement is infeasible. These are different certificates. The broader enforcement literature uses additional goals and operation classes (Baumann, Doutre, Mailly and Wallner, §3.1, Definition 3.1, pp. 457–458); the finite enumeration imports no efficiency guarantee for them.

11.5 Transition safety and alternative-set change

Proposition 11.10 (Immediate response, without completed history). *For every reachable public state of Definition 9.1 with pending $p = (i, u, \sigma)$, every legal next move is $\text{justify}(i, \Phi \vdash^\sigma u)$, $\Phi \neq \emptyset$. It clears the pending field and updates the ledger by (9.1). It is false that every legal finite record ends without a pending reply.*

Proof. CR2 sets a positive pending reply and CR4 a negative one. In either case the pending clause admits only the specified speaker, tagged target and polarity; its update clears the field and performs (9.1). No other reachable pending form exists, by induction on the transitions from the empty initial state. The two-move record $\text{assert}(A, \chi)$, $\text{challenge}(B, A, \chi)$, for distinct A, B , is legal and ends with $(A, \chi, +)$, refuting the stronger claim. $\square$

At this last state, withdrawal by A and justification by B are both illegal; A 's positive justification with $\{P\}$ is legal and changes the ledger to $\{(A, \{P\}, +), (B, \chi, -)\}$. Active assertion and challenge remain. Transition safety establishes neither a completed continuation nor adequacy of the grounds. The discharge remains the local convention, not Fatio's original positive effect.

Proposition 11.11 (Expansion of the alternative set). *Consider complete comparative instances of Definition 10.5 on $O \subseteq O'$. Suppose both totals $J(o, f), W(o, f)$ are unchanged for every unequal old pair. Then each old pairwise answer is unchanged and*

$$P_{O'} \cap O \subseteq P_O. \quad (11.7)$$

The reverse inclusion need not hold.

Proof. An old pair test uses just $J(o, f)$ and $W(f, o)$, both unchanged. An old option passing against every foil in O' passes in particular against every old foil, giving (11.7). In the dinner of Example 10.6, restriction to $\{R, C\}$ permits both options; adding T with the same old assessments leaves only R , since C fails against T . Restriction instead to $\{C, T\}$ permits only T ; adding R excludes it. These follow from the respective two surviving tests and all six full tests, and refute the reverse inclusion. $\square$

Takeaway is not permitted in the full instance but still excludes the café. No AF directionality theorem is involved. If old assessments may change, even (11.7) can fail: on fixed $O = O' = \{a, b\}$, sole reason favouring b over a with weight $(0, 1)$ excludes a ; replacing it by $(1, 0)$ permits a , while both assignments satisfy the weight domain. The unchanged-total hypothesis does real work.

11.6 Testing a principle and studying judgments

A finite principle tester takes a finite list of domain-valid records and a correct instance checker with values true, false, or unevaluated. A record may be an input, a changed pair, a targeted input or a restriction. Report 0 if some checked record violates the principle; 1 if the list is nonempty and every record is evaluated and passes; otherwise ?. Retain the sample and evaluation counts with this output.

Proposition 11.12 (Evidence supplied by the tester). *A 0 refutes the universal principle. A 1 proves satisfaction on the listed records, and proves the principle on its full domain only with a separate exhaustive-coverage argument. A ? establishes neither of these outcomes.*

Proof. A domain-valid checked failure is a witness to the negation of the universal quantifier. For 1, every listed instance is true; if every domain instance is listed, this covers that quantifier. Without coverage, any unlisted instance remains unchecked and could violate the principle.

For $?$, there is no checked failure and no completely checked nonempty list of passes, so neither certificate is present. $\square$

Thus one checked violation gives 0; six completed passes give 1 on that sample only; five passes and one unevaluated record give $?$; no records give $?$. The all-finite-BAF domain is not exhausted by six examples. In contrast, Theorem 11.9 proves exhaustion of its declared finite edit space.

An empirical study has another object:

$$Study = (\mathcal{I}, \sigma, H, \mathcal{P}, m),$$

recording presented inputs, a semantic prediction, a behavioral hypothesis, population and measurement. These are specifications of a study, not observations. The textbook’s Handbook vol. 2, Chapter 13, §2.2, pp. 860–865, reports three distinct designs:

- Rahwan and colleagues (2010): separate groups of 20 and 47 participants in Dubai offices/public spaces rated verbally presented conclusions on a seven-point confidence scale. The reported order was unattacked above reinstated above directly attacked; no simple/floating difference was detected. Intended graphs were not checked as participants’ reconstructions.
- Cramer and Guillaume (2018b): 130 Luxembourg undergraduates reconstructed attacks and gave acceptable/rejected/undecided judgments, with peer discussion and revised responses, on three graph patterns in news, science and calculation contexts. The reported majority responses across those patterns matched CF2, stage, stage2 and SCF2. Grounded and complete failed on floating and three-cycle cases; preferred and semi-stable failed on the three-cycle.
- Their 2019 design: 61 undergraduates judged twelve supplied AFs in an island/treasure setting after instruction about attacks. SCF2, CF2 and grounded were reported better predictors than preferred, semi-stable, stage and stage2, with no significant difference among the leading three. Stable was set aside when no extension existed, as in the earlier comparison.

These reports supply neither participant-level data nor a new analysis here. The latter two designs change presentation, graph families and response conditions; they cannot simply be pooled or treated as isolating a cause of changed rankings. The Handbook’s isolated spelling *SFC2* is retained as a source discrepancy; the surrounding definitions use SCF2. No extra semantic development of all those named predictors is needed to state what was compared.

For our running BAF, a prediction that participants choose $\{a, c\}$ because it has two internal support edges is hypothetical until a response task, support interpretation, population and observations are supplied. An alternative reading of a mismatch may concern reconstruction or task interpretation. Neither a mismatch nor a better predictive fit in one design refutes a theorem about the fixed mathematical function or establishes a universally best normative semantics. Runtime measurements, taken up next in the textbook, concern yet another specified object.

Related textbook exercises

- **11.1–11.4, 11.10:** stable candidates, all score ties, one-step reduction and permissive defense, with their distinct output types and monotonicity.
- **11.5–11.6, 11.9, 11.11:** exhaustive target-loss witness, numerical target requirements, output/change/methodological levels, and support-aware locality.

- **11.7, 11.12, 11.19:** finite enforcement and certificates; **11.19(i–iv)** distinguish a violation, a completed sample, an incomplete sample and no sample.
- **11.13–11.16:** counterfunctions and characterization, defense projection and grounded induction, the changed stable completion, and alternative defeat policies.
- **11.18(a):** deductive/inverse closure; **11.18(b):** inactive-support invariance; **11.18(c):** original-input versus iterated reduction (Study S11, *Mechanisms and change*).
- **11.20–11.21:** immediate-response safety and noncompletion, and fixed-old-pair alternative expansion.
- **11.8, 11.17:** hypothetical versus reported cognitive evidence and the three study designs; **11.L:** external source comparison. S11 has no separate numbered exercise bank.

12 Computing a specified answer

Verification, search and enumeration have different success conditions. A short witness can settle one question while leaving another open. We make those conditions explicit before treating attack graphs, public records and comparative reasons as computational inputs.

12.1 Tasks, representation and finite verification

Definition 12.1 (Tasks and certificates). *For a finite AF $F = (A, R)$, semantics σ , and $q \in A$, verification tests whether a supplied $E \subseteq A$ belongs to $\text{Ext}_\sigma(F)$; search returns such an E , perhaps subject to a query; enumeration returns the entire family. As in [Definition 6.12](#), put*

$$\begin{aligned}\text{Cred}_\sigma(F, q) &\iff \exists E \in \text{Ext}_\sigma(F) \quad q \in E, \\ \text{Skept}_\sigma(F, q) &\iff \text{Ext}_\sigma(F) \neq \emptyset \wedge \forall E \in \text{Ext}_\sigma(F) \quad q \in E.\end{aligned}$$

A certificate is an object with a specified checker for a specified claim. Checking a containing stable set certifies stable credulity. An omitting stable set refutes skepticism, but not credulity. A timeout certifies neither absence of a witness nor exhaustion.

The guard differs from unguarded skepticism on an empty family (Dvořák and Dunne, *Computational Problems in Formal Argumentation and their Complexity*, §3.2, pp. 647–648).

Number vertices $1, \dots, n$, store m distinct attacks as an edge list, and store set membership in an n -cell Boolean array. Indices fit a machine word; array access, index comparison and loop advance have constant word cost. Input storage is $O(n + m + 1)$ words and $O((n + m + 1) \log(n + 2))$ bits. A supplied candidate and edge list are assumed well formed. These representation assumptions, not the abstract set notation, license the following counts.

Procedure CheckStable. Initialize all n cells $\text{covered}[v]$ to false. Scan each edge (u, v) : reject if both endpoints lie in E ; otherwise, if $u \in E$, set $\text{covered}[v]$ true. After the scan, reject if some $v \notin E$ is unmarked; otherwise accept.

Procedure Defended. On membership array S , initialize attacked false and scan all (u, v) , marking v exactly when $u \in S$. Initialize a new array T true and scan all (b, a) , setting $T[a]$ false whenever $\text{attacked}[b]$ is false. Return T . All decisions read the same old S . Procedure **Grounded** starts with $S = \emptyset$, sets $T = \text{Defended}(S)$, returns S if $T = S$, and otherwise replaces S by T and repeats.

Theorem 12.2 (Correctness and cost). *CheckStable accepts exactly the stable candidates, in $O(n + m + 1)$ word operations and $O(n + 1)$ extra cells. Defended returns $\text{Def}_F(S)$ with the same bounds. Grounded returns G_F , using at most $n + 1$ defense evaluations, $O((n + 1)(n + m + 1))$ word operations and $O(n + 1)$ extra cells.*

Proof. After an edge prefix without rejection, no processed edge is internal to E , and a vertex is marked exactly when a processed edge from E reaches it. Initialization satisfies this invariant. The next edge either detects an internal attack or marks precisely its covered target. At termination, the two tests are exactly conflict-freeness and coverage of every outsider, the stable conditions of [Definition 6.4](#).

The first defense scan marks exactly $S^+ = \{v : \exists u \in S, uRv\}$. The second retains a exactly when every attacker bRa belongs to S^+ . This is defense, including vacuous defense of unattacked vertices. Each procedure initializes $O(n)$ cells and scans $O(m)$ edges plus at most n membership tests.

[Theorem 6.6](#) proves that the simultaneous empty-start defense iterates are admissible, increasing, and stabilize at the grounded extension. Each strict increase adds a vertex, so

at most n such increases precede one equality-detecting call. Each round, including array comparison, costs $O(n + m + 1)$. Only a fixed number of n -cell arrays is retained. For the empty AF one equality-detecting call suffices. $\square$

The semantic basis is Dung’s Definitions 6, 13 and 16 (pp. 326, 328–329); the explicit loops and their cost model are the declared textbook construction. A fixed point obtained from an arbitrary initial set is not thereby grounded.

Example 12.3 (A transfer with simultaneous updates). Let

$$H = (\{a, b, c, d\}, \{(a, b), (b, c), (c, b), (c, d)\}).$$

The candidate $\{a, c\}$ covers b, d and has no internal edge, so it passes. The candidate $\{a\}$ marks only b and fails at uncovered c . Grounded iteration is $\emptyset, \{a\}, \{a, c\}, \{a, c\}$: a first enters as unattacked, then counters c ’s attacker b . After deleting only (a, b) , iteration is $\emptyset, \{a\}, \{a\}$. The isolated a now counters nothing. In-place changes during an edge scan would not implement the simultaneous recurrence.

Coverage alone is insufficient for stable verification: the full set of a mutually attacking pair covers every outsider vacuously but conflicts internally.

12.2 An exact encoding and projected enumeration

For a CNF, empty disjunction is false and empty conjunction true. Introduce one original Boolean variable x_v per vertex and put

$$K_F = \bigwedge_{(u,v) \in R} (\neg x_u \vee \neg x_v) \wedge \bigwedge_{v \in A} \left(x_v \vee \bigvee_{uRv} x_u \right). \quad (12.1)$$

Conflict clauses alone lose attack direction; the coverage clause for v retains its *incoming* attackers.

Theorem 12.4 (Stable CNF contract). *The maps $I \mapsto \{v : I(x_v) = 1\}$ and $E \mapsto \mathbf{1}_E$ give inverse bijections between models of K_F and stable extensions. Before duplicate removal there are $n + m$ clauses and $n + 3m$ literal occurrences, counting repetitions. The encoding can be constructed in $O(n + m + 1)$ indexed word work. Consequently*

$$\begin{aligned} \text{Stable}(F) \neq \emptyset &\iff \text{SAT}(K_F), \\ \text{Cred}_{\text{stable}}(F, q) &\iff \text{SAT}(K_F \wedge x_q), \\ \text{Skept}_{\text{stable}}(F, q) &\iff \text{SAT}(K_F) \wedge \text{UNSAT}(K_F \wedge \neg x_q). \end{aligned} \quad (12.2)$$

Proof. A model selects no internal attack by the first clauses. For an unselected vertex its coverage clause requires a selected attacker. The selected set is therefore stable. Conversely a stable set satisfies each conflict clause and each coverage clause, the latter through membership or an attacker. There is exactly one total original assignment for each set, proving inversion. The m conflict clauses contribute $2m$ occurrences. The n coverage clauses contribute $n + m$. Build incoming lists with a single edge scan, then emit their clauses and the conflict clauses. The three query equivalences now follow by existence, containing membership and absence of an omitting model. $\square$

The final row has two obligations: a checked base model and a checked unsatisfiability argument for the omitting formula suffice. No polynomial-size UNSAT certificate is promised. For a self-attacking singleton, (12.1) is equivalent to $\neg x_a \wedge x_a$. Unguarded skepticism is vacuously true and guarded skepticism false. The empty AF instead has one model, the empty assignment, corresponding to its one empty stable extension.

Definition 12.5 (Exact argument block). *For $E \subseteq A$, define*

$$B_E = \bigvee_{a \in E} \neg x_a \vee \bigvee_{a \notin E} x_a.$$

For an encoding $Q_F(x, z)$ with auxiliary variables z , the projection contract is

$$\{\{a : I(x_a) = 1\} : I \models Q_F\} = \text{Stable}(F). \quad (12.3)$$

This requires soundness and surjectivity of extraction, not injectivity on full assignments.

Proposition 12.6 (Distinct-output enumeration). *Assume (12.3) and correct SAT/UNSAT answers. Starting with Q_F , report the argument set of each returned model, conjoin its block B_E , and repeat until UNSAT. Every stable set is reported exactly once. At most 2^n successful calls precede the final unsatisfiable query.*

Proof. Every literal of B_E is false at the argument assignment of E . Any different argument assignment flips a variable and makes its corresponding literal true. Thus the block removes exactly one projection, including all its auxiliary realizations. Inductively, the remaining projections are precisely the stable sets not yet reported. There are at most 2^n ; each successful call removes one. Final UNSAT says no realization remains, proving completeness as well as soundness and distinctness. $\square$

Blocking a full assignment of $K_F \wedge (z \vee \neg z)$ can report one extension twice, once for each value of z . For the empty AF, the empty extension is reported once; its empty block is false and removes all auxiliary realizations. The self-attacker is UNSAT initially and reports nothing.

The shorter clause $\bigvee_{a \in E} \neg x_a$ excludes every assignment selecting a superset of E , not exactly one argument assignment. For $E = \{a\}$ on a, b , it also excludes $\{a, b\}$, whereas $\neg x_a \vee x_b$ does not. This Boolean comparison does not claim that another stable set is necessarily lost: stable families are antichains. Indeed if stable $E \subsetneq E'$, E attacks each vertex in $E' \setminus E$, contradicting conflict-freeness of E' . The two-sided block provides the general exact-assignment contract.

In Example 12.3, the unit x_a forces a , conflict excludes b , coverage forces c , and conflict excludes d . The sole model is $\{a, c\}$, so c is guarded skeptical. Blocking it with $\neg x_a \vee x_b \vee \neg x_c \vee x_d$ proves exhaustion. After deleting (a, b) , a is forced independently; exactly one of b, c is selected. Choosing c excludes d , while choosing b forces d . Hence the new models are $\{a, c\}, \{a, b, d\}$: c remains credulous but not skeptical, and d becomes credulous. Deleting the graph edge changes both its conflict contribution and b 's coverage clause. Arbitrarily deleting one clause does not implement that update.

12.3 Verification, search and a query-specific lower bound

Exhaustively testing all 2^n subsets with Theorem 12.2 is a correct enumeration algorithm: after any candidate prefix it has retained exactly the stable sets in that prefix. This does not make search polynomial. Moreover, k disjoint mutual pairs have exactly 2^k stable sets. Each pair must contribute one endpoint, and every such choice is stable. For $k \geq 1$, writing all $n = 2k$ membership vectors requires $n2^k$ bit writes, superpolynomial in the explicit input length $O(k \log(k + 2))$. A compact symbolic family is a different requested output. At $k = 0$ one empty output record remains.

We use the usual encoded-input classes: P has deterministic polynomial decision procedures; NP has polynomial-length positive witnesses with polynomial checkers. A polynomial many-one reduction $f : L \rightarrow M$ satisfies $x \in L \iff f(x) \in M$. NP-hardness transfers every NP problem; NP-completeness combines this with NP membership. *Imported background:*

CNF-SAT is NP-complete. The general computation-to-formula proof is not developed here. This is the starting theorem in the textbook's Dvořák–Dunne route, §§2.1–2.2, pp. 637–643. The following argument proves the selected AF reduction in full.

Let $\varphi = \bigwedge_{j=1}^t C_j$ use variables $x_1, \dots, x_k$. Clauses are indexed even if equal; within each clause use a set of literals, and let $L = \sum_j |C_j|$. All following argument names are fresh and mutually distinct:

$$A_\varphi = \{q_\varphi\} \cup \{c_1, \dots, c_t\} \cup \{\ell_i^+, \ell_i^- : 1 \leq i \leq k\}.$$

The attacks are the mutual pairs $\ell_i^+ \leftrightarrow \ell_i^-$, the edges $\ell_i^+ Rc_j$ for $x_i \in C_j$ and $\ell_i^- Rc_j$ for $\neg x_i \in C_j$, and $c_j Rq_\varphi$ for every clause.

Theorem 12.7 (Selected SAT reduction).

$$\text{SAT}(\varphi) \iff \text{Cred}_{\text{stable}}(F_\varphi, q_\varphi).$$

The construction uses $2k + t + 1$ vertices and $2k + t + L$ attacks. Stable credulous acceptance is NP-complete, conditional on the imported CNF-SAT theorem.

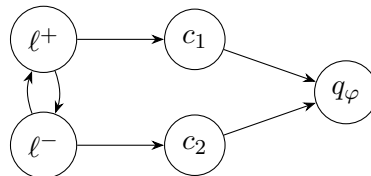
Proof. For a satisfying assignment I , select q_φ and its one true literal argument per variable. No clause or opposite literal pair is selected, so this set is conflict-free. Every omitted literal is covered by its selected opposite, and every clause by a selected true literal. These are all outsiders; the set is stable.

Conversely, a stable set containing q_φ selects at most one literal per pair by conflict-freeness, and at least one because each literal's only attacker is its opposite. It contains no clause argument, since each attacks q_φ . Coverage of every omitted clause therefore supplies a selected literal in it. The pair choices define a total satisfying assignment.

The displayed counts follow by the disjoint node and edge types. Constructing the normalized instance costs $O(k + t + L + 1)$ indexed operations and polynomial bit work. If duplicate literal occurrences are supplied, their normalization is charged to the raw input length and can also be done polynomially. Thus the construction is a polynomial reduction (Reduction 3.6 and the credulous-acceptance discussion in Dvořák–Dunne, pp. 651–652). A guessed n -bit set containing the query is polynomially checked by Theorem 12.2, establishing NP membership. Composing any NP-to-SAT reduction with this construction proves NP-hardness. $\square$

With no clauses, the query is unattacked and the empty conjunction is satisfiable. An empty clause gives an unattacked clause node which cannot be covered while the query is selected. An unused variable still requires a literal choice. A tautological clause is covered by either choice of its complementary variable. For $(x \vee y) \wedge \neg x$, the seven-vertex, nine-edge graph has sole query-containing stable set $\{q, \ell_x^-, \ell_y^+\}$; the second clause forces ℓ_x^- , then the first forces ℓ_y^+ . The candidate $\{q, \ell_x^+, \ell_y^-\}$ leaves the second clause uncovered.

Counterexample 12.8 (Dropping the query). For $\varphi = (x) \wedge (\neg x)$, the reduction graph is



Choosing ℓ^+ excludes c_1, ℓ^- , forces uncovered c_2 in, and excludes q_φ . This yields $\{\ell^+, c_2\}$; the other literal symmetrically yields $\{\ell^-, c_1\}$. Both are stable and these cases exhaust the literal choices. Thus an unsatisfiable formula gives stable existence, but no query-containing stable set. The theorem proves hardness of the designated credulous task. Stable existence is in NP by our checker; its separate NP-hardness proof is not inferred or imported here.

Preferred credulity has a different certificate: [Theorem 6.5](#) gives

$$\exists P \in \text{Pref}(F) \ (q \in P) \iff \exists S \in \text{Adm}(F) \ (q \in S).$$

Check admissibility by conflict-freeness and $S \subseteq \text{Def}_F(S)$, using [Theorem 12.2](#). This proves NP membership without verifying a maximal output. For H , $\{a\}$ is such a witness although the proper superset $\{a, c\}$ is admissible. Nor does testing only single-argument additions establish maximality: on aRb, bRc, cRd, dRa , no singleton is admissible, yet $\{a, c\}$ is, so $\emptyset$ is not preferred.

Define coNP by complementation of NP and

$$\text{DP} = \{L_1 \cap L_2 : L_1 \in \text{NP}, L_2 \in \text{coNP}\}.$$

On well-formed (F, q) , let L_1 assert stable existence and L_2 assert no stable set omits q . A stable set witnesses L_1 , and an omitting stable set witnesses the complement of L_2 , both with polynomial checks. Thus guarded stable skepticism belongs to DP. This is an upper bound, not DP-hardness or completeness; the credulous reduction does not prove the latter ([Study S12](#); the class definition is [Dvořák–Dunne, §2.1.2, pp. 638–639](#)).

12.4 Exact conditioning of components and updates

A directed path may have length zero. Mutual reachability partitions a finite AF into strongly connected components (SCCs). Their directed component graph is acyclic: a cycle of distinct components would make their vertices mutually reachable. Hence SCCs admit a topological order, by the finite acyclic ordering argument in [Lemma 9.9](#). The following stable-only construction is proved here; it requires no imported general SCC theorem for other semantics.

Theorem 12.9 (Stable residual components). *Let $C_1, \dots, C_s$ be a partition of A into nonempty blocks, with no attack from C_j to C_i when $j > i$. For a candidate $E \subseteq A$, put*

$$E_i = E \cap C_i, \quad E_{<i} = \bigcup_{j<i} E_j, \quad D_i = C_i \cap E_{<i}^+, \quad U_i = C_i \setminus D_i.$$

Then

$$E \in \text{Stable}(F) \iff E_i \in \text{Stable}(F|_{U_i}) \quad \text{for every } i. \quad (12.4)$$

In particular the right side requires $E_i \subseteq U_i$.

Proof. If E is stable, conflict-freeness forbids selecting a vertex in D_i . Restriction makes E_i conflict-free. For $x \in U_i \setminus E_i$, global stability supplies a selected attacker. It is not earlier, by the definition of U_i ; it is not later, by the ordering. It therefore lies in E_i . So E_i is stable in the residual.

Conversely suppose every residual choice is stable. Internal conflicts are excluded. For an edge from an earlier selected vertex into C_i , the target lies in D_i , hence outside E_i . Backward edges do not exist, so the union is conflict-free. For an unselected $x \in C_i$, membership in D_i gives coverage by the prefix; otherwise residual stability gives coverage by E_i . All outsiders are covered. $\square$

Enumerate every residual stable choice for each current prefix, then pass each extended prefix forward. After block i the prefixes are exactly the stable sets of the prefix-induced AF: apply (12.4) to that initial block partition and induct on i . At the final block this proves soundness and completeness of the enumeration. An empty residual has family $\{\emptyset\}$ and extends the prefix without adding a vertex; an empty local family kills that branch. When $A = \emptyset$, there are no blocks and the initial empty prefix is the one stable output.

Example 12.10 (Three conditioned branches). Use the separate computational graph of [Proposition 6.8](#):

$$A = \{a, b, c, d, e\},$$

$$R = \{(a, b), (b, a), (b, c), (c, d), (d, c), (d, e)\}.$$

Its ordered SCCs are $\{a, b\}, \{c, d\}, \{e\}$. The first block chooses a or b . After a , the second residual is the mutual pair c, d ; choosing c forces the last singleton e , and choosing d removes it. After b , $D_2 = \{c\}$, so the second residual forces d , which removes e . Thus (12.4) gives exactly

$$\text{Stable}(F_{\text{reuse}}) = \{\{a, c, e\}, \{a, d\}, \{b, d\}\}. \quad (12.5)$$

Each set is conflict-free and covers its outsiders. The grounded extension is empty since every vertex is attacked. Free combination with the isolated second pair would wrongly allow b, c , an internal attack. [Proposition 8.4](#)’s disconnected product and [Proposition 11.3](#)’s safe upstream projection do not justify that combination.

For reuse after a change, equality of a block’s vertex set, internal relation and current incoming attacked set is a sufficient condition for equality of its residual problem. Its *complete local family* may then be reused. This statement is branch-relative: enumerate every updated prefix, recompute downstream D_i , and check that the no-backward-edge partition remains valid. It does not authorize retaining only old global outputs or discarding formerly unsuccessful prefixes. Equality of the actual residual AF is the underlying sufficient condition; the listed conditions are not claimed necessary.

Add only (e, a) to F_{reuse} . The old $\{a, c, e\}$ conflicts; the other two sets survive. To prove there are no new sets, stability still requires exactly one of a, b : both conflict, and neither leaves b uncovered. With a , omit e , so d must cover e , excluding c . With b , exclude a, c , force d , and exclude e . Hence the new family is exactly $\{\{a, d\}, \{b, d\}\}$. The cycle $eRaRbRcRdRe$ merges all SCCs; the old partition fails the hypothesis of (12.4). Meanwhile additions can also *create* stable outputs: a self-attacking u and unattacked v have no stable set, but adding vRu yields the unique stable set $\{v\}$. Checking survivors is never a general completeness argument.

Modularity supplies the source context for this conditional treatment (Baroni, *Modularity in Argumentation*, §2.2, pp. 941–945). Neither the finite proof nor the example establishes an incremental speedup. A portfolio advantage likewise requires a declared task, input collection, solver configurations, hardware, time/memory and failure measures, selection overhead and comparison baseline. A worst-case hardness result does not settle that empirical claim.

12.5 Public records and bounded continuation

[Definition 9.1](#) supplies the full typed local public state $s = (A_s, K_s, H_s, D_s, p_s)$, CR1–CR6 and the deterministic update δ . [Proposition 9.2](#) supplies unique execution. The following checker makes the returned evidence explicit:

```

CheckPublic(initial, moves):
  s := initial; certificate := []
  for j := 1 to length(moves):
    if not WellFormed(moves[j]): return Malformed(j, s)
    rule := LegalRule(s, moves[j])
    if rule == none: return Illegal(j, s)
    s := Delta(s, moves[j], rule)
    append rule to certificate
  return LegalPrefix(certificate, s, s.pending)

```

Assume a supplied well-formed initial state; claims of reachability from the empty start additionally require its earlier trace. `LegalRule` retains CR2 priority, active witnesses and the pending-reply guard, not just visible commitment sets.

Proposition 12.11 (Trace and bounded-choice contracts). *CheckPublic returns the earliest failure and its prior state, or the exact final state, successful rule list and pending field. Given a finite offered collection of finite move lists under a fixed response policy, a move budget b , a decidable stopping test applied after each successful move, and an effectively computable score on completed prefixes with an effective total-order comparison, exhaustive simulation returns exactly all highest-scoring offered prefixes whose first stopping state occurs after $1, \dots, b$ moves.*

Proof. After each successful iteration, s is the state reached by exactly the processed prefix, and the certificate lists exactly its rules. The empty prefix satisfies this invariant; one legal update preserves it. Testing well-formedness before legality and applying no update on failure proves the first-failure statement. If no failure occurs, the invariant proves the returned fields.

For each offered list start afresh at the initial state, check successive moves, and test stopping after each success. Retain only its first stopping prefix if reached within budget, and compute that prefix's score. Illegal, exhausted or over-budget unfinished prefixes are not completed candidates. Finiteness of the lists and budget makes this procedure terminate. It checks every eligible prefix and admits only such prefixes. A maximum of a finite nonempty score set exists; comparison retaining all ties returns precisely all maximizers. If there is no completed candidate, report that bounded failure. $\square$

This contract neither searches unoffered histories nor promises an optimal strategy under unknown responses. No general dialogue representation cost or strategic complexity result is asserted.

For the restaurant state s_3 in [Example 9.3](#), retain exactly A 's assertion χ , its justification $\{P\} \vdash^+ \chi$, and B 's recorded challenge. There are no other active records; $p = \perp$, $\text{Com}_A = \{\chi, P\}$, $\text{Com}_B = \text{Com}_C = \emptyset$, and $D = \{D_A, K_B\}$, where $D_A = (A, \{P\}, +)$, $K_B = (B, \chi, -)$. The one-move question $\text{question}(A, B, \chi)$ is CR4, leaving these stores unchanged and pending $(B, \chi, -)$. A following $\text{retract}(A, \chi)$ is illegal at position 2. A following $\text{justify}(B, \{Q, S\} \vdash^- \chi)$ instead gives certificate (CR4, CR5), adds $\text{Com}_B = \{Q, S\}$, replaces K_B by $D_B = (B, \{Q, S\}, -)$, and clears the reply.

Offer precisely withdrawal and that inquiry/reply list, with the reply content fixed by policy. Stop at the first nonpending state after at least one move where the assertion is withdrawn or the countergrounds exposed. Maximize (exposed, retained, $-n$) lexicographically. Withdrawal yields $\text{Com}_A = \{P\}$, ledger $\{D_A, K_B\}$ and score $(0, 0, -1)$; inquiry yields the state above and $(1, 1, -2)$. Budget two selects inquiry; budget one completes only withdrawal. The intervening question is legal but unfinished. Neither outcome proves restaurant excellence. The local discharge remains distinct from full Fatio (McBurney–Parsons, §§3–4.1, pp. 4–7), as already fixed in Chapter 9.

12.6 Permission followed by an explicit selector

Use [Definition 10.5](#)'s complete distinct comparative reason set, with finite indexed option set O of size k . For execution, encode each nonnegative, nonzero weight pair by exact rationals. Let r be the number of distinct reason records. These are assessed inputs; an unknown coordinate is not a zero.

Procedure **Permitted** initializes $J[o, f] = W[o, f] = 0$ for all $o \neq f$, then adds each reason's JW, RW once to its own ordered-pair cells. For each option o , initialize $\text{passes} = \text{true}$, test every $f \neq o$, and set it false if $J[o, f] < W[f, o]$. Return all passing options. Procedure **Select** scans a supplied strict total order of O , best first, and returns its first permitted member; if

none, return NoSelection.

Proposition 12.12 (Practical computational contract). *Permitted* returns exactly [Definition 10.5](#)’s all-foil set. *Select* returns its unique best option under the supplied order, or NoSelection exactly when that set is empty. The displayed loops initialize $2k(k-1)$ cells, make $2r$ rational additions and $k(k-1)$ rational comparisons, and use at most k indexed membership checks for selection.

Proof. After any reason prefix, each cell is the corresponding coordinate sum over exactly that prefix. Initialization and one addition per coordinate preserve this invariant. At termination the cells are the defined J, W . After a foil prefix, an option’s flag is true exactly when every processed comparison passes, so the final flag is the required universal test, with reversed W indices and equality allowed. A strict total order lists each option once; its first permitted member outranks every remaining one, and exhausting it means there is none. Counting the initialized cells, the two additions per reason, all ordered-pair tests without short-circuiting, and the final scan gives the bounds. $\square$

This counts arithmetic operations, not unit-time rational bit arithmetic; numerator and denominator sizes contribute further cost. Input validation, assessment and any deduplication are not included. For empty O there is no selection; a singleton passes vacuously. A completed empty reason list yields zero sums; an incomplete assessment does not satisfy the procedure’s input contract.

[Example 10.6](#)’s dinner is reused without changing its six reasons. For the ordered tests

$$(R/C, C/R, R/T, T/R, C/T, T/C),$$

the margins are $(1, 1, 4, -1, -1, 2)$. The failed-foil sets are respectively $\emptyset, \{T\}, \{R\}$, so only R is permitted. Both cost order C, T, R and priority order R, C, T select it. Changing only $RW(\text{speed}, T, C) : 3 \mapsto 2$ changes only $W(T, C)$, hence only the C/T test, to equality. Permission becomes $\{R, C\}$; the cost selector returns C and the priority selector R . T/C remains unchanged. Counting repeated utterances as new reasons, or summing only justifying weights, would implement a different function. The source comparison is Yu et al., AAMAS 2026 author version, §3.1, Definitions 3.1–3.4, p. 3; the two selection orders are supplied policies. Neither procedure grants execution authority.

Related textbook exercises

- **12.1–12.6, 12.9–12.10:** stable witnesses, conflict/coverage clauses, directed information, the family guard and negative-certificate boundaries.
- **12.7–12.8:** specifying explanation optimization (developed in §13.3) and testing portfolio performance.
- **12.11–12.12:** simultaneous iteration and the four-node transfer, exact blocking, exhaustion and graph deletion.
- **12.13–12.15:** the selected SAT reduction with its boundaries, conditioned SCCs and update completeness, admissible preferred-query witnesses, and explicit output size.
- **12.16–12.17** (Study S12): DP membership only, two-sided blocks, auxiliary projection and empty-family boundaries.
- **12.18(a–c):** the public question, failed retraction and legal reply; the rest of **12.18** compares completed continuations at budgets two and one.
- **12.19:** pair sums, failed foils, selectors, one-coordinate revision and arithmetic versus bit cost.
- **12.L:** external comparison of encoding/implementation, Handbook and computational sources (including the ICCMA alternative); its literature task remains explicitly external.

13 Explaining a specified answer

A correct calculation answers the question encoded by its inputs. An explanation must also say which answer is being explained, which context is retained and which changes are allowed. None of these mathematical conditions establishes interpretive adequacy or human usefulness.

13.1 Targets, witnesses and three statuses

Definition 13.1 (Explanation target). *Write an AF-based target as $\mathcal{T} = (F, \sigma, q, o, \tau)$, specifying a framework, semantics, query, reported output and explanation task. The query includes its type. For an extension-relative status it includes a specified, verified $E \in \sigma(F)$; for credulous or guarded skeptical acceptance it quantifies over $\sigma(F)$ as in Definition 6.12. An explanation of an implementation's output additionally fixes that implementation's operation, input version and output representation. A contrast consists of two typed targets and a declaration of what is held fixed and what varies.*

A graph certificate addresses this specified reconstruction of the content. It does not validate the reconstruction itself. The distinctions between content, occurrence, act, target, issue and option from Chapters 3–5 remain operative.

By Theorem 6.11, a complete extension E determines

$$\text{in}_E = E, \quad \text{out}_E = E^+, \quad \text{undec}_E = A \setminus (E \cup E^+). \quad (13.1)$$

Thus an out-certificate includes an accepted attacker; mere absence from E is insufficient. If x is undecided, it has no in attacker and has an undecided attacker. Indeed, no in attacker follows from $x \notin E^+$. If all attackers were out, each would be attacked by E , so completeness would put x in E , a contradiction. This also rules out an unattacked undecided vertex.

Use precisely F_{reuse} from Example 12.10, with stable extensions

$$E_1 = \{a, c, e\}, \quad E_2 = \{a, d\}, \quad E_3 = \{b, d\}.$$

Here e is in relative to E_1 and out relative to E_2 , witnessed by dRe . The empty grounded extension instead labels every vertex undecided. The fixed graph and semantics do not turn an E_1 -relative explanation into a skeptical one.

A positive stable-credulous certificate is a verified stable E containing the queried argument. A negative guarded-skeptical answer has two logically different causes:

$$\text{Stable}(F) = \emptyset \quad \text{or} \quad \exists E \in \text{Stable}(F) (q \notin E). \quad (13.2)$$

The second has a directly checkable witness; the first requires a no-output argument or a checked unsatisfiability proof for the exact encoding, not a missing search result. On the directed three-cycle $uRvRwRu$, every pair conflicts, a singleton attacks only one of its two outsiders, and the empty set covers none. Hence there is no stable extension. On F_{reuse} , E_2 instead witnesses the second cause for e . Neither case says that the queried argument is out in every complete labelling.

13.2 Finite certificates in a retained context

Fix an order of the vertex identifiers for deterministic choices. The following procedure takes the *original* finite F , a verified admissible E , an $a \in E$, and an initially empty list of ancestor proponent occurrences:

```

ExplainAccepted(F,E,a,path):
  if a labels a proponent on path:
    return BackReference(to that ancestor occurrence)
  create proponent occurrence p labelled a
  nextpath := path with p appended
  for every b with (b,a) in F.R:
    create an opponent child labelled b
    choose the first c in E with (c,b) in F.R
    attach ExplainAccepted(F,E,c,nextpath) below that child
  return the tree rooted at p

```

Each opponent is an occurrence, not a new argument vertex. A back-reference identifies an actual ancestor occurrence with the same proponent label. It is not itself an attack edge. Every ordinary parent-child pair displays an original attack, directed from the child label to the parent label.

Theorem 13.2 (Contextual certificate contract). *The procedure terminates. Every expanded proponent belongs to the same E , exposes all of its attackers in F , and answers each by an actual attack from a member of E . Every reference returns to an identically labelled ancestor. Each branch expands at most $|E|$ proponent occurrences before ending or returning. The set T of its proponent labels is an admissible subset of E containing a .*

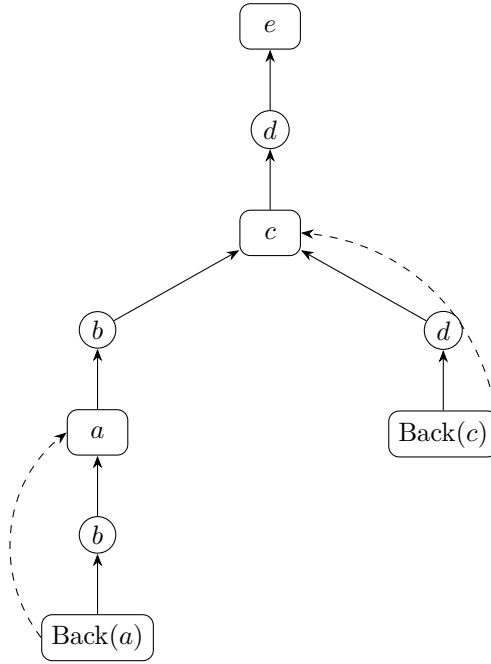
Proof. Admissibility of E guarantees a defender $c \in E$ against each attacker b of any chosen $a \in E$; thus every choice exists. The root is in E , and this choice rule inductively keeps all proponents in E . A repeated label returns at once. Before a return, expanded labels along a branch are distinct members of the finite set E . There are therefore at most $|E|$ expansions, with at most one further terminal proponent occurrence used for a reference. Each expansion has finitely many attackers, so the bounded-depth, finitely branching tree is finite. The loops and membership tests give exactly the stated local attack and reference properties.

For the last claim, $a \in T \subseteq E$, so T is conflict-free. Every label in T has an expanded occurrence: if an occurrence is a reference, its ancestor with the same label was expanded. At that expansion, every original attacker has a displayed defender whose label lies in T , whether the recursive call expands or returns. Thus T defends all its members, proving admissibility. $\square$

The returned tree can be checked by these finite conditions and original-AF adjacency tests. A checker must test *all original attackers*, not just the displayed ones. This certificate proves defensibility of its labels in that context; verifying the supplied E 's claimed complete, grounded, preferred or stable status is a separate obligation. Theorems 6.5–6.6 make each such E admissible, so the construction applies after that verification. Admissibility alone does not prove grounded acceptance, stability, skepticism or minimality. Repeated subtrees can make the output large; the branch-depth bound is not a polynomial total-size bound.

The definition of defence is Dung's Definition 6, p. 326. This finite presentation is the textbook's local certificate, not a wholesale import of an ABA dispute-tree calculus. Chapter 9's grounded game has its own rank-decreasing proof; a cyclic admissible certificate must not replace it.

For e in E_1 , the entire certificate is small enough to draw. Solid arrows below point in the direction of the original attack. Dashed arrows only identify the earlier occurrence.



The label behind $\text{Back}(c)$ is c , and similarly for a . Both attackers b, d of c are present. The selected accepted set was checked before the construction. In the isolated mutual pair $aRbRa$, the certificate for a relative to $\{a\}$ returns to a ; the grounded extension is nevertheless empty.

13.3 Preservation, deletion and contrast

Let $\mathcal{S}(F)$ be a declared family of admissible explanatory substructures, all retaining the queried object. Specify whether a substructure is evaluated independently or merely selects what is displayed while evaluation remains in F . Also specify the preserved query Q and the permitted deletion or change operation. For independent evaluation, call S sufficient when $Q(S) = Q(F)$. An inclusion-minimal sufficient object has no strictly smaller sufficient object in the declared inclusion order; a minimum-size object minimizes a specified size measure. Neither notion is defined until these choices are fixed.

For a finite, explicitly enumerable $\mathcal{S}(F)$, an exact decidable preservation test and a computable size with effective comparison, exhaustive filtering and comparison return all minimum-size sufficient objects, or report none. After any enumeration prefix the retained objects are exactly its sufficient members; comparison retains precisely the least size seen and all ties. These invariants prove the claim at exhaustion. Inclusion-minimal objects instead require checking that no retained sufficient object is strictly below the candidate in the declared, decidable inclusion order. Testing every pair does exactly that. This specifies a finite optimization contract, not a polynomial-time algorithm or a user-preference measure.

Proposition 13.3 (Query preservation is not attacker completeness). *A standalone subgraph can preserve positive stable-credulous acceptance while failing the contextual certificate contract. Even a minimum vertex explanation under unrestricted vertex deletion can lose all original defence obligations.*

Proof. In F_{reuse} , e is stable-credulously accepted. Delete the two reverse edges bRa, dRc . The remaining chain $aRbRcRdRe$ has unique stable set $\{a, c, e\}$: unattacked a is forced, b excluded, then c forced, d excluded and e forced. This preserves the query but omits an original attacker of a and of c . The induced subgraph on $\{c, d, e\}$ can also support e through the stable set $\{c, e\}$, while omitting bRc .

For unrestricted induced vertex deletions retaining e , the singleton $\{e\}$ has no self-attack and its unique stable set is $\{e\}$. It preserves the positive query and is minimum in number of vertices, since retaining e requires at least one. It displays no attacker, though dRe holds in the original AF. More generally the same argument works for any self-loop-free q originally stable-credulously accepted when arbitrary vertex deletions retaining q are allowed. It proves the loss of contextual obligations, not interpretive adequacy of the singleton. $\square$

A necessity claim is equally relative to the allowed change. Delete only cRd from F_{reuse} . Vertex d becomes unattacked and is forced in every stable set, excluding c, e . Exactly one of a, b is needed, and either choice now covers all outsiders. The new stable family is exactly $\{\{a, d\}, \{b, d\}\}$, so this deletion destroys credulous acceptance of e . In that precise deletion test, the edge cRd is necessary. This says nothing about every other possible replacement explanation.

Adding eRa has the same resulting stable family, by the exhaustive argument in §12.4, but a different mechanism: the old E_1 now conflicts. Its old tree may still display some valid attacks, yet its recorded accepted context is invalid. The new exact family proves e is no longer credulous and d is guarded-skeptically accepted. Checking only the two surviving old outputs would not establish the latter claim.

A contrast between e 's status in E_1 and E_2 fixes F_{reuse} and stable semantics and varies the witness. In E_1 , c answers d , permitting e ; in E_2 , d is selected and attacks e . This is a witness-relative contrast, not an assertion that E_1 is the uniquely justified choice. By contrast, deleting cRd fixes the query and semantics but varies the framework. Comparing the two targets without identifying that change would conceal the explanatory operation.

13.4 Requests about representation and public responsibility

The claim that content warrants cRd is different from the query computed using cRd . Let the opaque claim χ express that proposed attack; let ρ name its supplied interpretation and ω its proposed warrant. Use Definition 9.1, with system M and user U , initially only $\text{assert}(M, \chi)$ active, $\text{Com}_M = \{\chi\}$, $\text{Com}_U = \emptyset$, $\text{DOS} = \{(M, \chi, +)\}$, no challenges and no pending reply. Then

$$\text{question}(U, M, \chi), \quad \text{justify}(M, \{\rho, \omega\} \vdash^+ \chi) \quad (13.3)$$

has certificate (CR2, CR3). Substituting into those rules shows that the question leaves the ledger unchanged and makes $(M, \chi, +)$ pending. The reply replaces its obligation by $(M, \{\rho, \omega\}, +)$, adds both grounds to M 's commitments, clears pending, and preserves the assertion. Proposition 9.2 makes this result unique. A challenge instead records U 's opposition; a retraction cannot replace the required reply.

The move target is the claim χ , not an untyped edge token. This trace records who supplied grounds and owes their defence; it establishes neither the truth of ρ, ω nor adequacy of the extracted attack. Another extension of the unchanged AF cannot answer that reconstruction question. The restricted public rules retain Chapter 9's distinction from full Fatio; no new source correspondence is assumed here.

13.5 Permission, selection and an authorized revision

Use the complete six-row dinner input of Example 10.6, version v_1 , in fixed context d . The explanatory arithmetic is:

option	tests against its two foils	permission
R	$4 \geq 3$ (R/C) $7 \geq 3$ (R/T)	yes
C	$3 \geq 2$ (C/R) $2 < 3$ (C/T)	no
T	$4 < 5$ (T/R) $3 \geq 1$ (T/C)	no

(13.4)

All foils remain in the all-foil test, including nonpermitted T . The common company ground has different comparison triples for R/C and R/T ; these tests do not derive its stipulated weights from the ground's content.

Change only $RW(r)$ for $r = (\text{speed}, T, C)$ from 3 to 2. This changes the opposing $W(T, C)$ in C/T and no other test; $2 < 3$ becomes $2 \geq 2$. Thus the all-foil set changes from $\{R\}$ to $\{R, C\}$. The cost order C, T, R changes selection from R to C ; the priority order R, C, T keeps R . These conclusions follow from Proposition 12.12's exact contract. The two permitted options pass their mutual tests in both versions. A selected option need not be uniquely permissible or defeat its alternative. Policy alone cannot select the impermitted C under that contract in v_1 .

Now let coordinator K assert opaque claim η : in this context the requiring coordinate of r should be 2 under the agreed deadline policy. Let ρ be the supplied ground that both C, T can meet the end time, and ω the supplied warrant assigning requiring strength 2 in that circumstance. Starting with only K 's assertion, ledger $\{(K, \eta, +)\}$, commitment $\{\eta\}$, no challenges or pending reply, the question by U followed by K 's justification from $\{\rho, \omega\}$ again applies CR2 then CR3. The final ledger contains $(K, \{\rho, \omega\}, +)$; K 's commitment includes all three claims and its assertion persists. The numerical model is still v_1 . Legal explanation of a proposal does not apply it.

Definition 13.4 (Versioned one-cell revision). *A current model $I = (d, v, G, O, R_{cr}, w)$ is a complete input to Definition 10.5, with exact rational weights. A separate supplied relation $\text{Auth}(i, d, r, j)$ authorizes issuer i for coordinate $j \in \{JW, RW\}$ of row r . A proposal records context, base version, row, coordinate, expected old value, proposed value, issuer, and identified ground and warrant records.*

*Validate, without mutation, that context and base version match; the row and coordinate exist; the expected old value matches; the issuer is authorized for that cell; the proposed pair lies in $\mathbb{Q}_{\geq 0}^2 \setminus \{(0, 0)\}$; and the identified ground and warrant records are supplied. On any failure, return its recorded failed check and the unchanged model and outputs. On success, copy to a fresh version identifier, change exactly the designated coordinate, retain the proposal's provenance, and run *Permitted* and *Select* with the named order. Version identifiers are never reused.*

This is the textbook's stipulated interface, not a rule derived from Fatio, Libra or the balancing source. Identifying supplied records does not verify their truth or normative adequacy.

Proposition 13.5 (Revision preservation and replay). *For a well-formed current input and supplied authority relation, the interface either leaves the entire model and its outputs unchanged, or produces a complete valid weight input differing only in its designated coordinate, version and provenance, with the exact recalculated permission and selection. Replaying a previously accepted proposal against its immediate successor fails the base-version check.*

Proof. Validation precedes mutation, so every failed branch returns the original model and outputs. If all checks pass, the addressed cell exists and its replacement pair is in the weight domain. Every other row and coordinate is copied, and G, O, R_{cr}, d are preserved, so the new input is complete and well-formed. The successful branch's sole numerical assignment gives the one-coordinate property. Proposition 12.12 then proves correctness of both recalculated outputs for the specified selector; copying an old selected option is not part of the procedure. The successor has a fresh identifier different from the proposal's base version. Hence replay fails that check, regardless of any other matching fields. $\square$

Supply exactly $\text{Auth}(K, d, r, RW)$, but not $\text{Auth}(U, d, r, RW)$, for the dinner cell. The proposal $(d, v_1, r, RW, 3, 2, K, \rho, \omega)$ passes and creates v_2 . Three otherwise identical proposals, evaluated independently against v_1 , fail respectively when the issuer is U , the base is $v_0 \neq v_1$,

or the warrant record is absent. They leave $\{R\}$ and its selection unchanged. Replay of the accepted proposal at v_2 also has old value $3 \neq 2$, as well as a stale version. Public legality cannot repair these failures. Successful authorization of this numerical edit grants no authority to execute an option and does not show that 2 is normatively best.

13.6 Explaining through argumentation: what is guaranteed?

For explanation *of* argumentation the evaluated argument model is itself the target. For explanation *through* argumentation, an extracted or surrogate argument model purports to explain another system. Its relation to that system becomes an additional proof or measurement obligation.

Let $f, g : X \rightarrow Y$ be respectively the original output function and an argumentation-based output function. For a finite, nonempty test set $X' \subseteq X$, define

$$\text{Fidelity}_{X'}(f, g) = \frac{|\{x \in X' : f(x) = g(x)\}|}{|X'|} = 1 - \frac{|\{x \in X' : f(x) \neq g(x)\}|}{|X'|}. \quad (13.5)$$

The equality follows because the agreement and disagreement sets partition X' . For the four synthetic output pairs

(approve, approve), (approve, refuse), (refuse, refuse), (refuse, refuse),

the score is $3/4$. These are stipulated outputs, not observations of participants.

Proposition 13.6 (Finite agreement boundary). *If $X \setminus X' \neq \emptyset$ and Y has at least two elements, complete agreement on X' does not imply agreement on X . Agreement with f , even everywhere, does not by itself certify correctness against a separate target function.*

Proof. Choose $x_* \notin X'$ and distinct $y_0, y_1 \in Y$. Let f be constantly y_0 , and g be the same except $g(x_*) = y_1$. Their finite fidelity is one but they disagree at x_* . For the second statement take $f = g$ constantly y_0 , and a target function h constantly y_1 . Perfect agreement of f, g is compatible with every output being wrong relative to h . $\square$

There is likewise no human response variable in (13.5). A theorem about f, g cannot supply an unobserved comprehension or usefulness score. Such claims require an identified population, presentation, task, response measure and comparison conditions.

Source construction and proof status. Study S13 discusses AXPLR for a *pattern-based logistic-regression binary text classifier*, not an arbitrary black box. Lertvittayakumjorn–Toni, *Argument & Computation* 14(2), 2023, pp. 163–234, DOI 10.3233/AAC-220004, §4.2, Theorem 2, is the source locator for the reported correspondence. Its extracted quantitative bipolar structures and LR semantics are defined by its Definitions 2–3. The theorem’s probability identity relates the strength of its designated default argument to the probability of that argument’s assigned class through the logistic function $s \mapsto (1 + e^{-s})^{-1}$. That statement belongs to the source’s construction and domain; neither (13.5) nor our certificate theorem proves it for an arbitrary surrogate. We retain it as a source-attributed result with the source proof in Appendix A.3; the full extraction and proof are not developed or used as a premise here.

A bounded source check leaves a precise question open. In the available corrected-proof version, Theorem 2’s class-selection clause and Appendix A.3 do not consistently handle zero strength: the proof’s nonpositive branch concludes a probability strictly below $1/2$, although the logistic value at zero is $1/2$. Its positive- and negative-bias one-feature boundary cases also require reconciliation with a single class-based tie rule. The publisher HTML omits the decisive

inline mathematical glyphs. The official Imperial record identifies the published-version PDF; neither the earlier check nor renewed bounded retrieval on 27 September 2026 obtained it. The latter encountered a CAPTCHA response at the identified repository download and failed publisher PDF delivery. Exact final-version comparison therefore remains open. Consequently we do not import the full class-selection clause or silently repair it. Editorial evidence records the exact version, locators and witnesses for author disposition; this is not a claim that a defect has been verified in the final publisher PDF.

Evidence of usefulness. The source’s §§9.3–9.4 concern relations between explanation assessments and human judgments. Its §§10.1–10.4 and Table 10 concern tutorial and assistance conditions for deceptive-review detection. As reported in the approved Study S13, assistance improved scores against within-method baselines, without a significant between-method performance difference. Those empirical results have their own participants, task and measures. They neither follow from the correspondence theorem nor transfer to the four synthetic pairs. No new human-use result is asserted for this companion’s explanations.

Related textbook exercises

- **13.1–13.2, 13.9–13.10, 13.12:** original-context certificate construction, out versus undecided, typed outputs, cyclic references and the procedure’s precise guarantee.
- **13.3–13.5, 13.11:** deletion and recomputation, minimum standalone explanations, specified contrasts, and the changed accepted context after adding *eRa*.
- **13.6:** the question/reply and challenge alternatives for an attack-reconstruction claim.
- **13.7–13.8, 13.13** (the last uses Study S13): finite fidelity, explanation of versus through argumentation, and mathematical, plausibility and task-performance evidence.
- **13.14:** no stable output versus a stable witness omitting the query under the guarded convention.
- **13.15:** all-foil permission, the changed requiring coordinate, and policy-relative selection contrasts.
- **13.16:** public grounds, the four independent revision proposals, replay and the retained authority boundary.
- **13.L:** external source comparison of explanation targets, guarantees and evaluation; it is not an additional canonical solution collection.

14 Purpose-specific services and conditional release

An accepted argument, a checked public plan and a selected option have different output types. A system connects a computation to its intended use by declaring what it returns, what its checks establish and which caller enforces release. We first make that connection exact for a finite inferential service, then for public response and practical recommendation.

14.1 A typed inferential service

Write an inferential design as

$$\mathcal{S} = (I, \mu, \rho, \sigma, \delta, \epsilon, \nu, L).$$

For $x \in I$, extraction μ returns a finite identifier set $A_x \subseteq U$. Relation extraction ρ returns $R_x^-, R_x^+ \subseteq A_x \times A_x$; an absent endpoint fails the interface. Evaluation σ returns its declared type, here an accepted set $E \subseteq A_x$. The decision function δ consumes that type and the declared conclusion and con-argument maps; ϵ constructs a contextual explanation, ν supplies named checks, and L retains their record. This tuple describes one service. It imposes no attack-graph input on the other services below.

Example 14.1 (A supplied policy representation). The fictional late-filing adviser has argument summaries

$$\begin{aligned} a_1 : \text{emergency} &\Rightarrow \text{allow}, & a_2 : \text{deadlineMissed} &\Rightarrow \neg \text{allow}, \\ a_3 : \text{medicalEvidence} &\Rightarrow \text{emergency}, & a_4 : \text{publicInterest} &\Rightarrow \text{allow}. \end{aligned}$$

These are supplied summaries, with

$$\begin{aligned} A &= \{a_1, a_2, a_3, a_4\}, \\ R^- &= \{(a_3, a_2), (a_2, a_1), (a_2, a_4)\}, \\ R^+ &= \{(a_3, a_1), (a_4, a_1)\}. \end{aligned}$$

In particular $a_3 R^- a_2$ encodes the stipulated medical objection to applying the deadline reason; it is not derived from the summaries by an unspecified structured rule. Evaluate only (A, R^-) by grounded semantics. Then

$$\emptyset, \quad \{a_3\}, \quad \{a_1, a_3, a_4\}, \quad \{a_1, a_3, a_4\} \quad (14.1)$$

is the complete empty-start iteration: a_3 is the only unattacked vertex, it defeats the sole attacker of a_1, a_4 , and no set defends a_2 against unattacked a_3 . [Theorem 6.6](#) therefore gives $E = \{a_1, a_3, a_4\}$. It is also stable: each attack has one endpoint outside E , and a_3 attacks its only outsider.

With con set $K = \{a_2\}$, declare

$$\delta(E) = \begin{cases} \text{allow} & (\exists a \in E \text{ Conc}(a) = \text{allow}) \wedge K \subseteq E^+, \\ \text{withhold} & \text{otherwise.} \end{cases} \quad (14.2)$$

Here E^+ uses R^- . Thus the candidate is *allow*; failure of (14.2) would not itself recommend denial. Both support links are *explanatory*: removing R^+ changes neither the evaluator's input nor (14.2)'s inputs, hence neither E nor the candidate. It can change the displayed routes into a_1 .

The policy is synthetic, not the law of a jurisdiction. Dung's Definitions 13, 16 and 20 (pp. 328–329) are the semantic source; the service and release policy are local constructions. An explanatory support edge is neither an additional accepted argument nor automatically a semantic dependency.

14.2 Route certificates and the release caller

Fix immutable input, graph and decision versions u, f, d . They name their full retained payloads, not merely equal strings. Let the required premise pairs for Example 14.1 be

$$Q_{\text{prem}} = \left\{ (a_1, \text{emergency}), (a_2, \text{deadlineMissed}), \right. \\ \left. (a_3, \text{medicalEvidence}), (a_4, \text{publicInterest}) \right\}.$$

A supplied provenance entry is (u, a, p, ℓ, b) , with locator ℓ and support-check value $b \in \{0, 1\}$. Define $c_{\text{prem}} = 1$ exactly when every required pair has a matching entry at u with value 1. A missing or solely unsupported entry fails. This compares supplied evidence-check results; it neither authenticates their issuer nor decides natural-language truth.

Definition 14.2 (Bound grounded route). *A route certificate is*

$$c = (u, f, \text{grounded}, d, S_0, \dots, S_k, E, o), \quad k \geq 1.$$

Here $S_0, \dots, S_k, E \subseteq A$, and o has the decision rule's output type. Its route check binds it to the actual input, full attack graph and decision rule, and tests

$$S_0 = \emptyset, \quad S_{i+1} = \text{Def}_{(A, R^-)}(S_i) \quad (0 \leq i < k), \quad S_k = S_{k-1} = E, \quad o = \delta(E). \quad (14.3)$$

Its separate certificate check tests presence, well-formedness, version binding, and that E, o are exactly the returned accepted set and candidate. Call these values c_{route} and c_{cert} . The computation record retains

$$B = (u, f, d, F, E, o, e, c, \mathbf{c}, \text{audit}), \quad \mathbf{c} = (c_{\text{prem}}, c_{\text{route}}, c_{\text{cert}}),$$

where $F = (A, R^-, R^+)$ and e is the explanation. The caller returns **Certified**(o, E, e, c) precisely when all three checks are true on this record, otherwise **NoCertifiedRecommendation**, and records that choice.

The caller uses checks actually computed for the same retained record; writing three true bits in an unrelated log does not satisfy this contract. The computation can return a candidate without releasing it.

Theorem 14.3 (Route and release guarantees). *A successful route check proves that the certificate's E is the grounded extension of the bound attack graph and that $o = \delta(E)$. Every certified output of the declared caller has that property and passes the supplied premise and binding checks. The finite grounded procedure produces an iteration certificate using at most $|A| + 1$ characteristic-map evaluations.*

Proof. From $S_0 = \emptyset$ and the step equalities, induction gives $S_i = \text{Def}^i(\emptyset)$. The terminal equality is a fixed point on that exact sequence. Theorem 6.6 proves that this empty-start sequence consists of admissible increasing sets and reaches the least complete extension; a fixed stage is therefore the grounded extension. The last test in (14.3) gives its decision. Theorem 12.2 gives the stated evaluation bound, including the equality test; for $A = \emptyset$ the certificate is $(\emptyset, \emptyset)$. The caller's only certified branch requires all three actual check values. The other branch emits no certified recommendation. Hence every certified record satisfies the claimed conjunction. $\square$

No claim about the truth of every premise, normative adequacy of δ , or a component that bypasses the caller follows. A hash can identify retained bytes but cannot establish those claims. A certificate may be present and still name the wrong graph; existence alone is weaker than successful verification.

For the supplied u_1, f_1, d_1 , give one entry at each of e_1, e_2, e_3, e_4 , all supported, and bind (14.1). Then $\mathbf{c} = (1, 1, 1)$. Keeping the graph and certificate but changing only e_3 's value to zero, or removing its entry, gives $(0, 1, 1)$. The same candidate *allow* is withheld. Removing the certificate gives $(1, 0, 0)$; an audit line mentioning E does not replace it.

Counterexample 14.4 (A fixed point without the selected route). In the mutual pair $uRvRu$, $\{u\}$ is conflict-free and defends u , and $\text{Def}(\{u\}) = \{u\}$. Yet the empty-start sequence is $\emptyset, \emptyset$. Thus admissibility, even with a fixed-point test, cannot substitute for the selected grounded route.

A timeout before a complete checked route yields a marked partial diagnostic and withheld release. Changing the graph or semantics on timeout defines a different task.

A service promising sampled classifier faithfulness has another check, with paired outputs and a declared threshold. Three agreements in five cases give $3/5 < 0.8$; this fails that sampled claim. It does not refute a separately verified grounded route. If the service promises that extra claim, its release conjunction must include the extra check; it cannot silently attach the claim under the original three-check policy. [Proposition 13.6](#) supplies the limits of finite agreement. Likewise the local expression $B_i\text{accepted}(a_3)$ states an agency's belief; no truth-entailing modal axiom is assumed. Computed status, evidence, agency belief and release remain distinct.

14.3 Public-response and practical recommendation services

The branch argument extends to a finite list of declared checks: if a trusted caller emits a named certified type only under their conjunction for the same bound record, every such output passes each check. Any further semantic conclusion requires the relevant check's correctness theorem. We now give those conclusions for two different record types.

Definition 14.5 (A public plan record). *Fix the typed initial state s , offered finite move lists $\mathcal{H}$, response policy π , effective objective h , budget b , and decidable stopping condition T of [Proposition 12.11](#). A record contains these inputs and versions, a chosen first-stopping prefix η , its score, the checked simulated state trace and rule list, and its final pending field. Its checker reruns the finite bounded procedure, tests that η belongs to its complete maximizer set, and verifies that all returned fields match. A caller releases a **CertifiedPlan** only on successful checking. If there is no eligible prefix, it returns that bounded failure.*

This local service contract preserves all ties at computation; selecting one returned maximizer is not a claim of unique optimality.

Proposition 14.6 (A plan and an observed history). *A released plan is legal and completed within the stated budget under its supplied simulated response policy, and has maximal score among the offered eligible prefixes. It does not imply that its replies occurred. For an actual supplied history h_{obs} , the trace checker separately certifies only that history and its resulting state.*

Proof. The record checker recomputes precisely [Proposition 12.11](#)'s finite eligible family and maximum set. Membership proves legality, its first stopping boundary, budget and score claim. Its trace-field checks give the corresponding state sequence. None of these tests reads evidence that a person uttered the policy-supplied moves. Applying the first part of [Proposition 12.11](#) to h_{obs} instead yields exactly its own successful prefix or first failure and pending state. $\square$

The restaurant instance of [§12.5](#) makes the separation concrete. Budget two selects inquiry with simulated rules (CR4, CR5), score $(1, 1, -2)$, retained assertion and exposed countergrounds. Budget one completes only withdrawal, CR6, score $(0, 0, -1)$. If the actual history contains only $\text{question}(A, B, \chi)$, CR4 leaves $(B, \chi, -)$ pending. Both a future reply and

no future reply are compatible with that observed prefix. The plan therefore cannot establish that countergrounds have already been obtained. This service needs no Dung extension.

A practical service instead fixes the complete versioned comparative instance of [Definition 10.5](#), with exact rational weights as required in [§12.6](#), and a named strict total selector $\prec$. Its result records

$$(d, v, O, R_{cr}, w, \prec, P_O, o, \mathcal{T}, e),$$

with the finite ground set retained in the input envelope, where $\mathcal{T}$ is the explanation target and e its evidence. The selection o may be `NoSelection`. The checker binds the current input and policy, recomputes P_O and the selection by [Proposition 12.12](#), and compares the fields. Its release contract is a recommendation, not an execution command.

Proposition 14.7 (Practical service preservation). *Every recommendation released by this checker and caller has the exact all-foil set and its named selector’s best permitted option, or no selection exactly when that set is empty. After a successful one-cell revision, checking must use the new instance; after a failed revision the old instance and its outputs remain unchanged.*

Proof. Binding fixes the input to both computations. [Proposition 12.12](#) proves the recomputed permission and selector values, so equality tests establish the returned fields. [Proposition 13.5](#) proves that a successful revision preserves a complete valid input apart from the designated coordinate and metadata, and that failure changes neither model nor outputs. Applying the checker to the appropriate resulting instance proves each case. $\square$

At dinner version v_2 , permission is $\{R, C\}$. The cost record must select C , whereas the priority record selects R . The record $(v_2, \{R, C\}, \text{cost}, R)$ fails even though R is permitted. At v_1 , U ’s unauthorized version of K ’s weight proposal fails; replay at v_2 fails the version and old-value tests. Chapter 13’s CR2/CR3 exchange attributes grounds without changing either numerical version or authority. A further, explicitly supplied act authorization would be needed to commit resources or execute the choice. It is not supplied in this dinner service.

These examples distinguish a hard exclusion built into a defined transition system, a norm that conduct may violate, and a monitor policy mapping check results to a response. The release caller enforces its conjunction. Logging the candidate and its failures is a different operation.

14.4 When support changes the evaluated object

Study S14.1 changes the policy graph’s explanatory support into a numerical input. This is a declared pedagogical rule, not a canonical bipolar semantics or a probability model. Let A be finite, $R^- \cup R^+$ acyclic, $b : A \rightarrow [0, 1]$, and $\lambda \in [0, 1]$. Set

$$s(x) = \left(1 - (1 - b(x)) \prod_{z \in R^+ x} (1 - \lambda s(z)) \right) \prod_{y \in R^- x} (1 - s(y)), \quad (14.4)$$

with empty products one.

Theorem 14.8 (Acyclic scores and local comparisons). *Equation (14.4) has a unique evaluation in $[0, 1]^A$, independent of topological order. Holding the other target coordinates, base and parameter fixed, increasing a support coordinate cannot decrease the target value; increasing an attack coordinate cannot increase it.*

Proof. [Lemma 9.9](#) gives a topological order and uniqueness of evaluation for a declared finite acyclic network. At each vertex all predecessor scores are already defined. If they belong to $[0, 1]$, both products do also; the positive bracket is between $b(x)$ and one. The resulting score

lies in $[0, 1]$. Induction establishes the bounds, and the same lemma gives uniqueness and independence of the chosen order.

For an isolated support coordinate t , write the other support product as P and the attack product as Q . The target is $[1 - (1 - b)P(1 - \lambda t)]Q$. Increasing t by $\Delta \geq 0$ changes it by $(1 - b)P\lambda Q\Delta \geq 0$. For an isolated attack coordinate, writing the fixed positive bracket as B and the other attack product as Q' , the target is $BQ'(1 - t)$, with change $-BQ'\Delta \leq 0$. These computations keep the roles separate. $\square$

If one predecessor occupies both roles, its change alters two coordinates: for $b(x) = 0, \lambda = 1$ and that sole predecessor of score t , the target is $t(1 - t)$. It rises from zero to $1/4$ and falls back to zero. A propagated change may also arrive along paths with competing roles. Acyclicity is substantive: with two mutual attackers, no support and bases one, the cyclic equations are $s(a) = 1 - s(b)$, $s(b) = 1 - s(a)$. Every pair $(t, 1 - t)$, $0 \leq t \leq 1$, solves them, so the declared acyclic uniqueness proof cannot be extended.

On Example 14.1, bases 0.8 and $\lambda = 0.5$ give, in order a_3, a_2, a_4, a_1 ,

$$\begin{aligned} s(a_3) &= 0.8, & s(a_2) &= 0.8(0.2) = 0.16, \\ s(a_4) &= 0.8(0.84) = 0.672, & s(a_1) &= (1 - 0.2 \cdot 0.6 \cdot 0.664)0.84 = 0.7730688. \end{aligned}$$

Removing both support links leaves the first three values unchanged and gives $s(a_1) = 0.672$. An isolated vertex retains $b(x)$; an attacker of score one forces a target's score to zero. The output is a strength function, not an accepted set. Using it for recommendation requires a new declared map from $[0, 1]^A$ to decisions; (14.2)'s membership test cannot silently supply that map.

14.5 Admission selects formulas from a report

Use Chapter 0's classical consequence. A report A is a set of formulas; trusted T and distrusted D are also formula sets, not issuers or weights. The two alternative modes from *Revealed Epistemic Trust*, Definition 2 (pp. 463–464), Definitions 15,17 and Example 5 (p. 467), are

$$\begin{aligned} \alpha_T(A) &= \text{Cn}(T \cap \text{Cn}(A)) \cap A, \\ B \perp D &= \{X \subseteq B : \text{Cn}(X) \cap D = \emptyset, X \text{ inclusion-maximal with this property}\}, \\ \beta_D(A) &= \left(\bigcap (\text{Cn}(A) \perp D) \right) \cap A. \end{aligned} \quad (14.5)$$

The last expression is used here only when the remainder family is nonempty. No new empty-family acceptance convention is imposed. Remainders are taken over the *closed theory* $\text{Cn}(A)$, not only over the displayed report. The final intersection returns only formulas actually reported.

The finite construction below proves the remainder calculation directly. It is not the source's general representation theorem. Fix a finite propositional vocabulary with valuation set V . For $M \subseteq V$, let $\text{Th}(M) = \{\varphi : \forall v \in M, v \models \varphi\}$.

Lemma 14.9 (Finite model construction of remainders). *Let $B = \text{Cn}(A)$, $M = \text{Mod}(B)$, and let D be finite. The remainders $B \perp D$ are exactly the theories $\text{Th}(N)$ for inclusion-minimal $N \supseteq M$ satisfying*

$$\forall d \in D \quad N \cap \{v \in V : v \not\models d\} \neq \emptyset. \quad (14.6)$$

Such N exist iff D contains no tautology. When $D = \emptyset$, the sole remainder is B .

Proof. Every maximal avoiding $X \subseteq B$ is closed: $\text{Cn}(X) \subseteq B$, and replacing X by its closure changes no consequences or avoidance property. Maximality therefore requires $X = \text{Cn}(X)$. Every $N \subseteq V$ is definable by the disjunction of the complete valuation descriptions of its

members, with the empty disjunction false. Consequently $\text{Mod}(\text{Th}(N)) = N$. Classical semantic closure also gives $\text{Th}(\text{Mod}(X)) = \text{Cn}(X)$. These two maps reverse inclusion and are inverse between closed theories and valuation sets.

Thus closed $X \subseteq B$ correspond exactly to $N \supseteq M$. Avoiding each d is precisely having a falsifying model in N , which is (14.6). Strictly enlarging the closed theory strictly shrinks N ; this proves the claimed correspondence between maximal avoiding theories and minimal model sets. If some d is a tautology, (14.6) is impossible. Otherwise $N = V$ satisfies it; finiteness of V supplies a minimal satisfying superset of M . For empty D , the unique minimal such superset is M , giving B . $\square$

For exactly two independent atoms p, q , take $T = D = \{p \wedge q\}$ as two alternative interpretations. For the paired report, $M = \{11\}$; the minimal enlargements avoiding the conjunction add exactly one of 10, 01, 00. By Lemma 14.9 the complete family is

$$\text{Cn}(p, q) \perp \{p \wedge q\} = \{\text{Cn}(p), \text{Cn}(q), \text{Cn}(p \leftrightarrow q)\}. \quad (14.7)$$

The third remainder cannot be found by selecting subsets of the two written atoms. Neither atom lies in all three theories. The whole theory $\text{Cn}(p)$ already avoids the conjunction, as valuation 10 witnesses, so its only remainder is itself; similarly for $\text{Cn}(q)$. These facts give

A	$\alpha_T(A)$	$\beta_D(A)$
$\emptyset$	$\emptyset$	$\emptyset$
$\{p\}$	$\emptyset$	$\{p\}$
$\{q\}$	$\emptyset$	$\{q\}$
$\{p, q\}$	$\{p, q\}$	$\emptyset$
$\{p \wedge q\}$	$\{p \wedge q\}$	$\emptyset$

(14.8)

For the white-list column the singleton atoms do not entail the trusted conjunction, so trusted consequences contribute only tautologies. The paired and conjunctive reports do entail it, and extraction returns their respective displayed formulas. The last two reports have the same closure and the same remainders, but their white-list output sets differ. Merely deleting a written conjunction while retaining both atoms would rederive it.

Even these exact observations need not determine a hidden trust set. For instance $T' = \{p \wedge q, \neg p \wedge \neg q\}$ agrees with T on the first four reports in (14.8), since none entails the extra formula. On $\{\neg p, \neg q\}$, T' admits both reported negative literals while T admits neither. This is a finite observation counterexample about these functions, not a psychological inference about an actual person. Authentication, truth and temporal belief update are different tasks from (14.5).

14.6 What system evidence establishes

For CONSULT/EQRbot, the supplied Study S9.2 distinguishes the described formal account from its implemented, limited follow-up locutions (Handbook Volume 3, dialogue applications, §4.2, pp. 251–252). An encoding error calls for comparing the supplied measurement, unit and meaning with the represented premise; a missing follow-up calls for checking the implemented move set and transition; a claim of improved understanding requires human observations with a task, measure and comparison. A correct formal service supplies none of those observations by itself. We import no unprinted application theorem here.

Requirements, interchange, epistemic planning, causal, modal and institutional routes in the textbook's further reading address additional objects. Our typed service records are not claimed to implement AIF, a complete modal logic or a universal agent architecture. The mathematical contribution is the exact connection between each purpose, its checked result and its limited release guarantee.

Related textbook exercises

- **14.1(a–c), 14.2–14.8, 14.10:** decision and audit types, the policy graph, explanatory support, certificate necessity, finite fidelity, belief and failure responses.
- **14.9, 14.11(a–e)** (Study S14.1): changed score type, exact values, acyclic uniqueness and separate-coordinate versus joint-role comparisons.
- **14.12(a–e):** complete returned record, premise failure, admissibility versus groundedness, the empty-start witness and truthful timeout.
- **14.13(a–d):** bounded simulated plans versus actual public history; authorized numerical revision, selector binding and absent execution authority.
- **14.14** (Study S14.2): closed-theory remainders, presentation-sensitive extraction and limits of trust inference.
- **14.15(a–c):** encoding, implementation and human evidence are separate verification targets.
- **14.L:** the explicitly external comparison of requirements, application and cross-field or institutional sources.

15 Revision, current authority and bounded action

A new observation can change an accepted conclusion. A new receipt of the *same* observation can leave every semantic output fixed while invalidating an old authorization. This chapter develops both phenomena in the textbook’s finite research-assistant case. Its records distinguish a reason for an action, permission to perform it, a dispatch and an outcome.

15.1 A finite revision family and two routes

Let p_0 identify the only candidate paper. Write $p = \text{tracked}(p_0)$, $q = \text{retracted}(p_0)$; fixed metadata m_0 also supplies $\text{recent} = \text{relevant} = 1$. The initial record s_0 and notice s_1 are synthetic:

id	document version	locator	retracted
s_0	status-0	fixture:status-0#p0	0
s_1	notice-1	fixture:notice-1#status	1

(15.1)

Both name issuer **publisher-service** and paper p_0 ; s_1 ’s extracted literal is q . Receipt identifier and document version are different fields.

Retain the textbook’s assumption names. **A1** gives q its actual-status meaning and supplies record correctness at the respective times. **A2** supplies reliable issuer identity, paper resolution and literal extraction, with no conflicting hard observations. **A3** restricts memory and argument construction to the following finite family. These are substantive inputs, not consequences of field equality or the report-selection functions of §14.5.

For $i, j \in \{0, 1\}$, define

$$B_{ij} = (H, \mathcal{R}_s^i, D, \text{Obs}^j), \quad H = \{p\}, \quad D = \{d : \Rightarrow \neg q\},$$

$$\mathcal{R}_s^i = \begin{cases} \emptyset & i = 0, \\ \{p \rightarrow q\} & i = 1, \end{cases} \quad \text{Obs}^j = \begin{cases} \emptyset & j = 0, \\ \{q\} & j = 1. \end{cases} \quad (15.2)$$

The complete declared arguments are

$$R_p = \langle p \rangle, \quad R_d = [\Rightarrow \neg q], \quad R_s = [R_p \rightarrow q] \ (i = 1), \quad R_q = \langle q \rangle_{\text{obs}} \ (j = 1).$$

Premises and observations are axioms. Only an attack on R_d ’s defeasible top succeeds here:

$$R_{\text{rev}}^{ij} = \{(R_s, R_d) : i = 1\} \cup \{(R_q, R_d) : j = 1\}. \quad (15.3)$$

Attacks and defeats coincide in this restricted fragment. The default cannot rebut a strict top or undermine an axiom. These choices instantiate Chapter 5’s firm-premise and rebut distinctions (ASPIC+, Definitions 5, 8–9, pp. 366–368); they are not an enumeration of all arguments in an arbitrary structured theory. In particular $p \rightarrow q$ is a rule, not another conclusion node.

Let $\Gamma(B) = \{\text{Conc}(R) : R \in \text{Ground}(F_{\text{rev}}(B))\}$. The direct route uses the least closure $C_s(B)$ of $H \cup \text{Obs}$ under just the declared strict rules, then returns

$$D_{\text{dir}}(B) = \begin{cases} C_s(B) & q \in C_s(B), \\ C_s(B) \cup \{\neg q\} & q \notin C_s(B). \end{cases} \quad (15.4)$$

This is rule closure over the displayed literals, not the entire classical consequence theory.

Theorem 15.1 (Exact agreement on the declared family). *For all four B_{ij} , $\Gamma(B_{ij}) = D_{\text{dir}}(B_{ij})$. Both equal $\{p, \neg q\}$ if $i = j = 0$, and $\{p, q\}$ otherwise. No transition in this family deletes the default rule.*

Proof. If $i = j = 0$, R_p, R_d are both unattacked and the whole two-element set is the first empty-start defence iterate and a fixed point. Its conclusion projection is $\{p, \neg q\}$. The strict closure is $\{p\}$, so (15.4) adds $\neg q$.

Otherwise let Q_{ij} contain R_s when $i = 1$ and R_q when $j = 1$. This nonempty set and R_p are unattacked; R_d is attacked by every member of Q_{ij} . The first iterate is $\{R_p\} \cup Q_{ij}$. No set attacks a member of Q_{ij} , so R_d is not defended and that iterate is already fixed. [Theorem 6.6](#) makes it grounded. Its conclusions are $\{p, q\}$. Exactly in these same three cases either the observation or the strict rule supplies q to $C_s(B)$, so (15.4) returns $\{p, q\}$. The definition of all four memories retains $D = \{d\}$; acceptance of R_d 's conclusion, not rule storage, changes. $\square$

The four grounded sets are respectively $\{R_p, R_d\}$, $\{R_p, R_q\}$, $\{R_p, R_s\}$, $\{R_p, R_s, R_q\}$, in row order $(0, 0)$, $(0, 1)$, $(1, 0)$, $(1, 1)$. They project to fewer distinct outputs: receiving q in the strict-rule variant adds an accepted argument without changing the conclusion set. The rule $p \rightarrow q$ is an abstract comparison device, not a claim that tracking causes retraction.

The principal temporal case uses $B_0 = B_{00}$ and $B_1 = B_{01}$. Admitting the checked s_1 performs only $Obs := Obs \cup \{q\}$. The negative s_0 is *not* inserted as a hard $\neg q$: its job is to check the exported initial status. Omitting a notice is not a negative-status observation.

The supplied *Reasoning Alignment for Agentic AI* manuscript, §2, pp. 3–5, distinguishes exact, partial and approximate relations between typed routes. The theorem above is our fully proved finite equality, with explicit common input and output types. It is neither that manuscript's separate §3.4 revision result nor a consequence of its agentic-composition outlook (§4.3, pp. 17–18). A generated language/action trajectory supplies yet another object, not an equality proof.

15.2 A partial adapter with a retained receipt

A4 supplies trusted metadata, the worked safety result *true*, authorized reviewer **reviewer-1**, a checked visible warning and an append-only audit log. **A5** supplies a trusted caller's current admitted record pointer and an immutable new context version for every new receipt. The version binds the Boolean payload and its frozen provenance, even when the payload does not change. Let $v(s_0) = \mathbf{agentic-c0}$ and $v(s_1) = \mathbf{agentic-c1}$.

Definition 15.2 (Status adapter). *On $B, s, m_0, currentRecordId$, Adapt first requires*

$$s.id = currentRecordId, \quad s.issuer = \mathbf{publisher-service}, \quad s.paper = m_0.paper = p_0,$$

and that B belongs to the declared finite family. For a received positive notice it also requires $s.literal = retracted(p_0)$ and $q \in B.Obs$. It recomputes $F_{rev}, E_{rev}, \Gamma(B)$ and tests that exactly one of $q, \neg q$ belongs to $\Gamma(B)$, with

$$[q \in \Gamma(B)] = s.retracted. \tag{15.5}$$

On success it returns

$$\begin{aligned} C &= (v(s), p_0, m_0.recent, m_0.relevant, [q \in \Gamma(B)]), \\ P &= \text{Receipt}(\text{adapter version}, B, F_{rev}, E_{rev}, \Gamma(B), m_0, s, C). \end{aligned}$$

On any failed requirement it returns `handoff_error`, with no exported pair.

Proposition 15.3 (Agreement and partiality). *Under A1–A5, a successful adapter result exports the current record's paper and status, with exactly the recomputed accepted sign and the supplied metadata; P retains that route and source. On the principal matched inputs (B_0, s_0) and (B_1, s_1) it returns*

$$C_0 = (\mathbf{agentic-c0}, p_0, 1, 1, 0), \quad C_1 = (\mathbf{agentic-c1}, p_0, 1, 1, 1).$$

This does not make the adapter total on all memories and records.

Proof. Every successful branch satisfies the equality and membership tests. The exclusive sign defines one Boolean; (15.5) equates it to the source status. Paper equality and the current-pointer test bind that source to p_0 at this stage. A1–A2 supply its intended, reliable status meaning; the checker does not prove those premises. Copying m_0 and applying A5’s version assignment supplies the remaining context fields. The returned receipt records precisely these tested and computed objects. Theorem 15.1 gives the two principal signs, so all tests pass on their matching supplied records.

For partiality, pair B_{10} , whose projection is $\{p, q\}$, with the false-status s_0 . Even if all identifier tests pass, (15.5) fails. A missing negative snapshot cannot establish initial agreement either. A record naming another paper fails paper equality; if its literal also names that paper, the positive-literal test fails too. $\square$

The four-row algebraic agreement theorem therefore must not be read as four successful temporal exports using s_0, s_1 . The default’s accepted $\neg q$ alone supplies no reliable negative observation. The receipt P is also not an action exception certificate: the raw action monitor below does not inspect P or authenticate the current pointer.

15.3 Runtime reasons, purpose and the monitor

A6 fixes policy version `policy-1`, reviewer set `\{reviewer-1\}`, warning-allowed actions $W = \{\text{ask}\}$, and rule priority $r_2 \succ r_1$. All failed citation checks are blocking. The action domain is the single request `ask` and well-formed citation proposals

$$x = (id, cite, p_0, purpose, warning), \quad purpose \in \{\text{evidence}, \text{discussRetraction}\}. \quad (15.6)$$

The example covers no arbitrary search, writing or tool domain.

At C_0 the only runtime tree is

$$A_1 : recent(p_0), relevant(p_0) \Rightarrow citeEvidence(p_0).$$

At C_1 add

$$A_2 : retracted(p_0) \Rightarrow \neg citeEvidence(p_0), \quad A_3 : retracted(p_0) \Rightarrow discussRetraction(p_0).$$

The complete pre-preference attack relation is $\{(A_1, A_2), (A_2, A_1)\}$. With the declared local priority, only $A_2 \rightarrow A_1$ remains a defeat; A_3 has neither attack nor defeat. Evaluating defeats gives

$$E_{act,0} = \{A_1\}, \quad E_{act,1} = \{A_2, A_3\}. \quad (15.7)$$

Indeed these are the unattacked sets and first defence iterates; unattacked A_2 prevents any defence of A_1 in the second case, so both iterates are fixed. Theorem 6.6 applies. These A_i are distinct from revision trees R_p, R_d, R_q .

The support map at stage t is $S_t(x) = \{A_1\}$ for evidential citation, $S_t(x) = \{A_3\}$ for discussion, intersected with the current runtime vertex set; $S_t(\text{ask}) = \emptyset$. Set

$$\text{Just}_t(x) \iff E_{act,t} \cap S_t(x) \neq \emptyset.$$

Acceptance supplies a reason for an action; it does not supply the action’s purpose or authority.

Definition 15.4 (Bound discussion exception). *A certificate has fields*

$$c = (id, Cver, \Pi ver, p, purpose, warning, scope, reviewer).$$

The first field binds the proposal identifier. For current C_t, Π_t , $\text{ValidEx}_t(x, c)$ holds exactly when x is a citation of the context’s retracted paper; the proposal, context, policy and paper

*identifiers match; both purpose fields are **discussRetraction**; both warning fields are true; scope is retraction-discussion-only; and the reviewer is in the current policy's authorized set. An absent or malformed certificate makes this predicate false. Define the local citation permission*

$$\text{Perm}_t(x, c) = \begin{cases} 1 & x = \text{ask}, \\ 1 & x \text{ cites the current paper as evidence and it is not retracted}, \\ 1 & \text{ValidEx}_t(x, c), \\ 0 & \text{otherwise.} \end{cases} \quad (15.8)$$

Every conjunct is required. Matching a reviewer field does not authenticate its author; A4 supplies the trusted identity. This permission is the citation policy's predicate, not [Definition 10.5](#)'s all-foil comparative calculation. A discussion exception changes neither the defeat graph nor evidential eligibility.

For clarity write ι_t for the textbook's intent record r_t , separating it from the names of the priority rules:

$$\iota_t(x, c) = (x, C_t, \Pi_t, F_{\text{act},t}, E_{\text{act},t}, S_t, c). \quad (15.9)$$

The trusted caller appends it before monitoring. Let $\text{Audit}_t(x, c)$ mean that this exact current record belongs to the supplied log. Retaining the pre-preference attack relation as additional provenance preserves the distinction between attack and defeat.

Put $J = \text{Just}_t(x)$, $U = \text{Audit}_t(x, c)$. The monitor is

$$\text{Mon}_t(x, c) = \begin{cases} \text{block} & \neg \text{Safe}(x, C_t) \vee \neg \text{Perm}_t(x, c), \\ \text{block} & (\neg J \vee \neg U) \wedge x.\text{action} \notin W, \\ \text{warn} & (\neg J \vee \neg U) \wedge x.\text{action} \in W, \\ \text{pass} & \text{otherwise.} \end{cases} \quad (15.10)$$

Branches are tested in that order; for **ask**, $x.\text{action} = \text{ask}$. Safety is a separately supplied test, not an inference from acceptance.

Theorem 15.5 (Exact local monitor contract). *On a well-formed declared input,*

$$\text{Mon}_t(x, c) = \text{pass} \iff \text{Safe}(x, C_t) \wedge \text{Perm}_t(x, c) \wedge \text{Just}_t(x) \wedge \text{Audit}_t(x, c). \quad (15.11)$$

*A citation cannot receive **warn**. A safe ask with the declared empty support receives **warn**, not **pass**, whatever its audit value.*

Proof. Reaching the last branch excludes failure of safety or permission. If $\neg J \vee \neg U$ held, exactly one middle branch would apply according to membership in W , so both J, U hold. Conversely the four conjuncts make each earlier guard false, giving **pass**. A citation's action is outside W , so any middle-branch failure blocks. For safe **ask**, permission is one but its empty support makes $J = 0$; the first branch is skipped and the third applies. $\square$

The theorem concerns the supplied check meanings and inputs. It does not establish external tool safety, authentic identity, an untampered log or the actual currentness of a whole supplied bundle.

15.4 Dispatch needs a current receipt and an output contract

A7 restricts outputs to two symbolic templates. The separately requested evidential purpose emits $\text{EvidenceCitation}(p_0)$. Discussion emits

$$\text{RetractionDiscussion}(p_0, \text{RETRACTED}, s.\text{locator})$$

restricted to the withdrawal and its notice, without using the paper as scientific evidence. Warning true requires the supplied visible-output check, not a generator’s self-declared flag.

Fix

$$x_e = (\text{e0}, \text{cite}, p_0, \text{evidence}, 0), \quad x_d = (\text{d1}, \text{cite}, p_0, \text{discussRetraction}, 1).$$

Let k_1 bind x_d to C_1 , **policy-1**, paper p_0 , discussion purpose, true warning, discussion-only scope and reviewer 1. It is supplied by that authorized reviewer, not derived from q or A_3 .

The trusted caller’s **O1** permits citation dispatch only when the retained adapter receipt is current, the proposal and A7 template fields match, and the monitor returns pass for that same stage’s bound input. A changed proposal payload gets a fresh proposal identifier. The caller appends exact intent before monitoring, records the decision, and after any dispatch records the actual supplied outcome. The current-stage checks are conditions at dispatch; a pass saved before an intervening receipt is not such a check. This is the declared sequential handoff, not a concurrency or authentication theorem about an arbitrary implementation.

Theorem 15.6 (Conditional citation dispatch). *Under A1–A7 and O1, every dispatched citation has a current source-linked receipt, matching checked template, supplied safety, policy permission, grounded action support and exact current intent. At retracted C_1 , a dispatched citation is a purpose-bound marked discussion, not an evidential citation. These conclusions imply neither successful delivery nor completion of the original scientific goal.*

Proof. O1 gives current receipt, matching template and pass on the same bound stage. Proposition 15.3 gives the receipt’s recomputed status/source connection under A1–A5; Theorem 15.5 gives the four monitor conjuncts. The definition of J refers to the recomputed grounded set (15.7), and the definition of U gives exact intent.

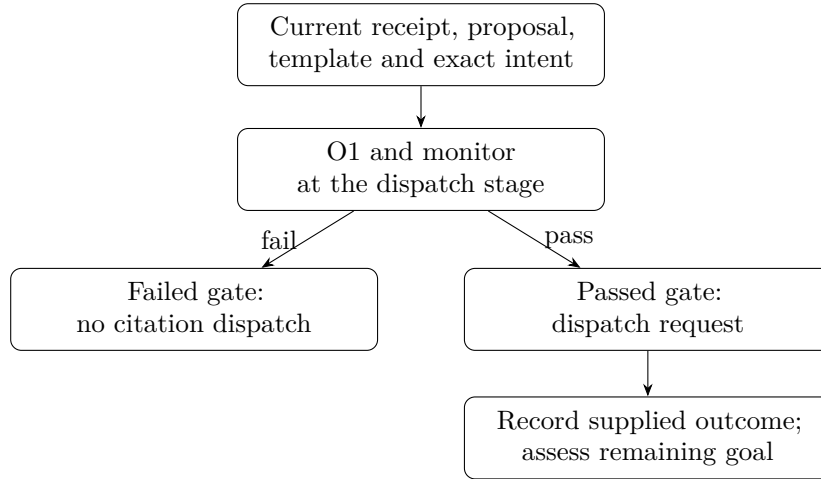
At retracted C_1 , a citation is not **ask**, and the nonretracted-evidence case of (15.8) is false. Its permission therefore requires **ValidEx₁**, whose conjunction fixes discussion purpose, warning and scope. A7 and O1 give the matching checked discussion template. For the limitations, the same passing pre-dispatch input can be followed by a supplied successful delivery or by a supplied delivery failure: no monitor predicate constrains that later outcome. Even the successful discussion need not contain eligible evidence, as the goal argument below makes explicit. $\square$

With **Safe** = 1, the relevant checks are

proposal, context, exception, intent	J	Perm	U	Mon	(15.12)
x_e, C_0 , none, exact	1	1	1	pass	
x_e, C_1 , none, exact	0	0	1	block	
x_d, C_1 , none, exact	1	0	1	block	
x_d, C_1, k_1 , exact	1	1	1	pass	
x_d, C_1, k_1 , none	1	1	0	block	

Each row follows by (15.7)–(15.11). Supplying k_1 to x_e cannot change its support and fails the exception’s proposal/purpose binding. Mutating only k_1 ’s context to C_0 , or only its paper to another paper, gives $(\text{Perm}, U) = (0, 1)$ when a fresh exact intent logs that mutated object. Recording an invalid certificate repairs neither equality nor permission.

For the successful discussion case, one supplied execution outcome is delivery of the visibly marked notice, with locator **fixture:notice-1#status**, explaining that the paper is not used as evidence. A failure outcome instead records non-delivery. The intent, pass and dispatch are identical up to that point; only the actual outcome distinguishes them.



The picture is the citation caller's finite control flow. It does not make inference, public dialogue and balancing mandatory consecutive stages for every service.

15.5 Unchanged conclusions and changed currentness

Before dispatch at stage 1, receive a new positive record s_2 for the same paper and issuer, with identifier s_2 , document `notice-2`, locator `fixture:notice-2#status`, and literal q . A5 gives the new context version `agentic-c2`. Set union is idempotent:

$$\{q\} \cup \{q\} = \{q\}.$$

A3 creates one observation argument per observation literal, not per receipt occurrence. Consequently the revision graph, its grounded arguments, $\Gamma = \{p, q\}$, runtime graph, grounded action set and support map are unchanged. The receipt identifier, document version, locator, provenance envelope, pointer and context version all change.

Let ι_1 be the old exact intent with k_1 . The renewed k_2 changes its context field to C_2 and is supplied anew by the reviewer. The agent is not authorized to renew that field on its own. At the actual current C_2 , with $J = \text{Safe} = 1$,

certificate and log content	Perm	U	Mon	(15.13)
k_1 , only ι_1	0	0	block	
k_1 , exact C_2 intent naming k_1	0	1	block	
k_2 , only ι_1	1	0	block	
k_2 , exact C_2 intent naming k_2	1	1	pass	

The first column of values is determined by context equality; the second by full tuple equality in (15.9). In the third row both context and certificate differ from the old intent. Only the last row passes (15.11), still subject to O1. Its current discussion template uses the new locator.

Proposition 15.7 (The complete old-bundle counterexample). *The stateless monitor cannot infer external currentness from internally consistent bound fields. An old complete bundle can pass it while O1 forbids dispatch at the actual new stage.*

Proof. After the new receipt, supply the raw monitor with the entire old tuple $x_d, C_1, \Pi, k_1, [\iota_1]$, its old recomputed graph and support map, and $\text{Safe} = 1$. It has exactly the same inputs as the passing fourth row of (15.12), so determinism gives pass again. There is no current-record pointer among these raw monitor inputs. The trusted caller, however, has current identifier s_2 . The old associated receipt names $s_1 \neq s_2$, so O1's current-receipt test fails and it forbids dispatch. $\square$

The mixed-field failures in (15.13) therefore do not prove universal stale-data detection. Nor does the assumed O1 caller prove that every external component must use it. The theorem states exactly which enforcement point supplies currentness. Chapter 13’s weight-revision interface similarly uses a current version, but its authority is for a different coordinate and purpose; neither authorization transfers.

15.6 Outcomes, remaining goals and a bounded progress test

An orchestration record can retain

$$O_t = (B_t, G_t, F_t, E_t, x_t, \text{Trace}_t, c_t, \Pi_t),$$

where Trace_t distinguishes receipts, intents, decisions and outcomes. One may add the adapter’s receipt and current pointer explicitly or retain them in the trace. This is a typed record, not by itself an enforcement theorem. For Chapter 14’s policy adviser, B would retain the premise evidence, F, E the policy graph and grounded set, x the decision candidate, and the caller must still prove Theorem 14.3’s release conjunction. For the research assistant, O1 supplies a different dispatch obligation.

Let the original goal require an answer using an eligible evidential source, and stipulate p_0 is the only supplied candidate. At C_1 or C_2 , (15.8) disallows its use as evidence; the eligible candidate set is empty. A successful authorized discussion can therefore achieve the separate status-information subgoal while leaving the original goal open. Formally, original-goal achievement requires at least one eligible source; that necessary condition is false. Failure of delivery does not even establish the status-information subgoal. A new source or an acknowledged information request is a new event, not an inferred result of the old pass.

For a finite observed event list $H = (e_1, \dots, e_n)$, fix a positive window k and a supplied observable predicate $P(e)$: the event records either acquisition of a newly eligible source or an acknowledged request for missing information. Define

$$\text{Progress}_k(H) = \begin{cases} 1 & \exists i \in \{\max(1, n - k + 1), \dots, n\} P(e_i), \\ 0 & \text{otherwise.} \end{cases} \quad (15.14)$$

The index set is empty for $n = 0$. This tests the recorded events; it does not authenticate them.

Proposition 15.8 (Window checking without stronger authorization). *Scanning the specified suffix computes (15.14) exactly. A supplementary progress policy that preserves every original monitor block and otherwise adds a progress warning when $\text{Progress}_k = 0$ never authorizes a blocked citation. Passing this progress test establishes neither goal completion nor eventual completion.*

Proof. Initialize a Boolean accumulator to false and disjoin $P(e_i)$ while scanning the suffix. After every scanned prefix it equals the existential test over that prefix; at termination it is (15.14). If the original monitor blocks, the supplementary policy returns block before testing progress. Otherwise it can only retain the original handling or add a warning. Thus it cannot create permission or O1 dispatch authority.

For the last claim, take a trace with an acknowledged request in every k -event window but no acquired eligible evidence. Every window passes. Stipulate that requested evidence never arrives, so the original necessary evidential condition is always false. The infinite continuation of this same pattern refutes an inference of eventual completion, while each finite prefix with a request refutes completion now. $\square$

For $k = 3$, the supplied events are receipt of s_1 , attempted marked discussion and recorded delivery failure. None qualifies, so progress is zero. Append an acknowledged information request: the new window contains that event and progress becomes one. A request acknowledgement is still not a delivered evidential answer. No long-run planning or liveness theorem is imported.

15.7 Public reasons and normative interpretation

The action case does not supply comparative weights or a selection rule among information-seeking alternatives. To use Chapter 12’s practical procedures for that purpose, one must supply those new inputs. Similarly a legal public request and reply can make a source or authority answerable, but a designer must specify which returned record fills which input field of this action policy. Public legality alone does not renew k_1 .

The local norm

$$N : \text{retracted}(X) \Rightarrow O(\neg \text{citeEvidence}(X)) \quad (15.15)$$

uses O only to express the stated obligation; no deontic axiom system is assumed. Given retraction, knowledge and a deliberate evidential citation as stipulated facts, the act is a knowing violation. Authorized marked discussion has a different purpose and does not violate this narrow evidential norm, although it uses an exception to the broader citation policy. The checker infers neither knowledge nor intent from the emitted words.

An institutional construction can instead supply a paper-only filing norm, an accessibility assistant’s knowledge of it, a digital filing that violates it, and an explicit request to reform the exclusionary format requirement. These supplied facts locate a proposed civil-disobedience interpretation. They do not establish its institutional acceptance or infer a reform intention from digital format alone. Reform, misleading intent and the appropriate response need further facts; they are not exhaustive disjoint monitor outputs.

The finite results in this chapter rely on A1–A7 and O1 at their stated interfaces. The A-BDI metamodel, epistemic planning, normative institutions and ReAct remain the textbook’s comparison and research routes, not unproved general guarantees transferred to this construction.

Related textbook exercises

- **15.1–15.3:** exact attack/defeat relations, purpose-specific support, permission failures and the contextual explanation of a blocked evidential citation.
- **15.4, 15.10:** supplied knowledge and conduct, the narrow evidential norm and the institutional reform interpretation without inferred intention.
- **15.5(a–d)** (Study S15.1): four restricted revision states, both routes, conclusion projection and retention of the defeated default.
- **15.6(a–b), 15.7–15.8:** current dispatch, successful versus failed delivery, typed orchestration and the distinct return to Chapter 14’s service question.
- **15.9(a–c)** (Study S15.1): complete exception binding, wrong-context/wrong-paper mutations and missing intent.
- **15.11:** the three-step progress test, new window, block precedence and noncompletion counterexample.
- **15.12(a–d), 15.13(a–d):** receipt, partial adapter, separate graphs and authorities, unchanged semantics with new provenance, all renewal combinations and the old bundle.
- **15.14(a–h):** the capstone’s typed handoffs, evaluations, contextual explanation, bounded guarantee, failed variant, further receipt and remaining-goal account.
- **15.L1–15.L2:** explicitly external source and research-capstone comparisons; no additional general alignment, planning or institutional theorem is imported.

Recurring-example index

These routes return to an existing input as the question changes. Each destination retains its own hypotheses and output type; the textbook owns the corresponding exercise statements and canonical answers.

Case	Where to return and what changes
Structured priorities	Propositions 5.10–5.11 (pp. 20–21) supply the Scot construction and local attacked-subargument comparisons; §7.2 (p. 30) compares them with source rule application. The shared-rule example then reappears in §7.5 (p. 35) as an AA2LP encoding: preserving a graph does not validate its source representation.
Two semantic graphs	Propositions 6.7 and 6.8 (pp. 26, 26) distinguish the linked-cycle graph from the registered reuse graph. Example 12.10 (p. 68) uses the latter for conditioned updates; Theorem 13.2 and Proposition 13.3 (pp. 72, 73) revisit it for contextual certificates and preservation.
Public response	Example 9.3 (p. 43) tracks public fields. Proposition 12.11 (p. 69) checks and selects bounded offered continuations; §13.4 (p. 74) targets explanatory requests; Proposition 14.6 (p. 80) distinguishes a certified plan from an actual history.
Comparative permission	Example 10.6 (p. 51) gives the dinner comparisons and selectors. Proposition 12.12 (p. 70) supplies the procedure; §13.5 (p. 74) supplies explanation and authorized weight revision; Proposition 14.7 (p. 81) binds the practical service to its version and named selector.
Release and revision	Theorem 14.3 (p. 79) binds a grounded route to release. Theorem 15.1, Proposition 15.3 and Theorems 15.5–15.6 (pp. 85–89) move through the four revision states, adapter, monitor and dispatch. Propositions 15.7–15.8 (pp. 90–91) separate renewed currentness, progress and achievement.

Sources and proof-status guide

This apparatus indexes the source and proof-status notes preserved in the chapters. It introduces no additional imported theorem. Page and theorem locators refer to the source version explicitly named at the point of use; supplied manuscripts and published articles need not share pagination. The approved textbook, its supplied studies and collected exercise mathematics are the common project sources.

Opening source notes

- **Source methods.** David Makinson, *Bridges from Classical to Nonmonotonic Logic*, 2005, Defs. 2.1, 2.4, 3.1, 3.8 and Lemma 3.2 at the pages cited in Chapter 1. The finite MCS guard, source priority and hard-premise filter are textbook restrictions.
- **Abstract argumentation.** Phan Minh Dung, *On the acceptability of arguments and its fundamental role in nonmonotonic reasoning, logic programming and n-person games*, Artificial Intelligence 77 (1995), Defs. 5, 6, 13, pp. 326–328. Local defense and stability tests are stated in the companion.
- **Base representation.** Rienstra, van der Torre and Yu, supplied reasoning-alignment manuscript, Sec. 3.2, Defs. 3–5 and Prop. 1, internal pp. 10–11. It states the result, credits Chen, Liao and van der Torre (2024), and does not prove it there. Theorem 1.7 gives a complete finite classical proof; the hard-premise corollary is restricted.
- **Agency.** Supplied 52-page agent-defense proof, Defs. 5–6, internal pp. 4–5, Example 5, pp. 6–7. Published pagination differs. Coalitions and enumeration are derived textbook constructions. The majority profile is attributed in textbook Chapter 2 to Baumeister, Neugebauer and Rothe, Handbook Vol. 2, Ch. 4, Sec. 3, pp. 261–262.

Later source and proof locations

Reconstruction and structured reasons (Chapters 3–5).

The stipulated reconstructions and question types come from the textbook and Study S3.1. Chapter 5 preserves the canonical Modgil–Prakken signature and the corrected 2018 preference clause, with exact local attack targets and finite-fragment boundaries. Its closing source note distinguishes imported definitions from the proved restricted minimum-rank result.

Semantics, representation and limits (Chapters 6–8).

Chapter 6 develops the finite semantic proofs with Dung’s source locators. Chapter 7 develops AA2LP and separately retains the exact attributed classical/SP–PDL conjunction as reported claim 7.5, refuted under the recorded construction. Lemmas 7.5a–7.5c and Counterexample 7.5d are accepted developed mathematics: no stable extension of the witness extracts its PDL full conclusion set. This refutes the conjunction without separately disproving existence or uniqueness. The source claims are AAMAS 2016, Theorems 5.2–5.3, pp. 632–633; arXiv:1609.05224v1, Theorems 3.13–3.14, pp. 24–27; and the TAFA-17 author manuscript, Theorem 1, p. 10. The separate Algorithm 1 procedure defect and equality obstruction are documented in the accepted source-discrepancy evidence; repairing that procedure alone does not settle equality. No raw-DWL or restricted-domain replacement is adopted. Chapter 8’s developed closure and bounded PDL/P2/P6 results are preserved. The author retains the wider Dung correspondences (§7.6), general ASPIC+ rationality results and redundant-default invariance as attributed conditional imports; the examples do not prove them.

Dialogue, issues and cross-input principles (Chapters 9–11).

The public-transition and Fatio operational/axiomatic comparisons, Carneades standards and comparative-reason constructions keep their point-of-use locators. The finite grounded-game and directionality proofs are developed in the companion. The Libra treatment follows approved Study S9.1 and its solutions. No fresh direct primary-paper comparison was completed in the recorded companion review. No general Libra soundness theorem is imported. The recorded JELIA example-list concern and SFC2/SCF2 discrepancy are retained without strengthening them.

Computation and explanation (Chapters 12–13).

The checker, encoding, reduction and contextual-certificate proofs are developed under their stated conditions. CNF-SAT NP-completeness is identified imported background. The AXPLR probability identity, classification-clause exclusion and version-specific boundary are recorded in Chapter 13. The exact published PDF remains identified but unretrieved after bounded repository and publisher attempts; decisive publisher HTML glyphs are missing. The renewed source report records exact routes and responses. No final-publication defect or version identity is inferred.

Services, revision and bounded action (Chapters 14–15).

The supplied system, score, report-selection and handoff studies are developed under their explicit finite assumptions. Chapter 14 proves the score and finite closed-theory remainder constructions; Chapter 15 states A1–A7 and O1 for route, adapter, monitor and dispatch results. No general revision, authentication, concurrency, institutional, planning or liveness theorem is imported.

Editorial evidence and author disposition

The editable source package retains the coverage map, source/exercise correspondence and full existing review evidence where available. The opening handoff summary is explicitly distinguished from the unavailable original review reports. Presentation-comparator evidence remains editorial evidence; it is not a mathematical premise. The wording-only completion checks the changed text, affected rendering and references, reusing unaffected evidence. The frozen whole-companion review remains intact. The author has granted whole-companion scholarly approval within the declared scope; publication readiness and the separate AXPLR source-verification item are not thereby settled.